MILITARY SPECIFICATION

RECORDER - REPRODUCER SET, DIGITAL DATA, AN/ASH-33

This specification is approved for use by The Naval Air Systems Command, Department of The Navy, and is available for use by all Departments and Agencies of The Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the design, construction, testing, performance, digital protocol and digital format requirements for the Recorder-Reproducer Set, Digital Data, AN/ASH-33, herein referred to as the Digital Magnetic Tape Set (DMTS). The DMTS consists of the following subunits (WRA's):

	Quantity Per Set
Control, Magnetic Tape, C-10553/ASH-33 (DMTC)	1
Recorder-Reproducer, Digital Data, RD-450/ASH-33 (DMTU)	1 to 4

The DMTU operates with a magnetic tape cartridge.

The DMTS provides computer controlled program load and auxiliary bulk storage memory for the AN/USQ-78(V) Display and Control System (DCS), the AN/ASQ-114, Computer Set, Digital (CP-901), and the AN/AYK-114(V) Computer Set, Standard Airborne.

Beneficial comments (recommendations, additions, deletions), any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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When configured to interface with only the AN/ASQ-114, Computer Set, Digital (CP-901), the nomenclature shall be:

AN/ASH-33, Recorder-Reproducer Set, Digital Data consisting of:

C-10553/ASH-33 Control, Magnetic Tape

RD-450/ASH-33 Recorder-Reproducer, Digital Data

When configured to interface with the CP-901 and the AN/USQ-78(V), Display and Control System (DCS), the nomenclature shall be:

AN/ASH-33A, Recorder-Reproducer Set, Digital Data consisting of:

C-10553A/ASH-33 Control, Magnetic Tape

RD-450/ASH-33 Recorder-Reproducer, Digital Data

Configured to interface with the CP-901, the DCS, and the AN/AYK-14(V) Computer Set, Standard Airborne the nomenclature shall be:

AN/ASH-33( ), Recorder-Reproducer Set, Digital Data consisting of the:

C-10553( )/ASH-33 Control, Magnetic Tape

RD-450/ASH-33 Recorder-Reproducer, Digital Data

1.2 Associated equipment. This equipment operates in conjunction with the associated equipment listed in 6.8.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified (see 6.4), the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-5	Capacitors, Fixed, Mica Dielectric, General Specification for
MIL-C-62	Capacitor, Fixed, Electrolytic (DC, Aluminum, Dry Electrolyte, Polarized), General Specification for
MIL-P-116	Preservation, Methods of

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DoD-D-1000	Drawings, Engineering and Associated Lists
MIL-C-3965	Capacitor, Fixed, Electrolytic (Nonsolid Electrolyte), Tantalum, General Specification for
MIL-B-5087	Bonding, Electrical, and Lighting Protection, for Aerospace Systems
MIL-W-5088	Wiring, Aerospace Vehicle
MIL-E-5400	Electronic Equipment, Aerospace, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-E-5558	Enamel, Wrinkle-Finish, for Aircraft Use
MIL-R-5757	Relay, Electromagnetic, General Specification for
MIL-C-6781	Control Panel, Aircraft Equipment, Rack or Console Mounted (ASG)
MIL-M-7793	Meter, Time Totalizing
MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aircraft and Missiles
MIL-C-14409	Capacitors, Variable (Piston Type, Tubular Trimmer), General Specification for
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-M-18012	Markings for Aircrew Station Displays, Design and Configuration of
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment, Including Ground Support Equipment
MIL-S-19500	Semiconductor Device, General Specification for
MIL-B-23071	Blower, Miniature, for Cooling Electronic Equipment, General Specification for
MIL-C-23269	Capacitors, Fixed, Glass Dielectric, Established Reliability, General Specification for
MIL-C-25050	Color, Aeronautical Lights and Lighting Equipment, General Requirements for (ASG)
MIL-P-28809	Printed Wiring Assemblies
MIL-M-38510	Microcircuits, General Specification for

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MIL-C-38999	Connectors, Electrical, Circular, Miniature, High Density, Quick Disconnect, Environment Resistant, Removable Crimp and Hermetic Solder Contacts, General Specification for
MIL-C-39001	Capacitors, Fixed, Mica Dielectric, Established Reliability, General Specification for
MIL-C-39003	Capacitors, Fixed, Electrolytic (Solid Electrolyte), Tantalum, Established Reliability, General Specification for
MIL-C-39006	Capacitors, Fixed, Electrolytic (Nonsolid Electrolyte), Tantalum, Established Reliability, General Specification for
MIL-C-39006/22	Capacitor, Fixed, Electrolytic (Nonsolid Electrolyte), Tantalum, (Polarized, Sintered Slug), 85 Deg C (Voltage Derated to 125 Deg C), Established Reliability, Style CLR79
MIL-R-39007	Resistors, Fixed, Wirewound (Power Type), Established Reliability, General Specification for
MIL-R-39008	Resistors, Fixed, Composition (Insulated), Established Reliability, General Specification for
MIL-C-39014	Capacitors, Fixed, Ceramic Dielectric (General Purpose), Established Reliability, General Specification for
MIL-R-39016	Relays, Electromagnetic, Established Reliability, General Specification for
MIL-H-46855	Human Engineering Requirements for Military Systems, Equipment and Facilities
MIL-R-55182	Resistors, Fixed, Film, Established Reliability, General Specification for
MIL-C-83421	Capacitors, Fixed, Supermetallized, Plastic Film Dielectric (DC, AC, or DC and AC), Hermetically Sealed in Metal Cases, Established Reliability, General Specification for

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-198	Capacitors, Selection and Use of

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MIL-STD-210	Climatic Extremes for Military Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electronic Emission and Susceptibility Requirements for the Control Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics Measurement of
MIL-STD-463	Definitions and System of Units, Electromagnetic Interference Technology
MIL-STD-470	Maintainability Program Requirements (for Systems and Equipment)
MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-740	Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment
MIL-STD-756	Reliability Modeling and Prediction
MIL-STD-781	Reliability Test Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-794	Parts and Equipment, Procedures for Packaging of
MIL-STD-810	Environmental Test Methods
MIL-STD-883	Test Methods and Procedures for Microelectronics
MIL-STD-965	Parts Control Program
MIL-STD-1397	Input/Output Interfaces, Standard Digital Data, Navy Systems
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-STD-1553	Aircraft Internal Time Division Command/Response Multiplex Data Bus
MIL-STD-1562	Lists of Standard Microcircuits

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MIL-STD-1635	Reliability Growth Testing
MIL-STD-1679	Weapon System Software Development
MIL-STD-2068	Reliability Development Tests
MIL-STD-2074	Failure Classification for Reliability Testing
MIL-STD-2084	General Requirements for Maintainability of Avionic and Electronic Systems and Equipment
MS 17321	Meter, Time Totalizing, Miniature, Digital 28 Volt DC
MS 17322	Meter, Time Totalizing, Miniature, Digital, 115 Volt, 400 Hertz

HANDBOOKS

MILITARY

MIL-HDBK-217	Reliability Prediction of Electronic Equipment
MIL-HDBK-472	Maintainability Prediction

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

OTHER GOVERNMENT DOCUMENTS

DEPARTMENT OF THE NAVY

SECNAVINST 3560.1	Tactical Data Systems Documentation Standards

NAVAL AIR SYSTEMS COMMAND

NADC Proteus/LAB 007	Proteus Digital Channel (PDC), Device Interface Description of NADC Sink Logic
NADC Proteus/LAB 008	Proteus Digital Channel (PDC) Theory of Operation of NADC Sink Logic
NADC Proteus/LAB 013	Proteus Digital Channel (PDC), Device Interface Description of NADC Source Logic
NADC Proteus/LAB 015	Proteus Digital Channel (PDC) Theory of Operation of NADC Source Logic
AR-106	Commonality of Digital Computer Hardware and Programming Language within Weapon Systems, Requirements for

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EI-680	Avionics Installation Instructions for Recorder-Reproducer Set, Digital Data AN/ASH-33
AD-1350	Engineering Drawings and Associated Data
AD-1404	Interface Design Specification (IDS) Digital Recorder-Reproducer Set AN/ASH-33 for P-3C UPDATE III Aircraft Equipment
AS-4613	Application and Derating Requirements for Electronics Components, General Specification for
A30-15590	Proteus Digital Channel

NATIONAL SECURITY AGENCY

NACSEM 5100	Compromising Emanations Laboratory Test Standard Electromagnetics (U)

PUBLICATIONS

NAVAL ELECTRONIC SYSTEMS COMMAND, THERMAL TECHNOLOGY LABORATORY

0967-437-7010 Vol 1-4	NAVELEX Reliability Design Handbook, Thermal Applications

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

ANSI X 3.39-1973	American National Standard, Recorded Magnetic Tape for Information Exchange (1600 CPI, PE)
	(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 First article inspection. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Parts and materials. In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. The requirements of MIL-STD-965 shall apply (see 6.2.2). In so doing, the following shall govern:

a. Microelectronic technology shall be considered and microelectronic items shall conform to requirements specified herein.

b. Other parts and materials requirements shall conform to MIL-E-5400.

c. Nonrepairable subassemblies shall be used in accordance with MIL-STD-2084 and as outlined in MIL-E-5400.

d. When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

e. Microelectronic integrated circuits and large scale integrated circuits shall be used for all arithmetic and logic functions.

f. Selection and application of microelectronic devices shall be in accordance with MIL-STD-454, Requirement 64. All second source microelectronic devices and components shall be mechanically and electrically interchangeable. Precap inspection is required for all microcircuits. Devices shall be soldered or welded into the circuit.

g. Circuits shall be designed to use a minimum of different materials, parts, subassemblies, cards and modules.

h. All passive circuit components shall be established reliability (ER) devices, failure rate level "M" or better.

i. Semiconductors shall be JAN TX, or better.

j. The use of magnesium, metallic oxide rectifiers, nonhermetically sealed wet tantalum capacitors (MIL-C-3965 Types), aluminum capacitors (MIL-C-62 Types), and plastic encapsulated semiconductors and microcircuits is prohibited.

k. Purchased DMTC modular power supplies and mode display shall not be required to meet the parts selection and derating requirements as contained herein. It shall be the responsibility of the DMTS contractor to assure compliance with MIL-E-5400, MIL-STD-454 and MIL-STD-704.

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3.2.1 Nonstandard parts and materials approval. Approval for the use of nonstandard parts and materials, including electron tubes, transistors, diodes and microelectronic devices, shall be obtained (see 6.2.2) as outlined in MIL-E-5400. Microelectronic devices shall be approved as outlined in MIL-STD-965, except that approval shall not be required for devices which are:

a. Listed in MIL-STD-1562

b. Available from at least two sources which are listed in the appropriate QPL of MIL-M-38510, and

c. Procured to the requirements of MIL-M-38510 from a qualified source.

3.2.2 Microelectronics modular assemblies. When used, Microelectronic Modular Assemblies shall meet the requirements of MIL-STD-965. Conformal coatings, encapsulants, embedments or potting materials used with modular assemblies containing circuits and discrete parts shall be easily removable without damage to the assembly.

3.2.3 Modules. The electronic portions of the equipment shall be functionally modularized in accordance with MIL-STD-2084 and AR-106. Detailed modularity requirements are specified in 3.3.7.4.

3.2.4 Electronic parts screening and derating.

3.2.4.1 Semiconductor screening. Integrated circuits, including hybrids, shall be screened to MIL-STD-883, Class B. In addition, the contractor shall provide for 100 percent functional testing of all semiconductors to the Group A requirements of their respective detail specifications, excluding AC testing. The functional testing shall be done at room ambient, -55°C and +125°C temperatures at a facility other than the part manufacturer's. For nonstandard semiconductors, the contractor shall prepare specifications, in accordance with his established format, which include Quality Assurance provisions and these screening requirements (see 6.2.2). (Non-JANTX and non-MIL-M-38510 semiconductors shall be considered nonstandard parts for this purpose. In addition, they require approval in accordance with MIL-E-5400). PROM's, RAM's and microprocessor integrated circuits are excluded from the 100 percent functional testing requirement of this paragraph, as are the DMTC power supply, mode display and alphanumeric display semiconductors.

3.2.4.2 Electronic parts derating. The derating of parts shall be governed by AS-4613 as modified by Table I herein. Temperature requirements shall be met without using thermal grease at the mounting surfaces. Once this requirement is met, thermal grease may be used to achieve even lower temperatures. The contractor shall conduct a part level stress analysis to verify derating compliance. This analysis shall verify that all parameter deratings are compliant under worst-case conditions of piece part stress (based on operating duty cycle and maximum allowable input power levels) occurring simultaneously with worst-case equipment thermal environment

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TABLE I. Electronic part derating

Part Type	MIL-Spec	Parameter	Maximum Stress	Additional Requirements
		Capacitors		
Ceramic	MIL-C-39014	Working Voltage	70%	
Tantalum Solid	MIL-C-39003	Working Voltage Effective Series Resistance Reverse Voltage Ripple Voltage	50%	 Minimum 3Ω/Volt 1 Volt Maximum See MIL-STD-198 Figures 801-1 & 801-2
Mica	MIL-C-5 MIL-C-39001	Working Voltage	70%	MIL-C-39001 Capacitors are preferred
Variable	MIL-C-14409	Voltage	50% of W V dc (Working volts direct current)	
Tantalum Non Solid	MIL-C-39006	Working Voltage	70% of W V dc	No reverse voltage permitted
Metalled Plastic Film	MIL-C-83421	Working Voltage	70% of W V dc or ac	
Glass	MIL-C-23269	Working Voltage	70% of W V dc	
All Others		Voltage	70% of W V dc	
		Resistors		
Carbon	MIL-R-39008	Power Voltage	50% 80%	
Wire Wound	MIL-R-39007	Power	50%	
Metal Film	MIL-R-55182	Power	80% of 125°C Rating	
All Others		Power	50%	

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TABLE I. Electronic part derating (continued)

Part Type	MIL-Spec	Parameter 1/	Maximum Stress	Additional Requirements
		Diodes		
Silicon, General	MIL-S-19500	Storage/Jct Temperature Current Reverse Voltage	60% 75% 75%	
Silicon, Power	MIL-S-19500	Forward Current Storage/Jct Temperature Reverse Voltage	75% 60% 75%	
Silicon, Zener	MIL-S-19500	Storage/Jct Temperature Zener Current	60% 80%	
		Transistors		
Silicon	MIL-S-19500	Storage/Jct Temperature Voltage	60% 70%	
		Integrated Circuits		
Op Amps	MIL-M-38510	Power Diff. Input V Storage/Jct Temperature	80% 80% 60%	
Regulators	MIL-M-38510	Power Diff. Voltage Storage/Jct Temperature	80% 80% 60%	
Digital	MIL-M-38510	Fan-Out Storage/Jct Temperature	90% When not Interfacing with TTL, 100% when interfacing with TTL 60%	

1/ Maximum stress junction (Jct) temperatures are in reference to equipment operating at 55°C ambient

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TABLE I. Electronic part derating (continued)

Part Type	MIL-Spec	Parameter	Maximum Stress	Additional Requirements
Relays				
	MIL-R-5757 MIL-R-39016	DC Resistive Contact Current Inductive Load Lamp Capacitive	80%2/ 33%3/ of Resistive Rating 12.5% of Resistive Rating 12.5% of Resistive Rating	These values shall not be used if they exceed the values specified on the detail specification sheet
Switches				
		DC Resistive Contact Current Inductive load Lamp Capacitive	80%2/ 33% of Resistive Rating 12.5% of Resistive Rating 12.5% of Resistive Rating	
Inductive Devices				
		Current	60% of Rated, 70% of Rated for RF Coils, General Purpose Inductors	
Connectors				
		Contact Current Working Voltage Ambient Temp.	70% of Ratings per Contact 75% of Flash-over voltage 75% of Rated Oper. Temp.	

2/Current through two parallel contacts is limited to one contact rating.
3/Use derated resistive rating if a diode is used to shunt the load.

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(±55°C). When specified in the contract, this analysis shall be included as part of a Part Level Stress Analysis and Prediction Report (see 6.2.2). Any noncompliance to the derating criteria shall require approval by the procuring activity prior to final design acceptance. The following types of parts shall require the approval of the procuring activity prior to being used in new and modified designs.

a. Hot carrier (Schottky) rectifiers

b. PROM's

c. Wet (sintered slug) tantalum capacitors, except MIL-C-39006/22 parts, which do not require prior approval.

d. Germanium semiconductors

3.2.4.3 End-of-life design allowance. The equipment shall be designed such that specified performance is achieved in spite of the component end-of-life drift characteristics specified in Table II.

3.2.5 Printed wiring. Requirement 17 of MIL-STD-454 shall apply for all printed wiring, except that MIL-P-28809 shall apply for repair and modification of printed circuit boards. Should the requirements of Tables I, II, III and IV of MIL-P-28809 be exceeded, the contractor shall notify the procuring activity (see 6.2.2). The procuring activity will determine whether the excess is acceptable or if a new board is required as a change in scope.

3.3 Design and construction. The equipment shall conform with all the applicable requirements of MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total weight. The weight of the DMTU shall be not greater than twenty-eight pounds, including cartridge, and the DMTC shall be not greater than seventy-eight pounds.

3.3.2 Reliability. The contractor shall conduct a DMTS reliability program in accordance with MIL-STD-785 and attachments to the contract (see 6.2.2). On a reorder from a contractor who has previously produced the equipment, the program previously used may be continued unless otherwise specified in the contract (see 6.2.1). (See 4.3, 4.4 and 4.8.)

3.3.2.1 Operational stability. The DMTS in any configuration shall operate as specified herein, continuously or intermittently, for a period of at least 3,000 hours in the operational environment without the necessity for readjustment or scheduled maintenance (excluding air filter replacement and head cleaning).

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TABLE II. Electronic part end-of-life

CHARACTERISTIC DRIFT OR DEGRADATION		
Part Type	Characteristic	Drift or Degradation
Integrated Circuits		
Linear	Input Offset Voltage	±0.5 MV
	Input Bias Current	200% of Rated
	Input Offset Current	200% of Rated
	Open Loop Gain	±20% of Rated
	Slew Rate	±20% of Rated
	Common Mode Rejection	±20% of Rated (in dB)
	Power Supply Rejection	±20% of Rated (in dB)
Transistors		
Silicon Bipolar	Current Gain	±10% of Rated
	Leakage Current	200% of Rated
Junction Field Effect	Transmittance	±15% of Rated
Diodes		
All	Leakage Current	200% of Rated
Switching	Reverse Recovery Time	120% of Rated

3.3.2.2 Operating life. The DMTS shall have a minimum operating life of 10,000 hours with minimal servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor (see 6.2.2) and shall be approved by the procuring activity.

3.3.2.3 Reliability in mean time between failure (MTBF). The DMTS, including Built-in-Test equipment (BITE) and two DMTU's, shall have a MTBF of 650 hours (series logic (i.e., independent of redundancies or degraded operation) when calculated via the part stress analysis method of MIL-HDBK-217 using values of electric stress and component temperatures based on an external ambient temperature of 55°C). The MTBF of 650 hours shall be demonstrated for production equipment in accordance with the Reliability Assurance tests specified in 4.8.3.

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3.3.2.3.1 Reliability prediction and analysis. Analytical reliability models shall be prepared in accordance with MIL-STD-756 and quantified using data in MIL-HDBK-217.

3.3.2.3.2 Reliability tracking. When specified by the contract, the contractor shall review causes for rejection (R_T) of the equipment, as directed by the procuring activity, and shall analyze the data for trends.

a. MTBR (mean time between rejection) =

$$\frac{\text{operating hours}}{\text{total rejection } (R_T)}.$$ All WRA rejections, regardless of cause, shall count.

b. Required MTBR = $\frac{\text{Spec MTBF}}{4}$

c. The required MTBR shall exceed one fourth the specified MTBF (see 3.3.2.3). Otherwise an investigation as to cause shall be initiated and conducted by the contractor.

3.3.2.3.3 Thermal design. Equipment designers shall view thermal design fully as important as circuit design. Criteria for equipment thermal design shall be governed by requirement 52 of MIL-STD-454. Also, 0967-437-7010, Reliability Design Handbook, Volumes 1 through 4 on thermal applications shall be used as a guide for thermal design.

3.3.2.4 Time Totalizing Meter. Each Weapons Replaceable Assembly (WRA) of the DMTS shall have a time totalizing meter which meets the requirements of MIL-M-7793, type MS17322-9 or -10 (for AC) or type MS17321-9 or -10 (for DC). The meters shall be located such that they may be read without the removal of the cover or any part from the unit. Each meter shall indicate "ON" time which is any operating mode including standby, shall have a range from 0 to 9999 hours, and shall not be capable of being reset. In addition, each DMTU shall have a time totalizing meter which shall function whenever tape is moving past the magnetic heads.

3.3.2.5 Storage life. The DMTS shall be designed to meet the performance and service life requirements of this specification after 5 years storage under world-wide climate conditions specified in MIL-STD-210, when packaged in accordance with MIL-STD-794 Level A and MIL-P-116 Method 11A (see 5.1).

3.3.3 Cabling and connections.

3.3.3.1 Cables and connectors. The DMTS shall provide for the use of cables and connectors in accordance with MIL-E-5400, except that high density connectors shall be in accordance with MIL-C-38999, Series 1. All external connectors shall be scoop proof. Input power for the DMTC and DMTU's shall be on separate connectors.

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3.3.3.2 Interconnection cabling. The equipment shall be capable of operation within specification limits using external wiring in accordance with MIL-W-5088.

3.3.4 Control Panels. All rack or console mounted control panels shall be in accordance with MIL-C-6781. The configuration of all control panels shall be approved by the procuring activity prior to first article testing (see 6.2.2).

3.3.5 Interchangeability. The equipment shall meet the interchangeability requirements of MIL-E-5400. In addition, all WRA's having the same part number shall be interchangeable with each other with respect to installation and performance. Parts interchangeability shall be in accordance with MIL-I-8500 and MIL-STD-454, Requirement 7.

3.3.6 Electromagnetic interference.

3.3.6.1 General Requirements. The generation by and the vulnerability of the DMTS to electromagnetic interference shall be in accordance with MIL-STD-461, MIL-STD-462, and MIL-STD-463 as modified by the following paragraphs (see 6.2.2).

3.3.6.1.1 CE02 and CE04. Interconnecting leads, including common returns, may be probed as a bundle as they will be installed in the aircraft.

3.3.6.1.2 CE03. Broadband conducted interference on AC power lines shall be in accordance with the limits shown in Figure 1. Narrowband conducted interference in AC power lines shall be in accordance with the limits shown in Figure 2.

3.3.6.1.3 RE01. The limits shall be in accordance with Figure 3.

3.3.6.1.4 RE02. The Narrowband and Broadband limits shall be in accordance with Figures 4 and 5, respectively.

3.3.6.1.5 CS01. This test shall be performed on all power lines, including returns and panel lighting.

3.3.6.1.6 CS02. A 10 microhenry choke shall be inserted in series with the power line at the point where the line stabilization capacitor is removed. This test shall be performed on all power lines, including returns and panel lighting.

3.3.6.1.7 CS06. Both pulses shown in Figure 6 are applicable.

3.3.6.1.8 RS02. For the case test, the test limit shall be relaxed 12 dB at 400 Hz (5 amperes). For both the case and cable test, the applied susceptibility current frequency shall be varied from 390 Hz through 410 Hz and the AC power to the DMTS maintained at 400 Hz.

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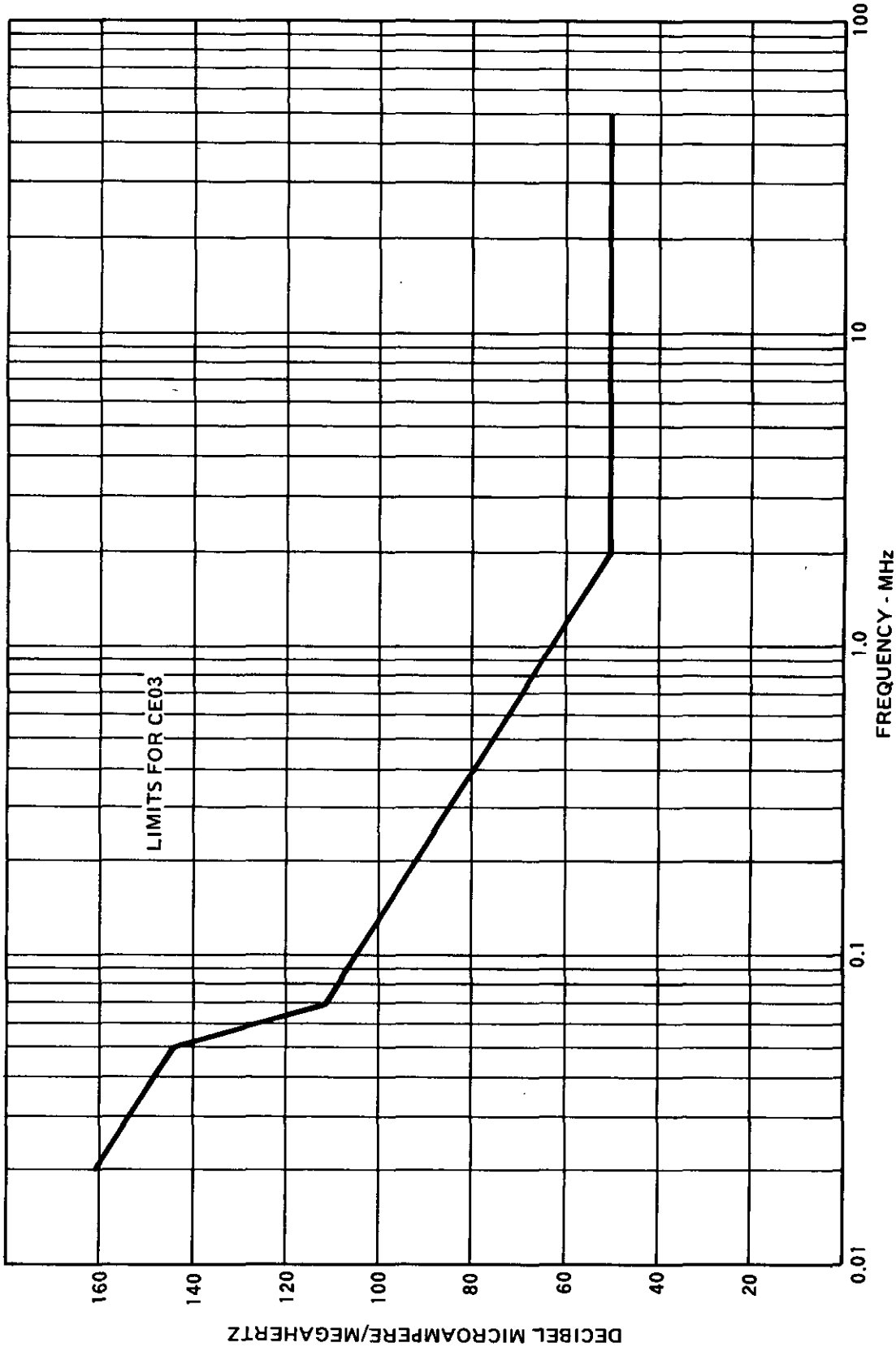


FIGURE 1. Broadband conducted interference on AC lines

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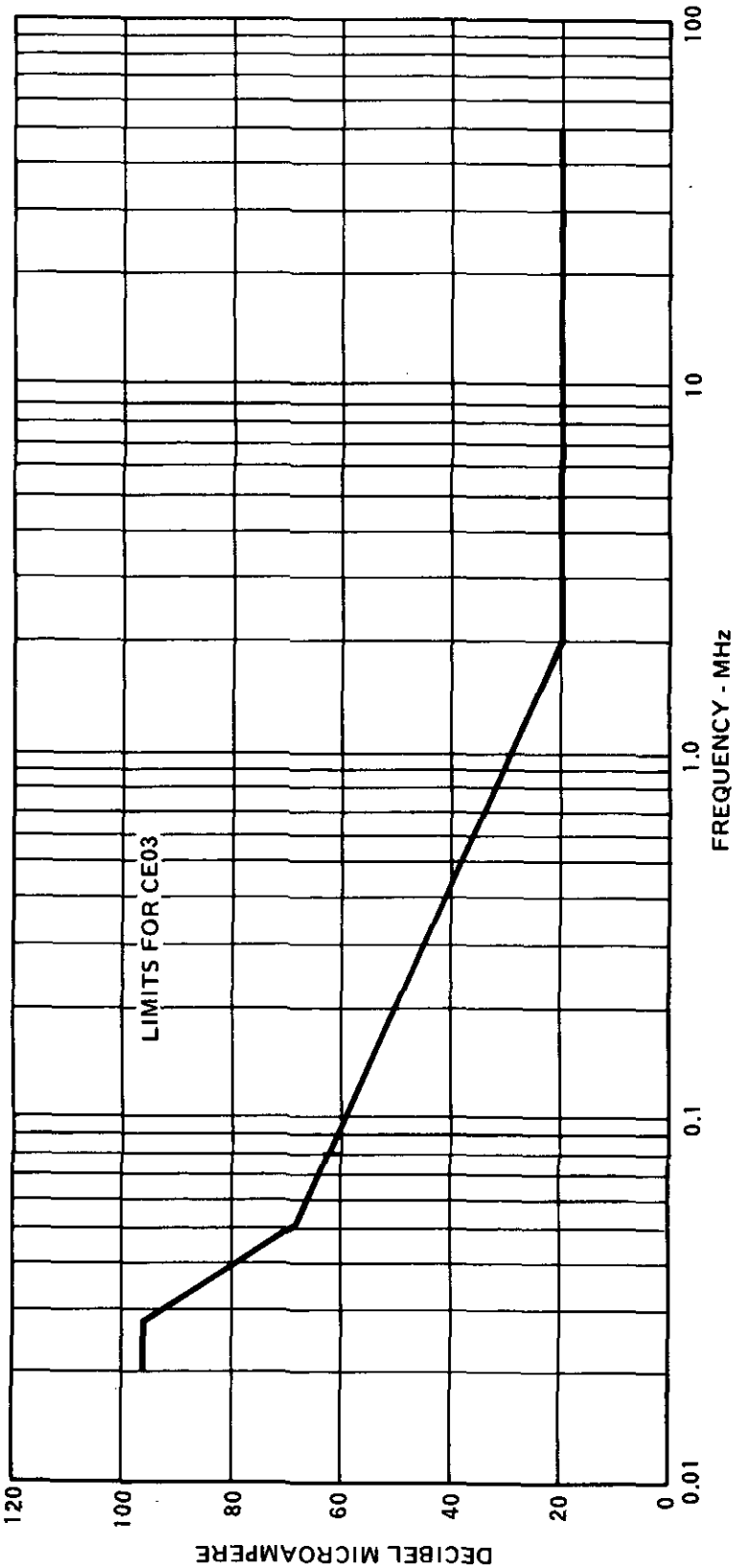


FIGURE 2. Limits for CE03 narrowband conducted interference on AC lines

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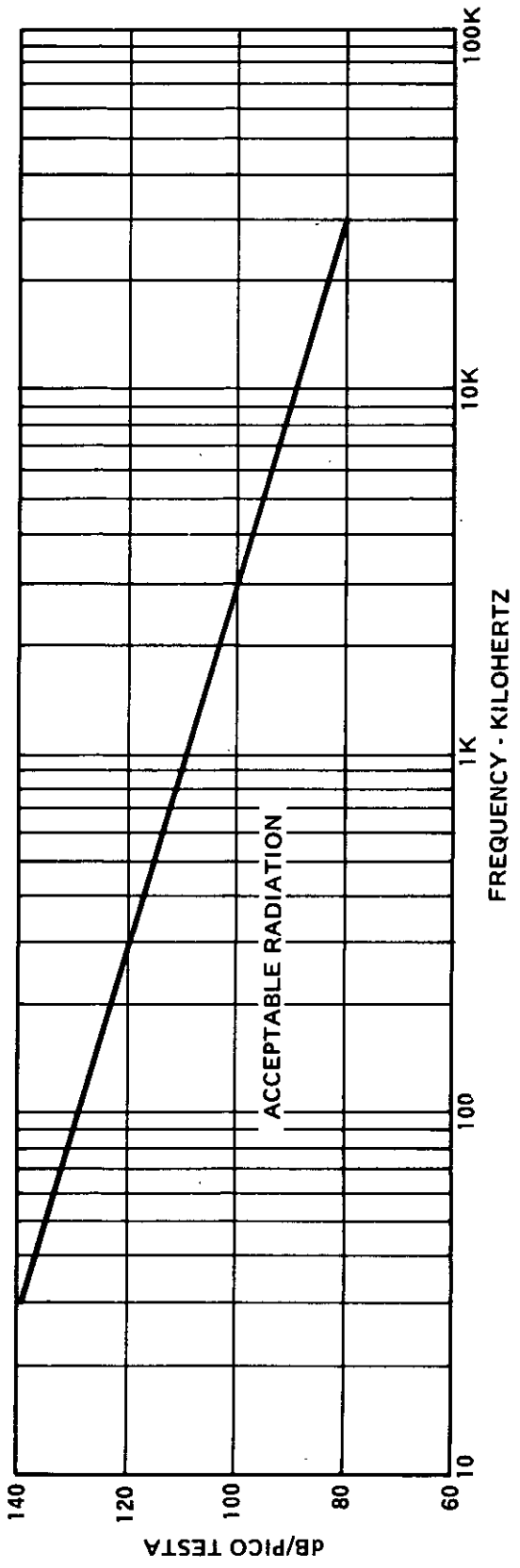


FIGURE 3. Limit for REO1 acceptable susceptibility (1 Hz bandwidth)

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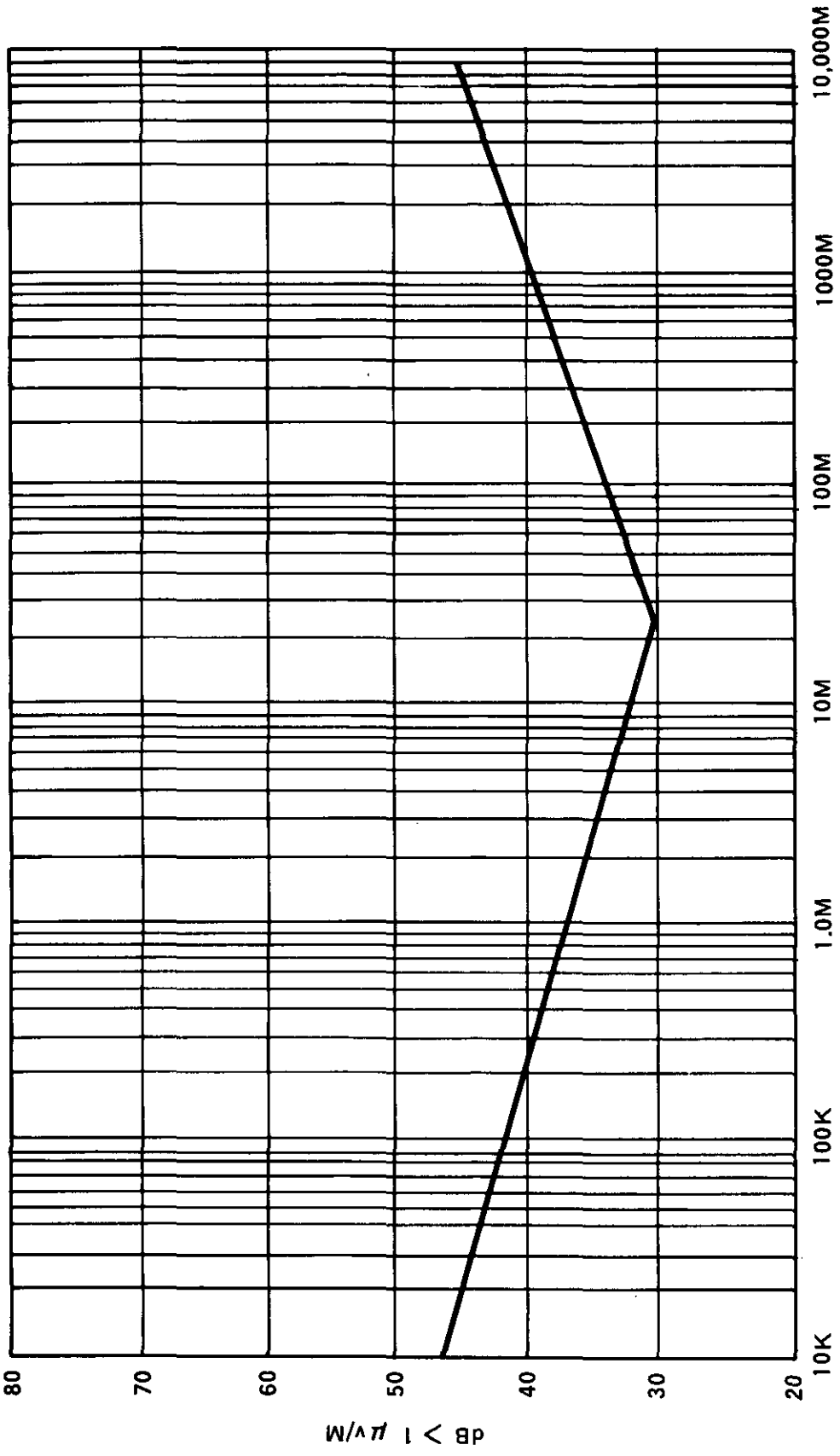


FIGURE 4. Limit for RE02 narrowband emissions (10 kHz to 10 GHz

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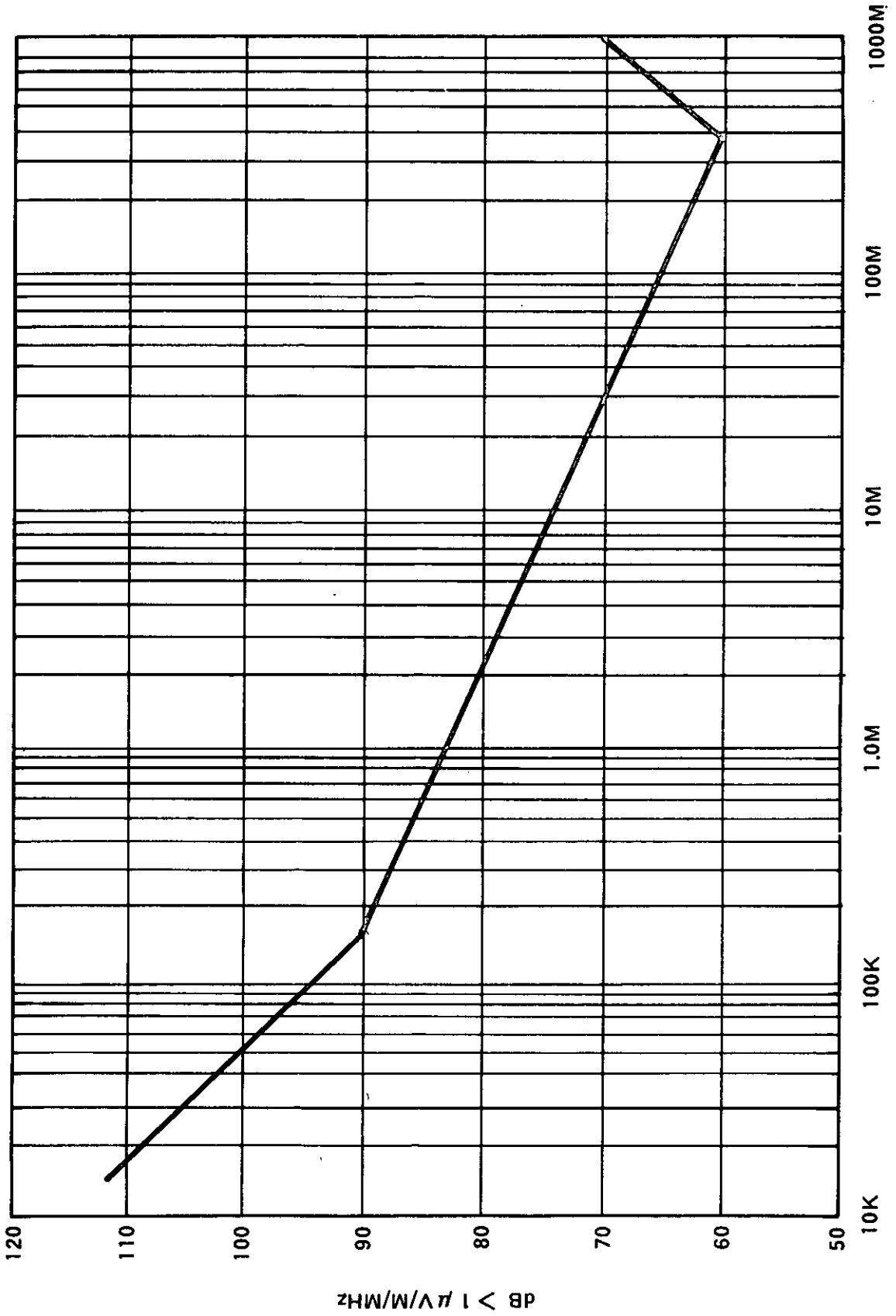
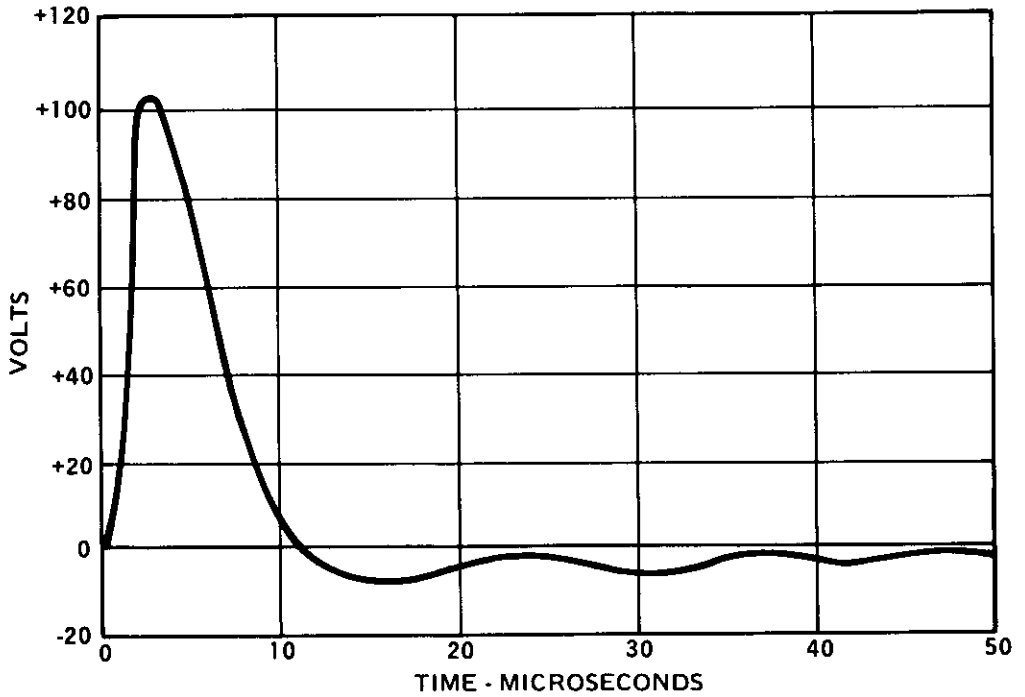


FIGURE 5. Limit for RE02 broadband emissions (14 kHz to 1.0 GHz)

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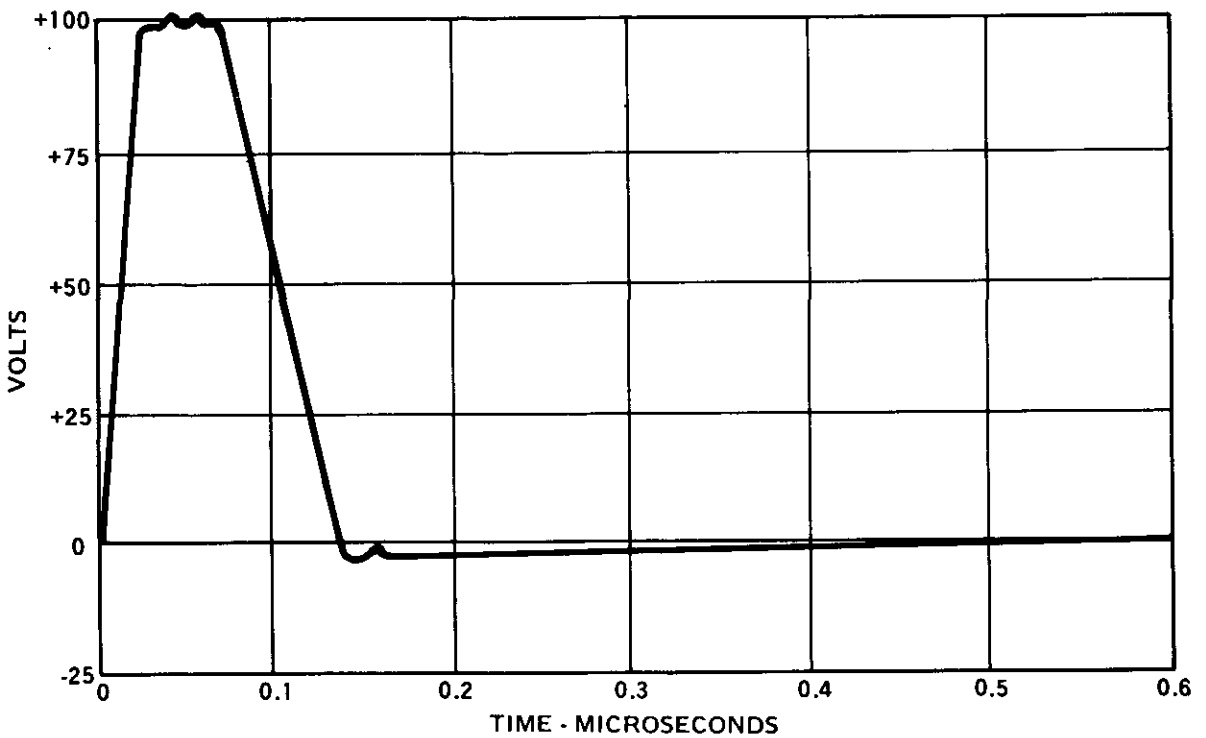


FIGURE 6. CS06 pulse characteristics

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3.3.6.1.9 RS03. The test limits shall be as specified herein:

Frequency	Field Strength (Volts/meter)
14 kHz to 2 MHz	10
2 to 30 MHz	40
30 to 200 MHz	5
200 to 400 MHz	10
400 MHz to 10 GHz	5

3.3.6.2 Specific requirements. In addition to the requirements specified in 3.3.6.1 and 3.3.6.3, the DMTS shall meet the requirements specified below:

3.3.6.2.1 Magnetic fields. The emanation of, and susceptibility to, steady state and dynamic magnetic fields shall be controlled to the extent specified below:

3.3.6.2.1.1 Magnetic susceptibility. The DMTS shall operate without degradation of performance specified herein when subjected to a near linear steady-state magnetic field which has a magnitude of one Oersted. This field shall be applied alternately in two geometrically opposite directions to preclude the possible cancellation of the test field by the earth's field.

3.3.6.2.1.2 Static magnetic fields. A static magnetic field is defined as a field which does not change in magnitude or direction with time. The DMTS shall not generate a static magnetic field due to DC greater than 1.0 Oersteds of Magnetic Field Intensity (H) measured 12.0 inches from the center of the magnetic source.

3.3.6.2.1.3 Dynamic magnetic fields. A dynamic field is defined as a field which can change with time. The DMTS shall not generate a dynamic field greater than 0.05 Oersteds of Magnetic Field Intensity (H) measured 12.0 inches from the center of the magnetic source.

3.3.6.2.2 Electromechanical end items. The design, development, and resulting hardware of electromechanical items (e.g., motors, relays, solenoids, and valves) shall be in accordance with MIL-STD-461 for Class 1D equipment. When the end item contains switches or solenoids, suitable EMI suppression devices shall be included. Compliance with the MIL-STD-461 requirements shall be demonstrated by tests performed in accordance with MIL-STD-462. Susceptibility tests may be omitted, subject to procuring activity approval, where the contractor provides conclusive written proof that the end item is nonsusceptible (see 6.2.2). Requirements calling for an EMI Control Plan are not applicable to electromechanical end items.

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3.3.6.3 Design requirements.

3.3.6.3.1 Interface design. Interfaces between subsystem equipment and aircraft wiring are defined as follows:

a. External Interface — The subsystem interface with aircraft wiring to other subsystems or discrete components

b. Internal Interface — The interface of a WRA of the subject subsystem with aircraft wiring to another WRA of the same subsystem.

3.3.6.3.1.1 Input circuits.

3.3.6.3.1.1.1 Balanced or differential. Balanced or differential input circuits shall be used at all WRA interfaces in order to reduce susceptibility, minimize the need for shielded aircraft wiring, and to prevent the possibility of creating ground loops.

3.3.6.3.1.1.2 High impedance. Input circuits operating at frequencies below 1 MHz where the wave length (lambda) is long compared to cable length (lambda/10) shall have a high input impedance, on the order of 10^4 Ohms, commensurate with the required common mode rejection. Circuits associated with coaxial cable inputs may be single-ended and shall be terminated in the characteristic impedance of the cable.

3.3.6.3.1.2 Output circuits. Output signals shall be presented from a low impedance source, on the order of 100 Ohms, commensurate with the required common mode rejection and both the output and return leads shall be balanced with respect to ground. Circuits associated with coaxial cables may be single-ended and shall provide a compatible source impedance to the cable as required by the input circuit.

3.3.6.3.2 Signal levels and rise time. Where possible, constant amplitude signals routed between WRA's shall be transmitted at a level of at least 4 volts but not more than 10 volts, unless such a level is absolutely necessary to accomplish the intended function and is approved by the procuring activity. A nominal value of 5 volts is required for data or logic pulses associated with circuits that have parity checks. The voltage magnitudes are peak-to-peak differential.

3.3.6.3.2.1 Pulses and step functions. The rise and fall time of pulses and step functions shall be as long as possible consistent with the function their circuits perform. The use of circuits that require fast rise times shall be avoided except where the rapid rise time is essential to the basic purpose the circuit serves.

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3.3.6.3.2.2 28 volt pulses or step functions. The use of 28 volt pulses or step functions shall be minimized. When such pulses are used, they shall have a rise time and a fall time no shorter than 3 milliseconds. (This constraint on rise and fall times is not mandatory with respect to the contacts of electromechanical switches or relay contacts; however, the equipment containing such devices shall meet the requirements of MIL-STD-461 and MIL-STD-462). Where practical, Gaussian waveshapes should be used in preference to square, trapezoidal, or triangular waveforms.

3.3.6.3.2.3 Analog signals. Analog signals between WRA's should be at a practical level, consistent with the constraints of power dissipation and this specification. A nominal value of 5 volts is recommended.

3.3.6.3.3 Grounding, bonding, and shielding. Grounding and bonding shall be in accordance with MIL-B-5087, except as otherwise specified herein. Shielding shall be as specified in 3.3.6.3.3.3 herein.

3.3.6.3.3.1 Power and signal returns. Primary AC and DC power returns and the return leads of incoming signals shall be isolated from each other and from the overall WRA case ground on the subsystem side of the interface. These returns and grounds shall be connected to separate pins at the appropriate WRA connector. The chassis of subassemblies within the WRA may be connected to the various grounds and returns, provided the WRA meets the above requirements at its interface with aircraft wiring.

3.3.6.3.3.2 Three phase power returns. The design of equipment using three phase power shall be such that the unbalance current (i.e., the current in the return or neutral lead) is less than one ampere, rms. Furthermore, the neutral or return lead shall be treated in the manner specified in 3.3.6.3.3.1 for primary AC power returns. Where practical, inherently noisy devices (relays, solenoids, etc.) shall use Phase B and C, and AC reference voltages should be derived from Phase A. However, the requirements pertaining to unbalanced current shall take precedence over the requirements of this paragraph.

3.3.6.3.3.3 Shielding. The design, construction, and finish of equipment enclosures shall utilize the case to provide at least 60 dB of shielding effectiveness for the frequencies above 150 kHz.

3.3.6.3.3.4 Bonding. All faying surfaces, case joints and case to mount joints shall provide faying surface conductivity in accordance with MIL-B-5087 Class R (2.5 milliohms or less). Aluminum alloys in accordance with MIL-E-5400, may be used with clean bare faying surfaces. Other aluminum alloys requiring contact resistance in accordance with MIL-C-5541 Class 3 (5 milliohms or less per square inch) may be treated with Alodine 200 or equal oxide retardant conductive coating. The contractor shall be responsible for the conductivity of the unit side of the installation faying surfaces. Dagger pins (used for physical alignment) shall not be a series element in the bonding circuit. The use of bonding jumpers within a WRA shall be avoided wherever possible, and, if used, their length shall not be greater than two inches or five times their width. The use of bonding jumpers external to a WRA is prohibited.

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3.3.6.3.3.5 Grounding. Refer to Installation Instruction EI-680 for grounding requirements.

3.3.6.3.4 Coaxial cables. The use of circuits requiring coaxial cables in the aircraft wiring shall be held to a minimum with preference being given to the use of twisted, shielded pairs. The decision to use such circuits shall be in accordance with the applicable equipment specification. The use of coaxial cables shall be limited to circuits handling signals which are necessarily broadband in nature (video, data pulses, etc.) or circuits handling multiplexed circuits. Signals which are only incidentally broadband (e.g., switching and control circuits) shall not be designed to be routed on coaxial or shielded cables. Each input or output coaxial cable shall have its shield connected to the case ground by the terminating WRA connector. This is a recognized exception to the restriction on the grounding of signal returns to the WRA case. However, coaxial cable shields shall remain isolated from primary power returns and other signal returns wherever possible.

3.3.7 Maintainability. The maintainability program (see 6.2.2), built-in test features, construction and packaging, provisions for test points, and other maintainability parameters shall be as specified in MIL-STD-2084. Maintainability shall receive equal consideration with such factors as cost, physical parameters and reliability. The maintenance concept to be implemented in the maintainability program shall be to perform corrective maintenance by utilizing computer or manually initiated built-in-test or test loops to fault isolate to the appropriate SRA, which will then be removed and replaced in the aircraft. The maintenance concept attempts to minimize or eliminate the depot level of maintenance. On board oscilloscope and multimeter may be utilized for fault isolation to the SRA level. No special support equipment will be permitted for fault isolation at the organizational level. The DMTU is considered an SRA for fault isolation and organizational maintenance purposes.

3.3.7.1 Maintainability program. The contractor shall establish and conduct a maintainability program using MIL-STD-470 as a guide. All organizational maintenance actions shall be capable of being performed by a technician with a comprehensive ability level as follows:

a. Civilian education - Ninth grade education and reading level or equivalent.

b. Technical training - Navy Class A and Class C technical schools appropriate to the maintenance task.

c. Experience - One year of technical experience in addition to technical training.

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3.3.7.2 Provisions for maintainability. The maintainability program, built-in-test features, construction and packaging provisions for test points and other maintainability parameters shall be as specified in MIL-STD-2084 except as modified herein. References to system BIT/BITE/ORT requirements in MIL-STD-2084 shall be interpreted as applying to the Digital Magnetic Tape Set as a standalone system.

3.3.7.3 Modular packaging. The DMTS shall be packaged in modular form. All electrical/electronic circuits and parts shall be packaged as replaceable and repairable plug-in modules and easily removable and replaceable when the equipment(s) is in its installed position in the aircraft. Circuit breakers, indicators and other items may be mechanically fastened to an equipment housing and shall not be considered modules. Nonmodule items shall be kept to an absolute minimum. All parts and assemblies which the contractor desires to install as nonmodule items shall require the prior written approval of the procuring activity (see 6.2.2). Special tools shall not be required for module removal.

3.3.7.3.1 Module commonality. The number of different types of modules shall be kept to a minimum. Unused functions on a module are permitted if commonality is improved. Such instances must be approved by the procuring activity (see 6.2.2).

3.3.7.3.2 Module interchangeability. Modules which perform identical functions shall be identical and interchangeable throughout the equipment.

3.3.7.3.3 Module access. Each module shall be normally removed and replaced at Organizational Level. Access to any module shall be achieved by the removal of, at most, one protective cover or enclosure. Hinged covers/doors shall be capable of being locked into both open and closed positions. Each module or hardware subunit shall be capable of being removed and replaced without prior removal or replacement of any other module or hardware subunit. Each module shall have guides for insertion and removal which prevent accidental damage to adjacent modules and connectors.

3.3.7.3.4 Module retention. Equipment designed to use screws or other similar devices for module retention shall be so designed that the retention devices are made captive to the module or equipment.

3.3.7.3.5 Keying. Modules and connectors shall be keyed to prevent the insertion of a connector or module into an incorrect location or position within or on the equipment. Specific ANEW channel requirements are specified in 3.5.2.3.2. An exception to the keying requirements herein is the intermixing of PDC and 1553 I/O channels as specified in 3.4.2.1.

3.3.7.3.6 Nonrepairable assemblies. All designs for nonrepairable assemblies shall be approved by the procuring activity prior to release for fabrication (see 6.2.2).

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3.3.7.3.7 Organizational maintenance adjustments. A WRA or SRA comprising a system/subsystem shall not require any adjustment after a single WRA or SRA has been replaced. For organizational maintenance purposes, the DMTU shall be considered an SRA.

3.3.7.3.8 Protective devices. Circuit breakers, when deemed necessary, shall be used in lieu of fuses. Circuit breakers shall be accessible from the exterior of the equipment, plainly marked as to rating and size, and shall be of the fault indicating type and visible when installed in aircraft. Circuit breakers shall meet the requirements of MIL-STD-454, Requirement 37.

3.3.7.3.9 Retention devices. Mechanical retention devices used for securing rack mounted equipment, if required, shall be the self-locking type and shall not require use of lock wire.

3.3.7.3.10 Periodic maintenance items. Equipment such as air filters and protective devices removed for head cleaning, requiring periodic removal for inspection or servicing, shall be retained by captive quick disconnect fasteners and shall be accessible for inspection or servicing with the equipment in its installed position in the aircraft. All items requiring scheduled maintenance shall be specified by the contractor and approved by the procuring activity within 60 days after contract award (see 6.2.2).

3.3.7.3.11 Access panels. Access panels shall be retained by captive quick disconnect fasteners.

3.3.7.3.12 Test points. Test points shall be provided for all levels of maintenance and shall be compatible with the test equipment designated and planned for fleet use. All test points shall be short circuit protected, i.e., no component failure will result if any test point is shorted to ground.

3.3.7.3.13 Compatibility with automatic test equipment (ATE). The DMTS shall be compatible with ATE. The equipment shall conform to the maintenance philosophy specified herein.

3.3.7.4 Organizational maintainability. The characteristics of the equipment shall be such that it shall be possible to perform all corrective organizational maintenance actions, other than combat damage repair, within a median Equipment Repair Time (ERT) of 0.20 hours utilizing BIT/BITE or loop tests as described herein. As applied herein, an organizational corrective maintenance action includes the following:

a. Detection of a fault in the WRA.

b. Isolation of a fault to the Shop Replaceable Assembly (SRA) within that WRA.

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c. Remove and replace faulty SRA[1/].

d. Check out of DMTS.

For measurement purposes as applies to the Maintainability tests specified in 4.5.6 herein and as specified below, the following types of failures shall be excluded:

a. Intermittent failures

b. Wiring/Pin problems

c. Interface failures on busses

These types of failures shall be automatically detected utilizing BIT/BITE or loop tests but may report ambiguous results and may not be necessarily isolated to the correct WRA or SRA without utilizing standard test equipment on board the aircraft (oscilloscope and meter).

3.3.7.4.1 Fault isolation requirement. MIL-STD-2084 shall apply as a minimum to the Organizational level of maintenance with the following modifications; in at least ninety-eight percent of the cases of probable malfunction of an SRA, the fault shall be automatically isolated to that sole SRA; in one hundred percent of the cases of probable malfunction of an SRA, the fault shall be automatically isolated to that SRA and no more than one other SRA: as a minimum, fault isolation to the correct WRA (i.e., DMTU, DMTC) shall be accomplished in ninety-eight percent of cases of WRA fault.

3.3.7.4.2 Built-in-test (BIT) provisions. BIT provisions shall be incorporated into the DMTS to allow automatic fault detection to a WRA of the DMTS and automatic fault isolation to a faulty module (SRA) within that WRA. BIT can be manually initiated from the Operator's Panel (OP) panel in Off Line Mode or any interface computer in On Line Mode. Manually initiated BIT shall check out only that functional loop selected on the OP panel (i.e., ANEW Interface/Controller 1/DMTU A, etc). BIT shall check all functions within that loop so far as possible and shall report GO/NO GO status for the entire path and provide expanded messages only to the BIT originating peripheral device. If a NO GO condition exists, the message shall isolate the fault to the faulty SRA within the DMTC or attribute the fault to the selected DMTU. If the DMTU is at fault, expanded messages shall be provided to indicate the fault condition(s) detected. Provisions shall be made for all fault messages to the OP panel to be displayed in English text without having to refer to any document in order to interpret any fault localization indications. There shall also be an OP panel BIT test which, when initiated from the OP panel, shall provide a functional checkout of all the logic elements within the OP panel. The BIT design for the DMTS shall be approved by the procuring activity prior to implementation (see 6.2.2).

1/The DMTU and Operator's Panel are considered SRA's in this context.

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3.3.7.4.2.1 GO/NO GO indicators. If latching type GO/NO GO indicator(s) are used, they shall be resettable without disassembly of the equipment and without special tools.

3.3.7.4.3 On Line loop test. Test loops shall be designed into each Logic Subunit of each WRA of the DMTS. The test loops shall permit the computer program to exercise the selected Logic Subunits of the DMTS and monitor the performance for possible logic malfunctions within that functional loop. The test loops shall be comprehensive; i.e., a selected test loop shall exercise every function in the selected Logic Subunit of the DMTS so far as possible. The selected test loops shall be utilized for in-flight performance monitoring (IFPM) and diagnostic programs. Loop tests are initiated from a peripheral computer and specific test requirements are specified in AD-1404. A loop test, when initiated, shall check out the functional loop selected for that computer from the configuration setup on the OP panel, i.e., PDC/Controller 2/DMTU B, and the originating computer will monitor the results to determine GO/NO GO status. This shall be a fault detection test and shall have no fault isolation provisions.

3.3.8 Nomenclature, nameplates and identification marking. Nomenclature and serial number assignment, nameplate approval and identification marking shall be in accordance with MIL-N-18307.

3.3.9 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

a.	Temperature	Room ambient (25° ±10°C)
b.	Altitude	Normal ground (0-10,000 ft)
c.	Vibration	None
d.	Humidity	Room ambient up to 90 percent relative humidity
e.	Input Power Voltage	115 ±5.0 VAC, 400 Hz ±20 Hz, 3 phase and 28 ±0.5 VDC

3.3.10 Service conditions. The DMTS shall operate within limits specified herein under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 1A equipment, except as modified herein.

3.3.10.1 Vibration. The DMTS shall operate within the accuracies, limits and tolerances specified herein when subjected to sinusoidal vibration levels of 0.08 inch double amplitude from 5 Hz to 10 Hz, 0.42 g from 10 to 15 Hz, 0.036 DA from 15 to 32 Hz, and 2 g (zero to peak) from 32 Hz to 500 Hz in any axis. If equipment is individually supported on vibration isolators, the equipment shall, in addition, be subjected to

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Curve II, Figure 2 of MIL-E-5400, with isolators removed. Resonant modes uncovered during test shall be reported to the procuring activity for design guidance (see 6.2.2).

3.3.10.2 Temperature. Degraded performance shall be allowed during operation at temperatures below -25° and during the warm-up period specified in 3.3.11 at temperatures above -25°C. However, no equipment damage shall occur during those periods.

3.3.10.2.1 Temperature and humidity limitations. The DMTS shall be designed to operate as specified herein within the temperature and humidity limits specified below and to withstand storage temperatures as specified:

a. Operating Limits (with dummy cartridge) -25°C to +55°C with intermittent operation at +71°C

b. Operating Limits (with operational tape in cartridge) 0°C to +50°C

c. Storage Limits (without tape) -54°C to +85°C

NOTE: Magnetic Recording Tapes for demonstrating DMTS performance shall not be exposed to temperatures below 0°C or above 50°C, nor to relative humidities below twenty percent or above seventy percent. The DMTS, including operational tape, shall operate within specified limits within these temperature and humidity limits. The DMTS, less tape, shall operate within specified limits, excluding read/write operations off tape, over the temperature range of -25°C to +55°C and over the full humidity range specified by MIL-E-5400 for Class 1A equipment.

3.3.10.3 Altitude. The DMTS shall operate within limits specified herein at pressure altitudes up to 15,000 feet.

3.3.10.4 Explosive conditions. Explosion proofing of the DMTU is not required.

3.3.10.5 Sand and dust. The DMTU with cartridge installed is not required to meet the sand and dust requirements of MIL-E-5400.

3.3.10.6 Salt atmosphere. The DMTU with cartridge installed shall not be required to meet the salt atmospheres requirements of MIL-E-5400.

3.3.11 Warm-up time. The time required for the DMTS to warm up prior to operation shall be kept to a minimum and shall not exceed one minute at temperatures of -25°C and above (does not include tape cartridge from -25°C to 0°C). Degraded performance may occur when operating prior to warm-up, but no damage to the equipment shall occur. Normal DMTS operation shall resume when temperature returns to the limits specified in 3.3.10.

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3.3.12 Input electrical power

3.3.12.1 Operating power. The DMTS shall perform as specified when energized from power sources with the characteristics and limits of Figures 3 and 5 (AC power) and Figure 9 (DC power) of MIL-STD-704 as modified herein. The DMTS shall operate with limit 3 of Figure 3 and limits 1, 2 and 3 of Figure 9 of MIL-STD-704 as modified below.

3.3.12.1.1 AC power limits

a. Curve 3 in Figure 3 of MIL-STD-704 shall be modified to begin at a point defined by 500 microseconds and 80 volts and extending along the 80-volt line to 0.05 second, then rising to intercept the NSSL line (108 volts) asymptotically at 0.13 second. The rising curve shall pass through 92 volts at 0.07 second and 100 volts at 0.096 second.

b. Operational malfunction may occur when the input voltage exceeds the above limits, but remains within limits 1 and 4 in Figure 3.

3.3.12.1.2 DC power limits

a. Curve 3 in Figure 9 of MIL-STD-704 is modified to begin at a point defined by 500 microseconds and 18 volts and extending along the 18-volt line to 0.05 second, then rising to intercept the NSSL line (24 volts) asymptotically at 0.13 second. The rising portion of the curve passes through 20 volts at 0.07 second and 23 volts at 0.10 second.

b. Curve 2 in Figure 9 of MIL-STD-704 is modified to begin at a point defined by 500 microseconds and 50 volts and extending along the 50-volt line to 0.025 second, then falling to intercept the 40-volt line at 0.10 second. The falling portion of the curve passes through 45 volts at 0.05 second. It continues along the 40-volt line to 0.29 second, then falls to intercept the NSSL (28.5 volt) line asymptotically at 0.9 second.

c. Curve 1 in Figure 9 of MIL-STD-704 is modified to begin at a point defined by 500 microseconds and 50 volts, then extending along the 50-volt line to 0.025 second, then falling to intercept the 40-volt level at 0.10 second. The falling portion of the curve passes through 45 volts at 0.05 second. It continues at the 40-volt level to 1.7 seconds, then falls to intercept the ASSL (30 volts) line asymptotically at 5 seconds.

d. Operational malfunctions may occur when the input voltage exceeds the above limits but remains within limits 1 and 4 in Figure 9 as modified herein.

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3.3.12.1.3 Equipment damage. The DMTS shall not be damaged as a result of exposure to the transient conditions defined by MIL-STD-704 as modified herein. Normal operation in the same configuration of channel and DMTU assignments as before the transient shall be resumed in the initialized state upon return of the input voltage to levels within limit 3 in Figure 3 and limits 2 and 3 in Figure 9 of MIL-STD-704 as modified herein and the receipt of a new computer command.

3.3.12.1.4 Power requirements. The DMTC shall be designed to operate with AC power within the limits specified in 3.3.12.1. The DMTU shall be designed to operate with DC power within the limits specified in 3.3.12.1. Primary input to the DMTC shall be controlled by a POWER ON switch on the operator's panel. In addition, each DMTU shall have a POWER ON switch. The total power requirements shall not be greater than specified below:

a. AC power (three phase) - 200 VRMS line-to-line, Category B, 3-wire system (delta input), 450 VA.

b. DC power - 28 VDC at 2 amperes (continuous maximum for each DMTU) and 4.0-ampere peak (during tape motion start/stop intervals).

3.3.12.1.5 Initialized state. Initialization shall place the DMTS in the configuration last selected on the operator's panel (OP) switches. Upon restoration of power, the DMTS shall be ready to receive commands from the computers, in the On Line Mode, or from the OP if in the Off Line Mode.

3.3.12.2 Loss of power. No physical damage shall result to tape or DMTS from accidental or deliberate stoppage of electrical power to the equipment regardless of the time in its operating cycle that such a stoppage should occur, and regardless of the time interval that elapses before power is restored.

3.3.12.3 Abnormal power supply operation. Each WRA of the DMTS shall provide for protection against any abnormal operation of any power supply in the DMTS, including undervoltage, overvoltage, overload and short circuit conditions. The protective design within the DMTS must be such that any failure mode of any DMTS power supply does not cause any component failure or chain reaction failures within the equipment. When all voltage levels return to normal, the DMTS shall assume an initialized state, ready to resume normal operation.

3.3.13 Cooling. Typical avionics cooling provisions are specified in Figure 7. Cooling of the DMTS by convection shall be a design goal. If internal fans are required to provide proper cooling, no fan-induced cooling air shall come in direct contact with the module components. All fans shall be in accordance with MIL-B-23071. Removable external cooling air filters, if required, shall be accessible for inspection or servicing with the DMTS installed in the aircraft. Fan selection criteria shall include the noise generation requirements of 3.3.18.

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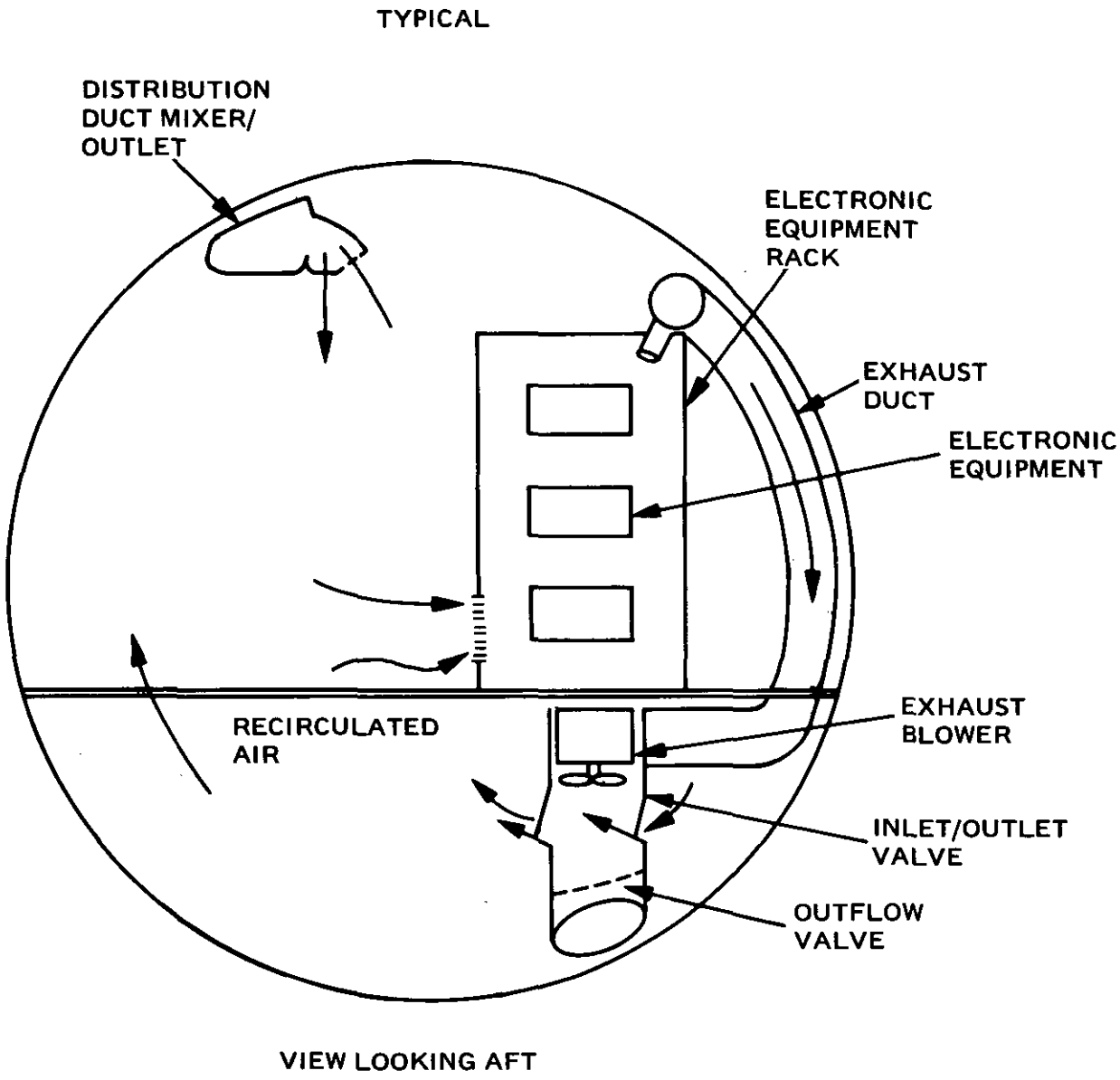


FIGURE 7. Avionic rack cooling inside pressurized cabin

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3.3.13.1 Overtemperature sensor. The DMTC power supply shall contain thermal sensing circuitry to detect the maximum safe operating temperature and provide an external signal. This signal line shall be used in the aircraft to warn the equipment operator of the overtemperature condition. The DMTC shall not automatically shut down as a result of the overtemperature sensor. If fans are used for cooling, fan failure shall trigger the overtemperature sensor (see 3.5.1.2.2.3).

3.3.14 Dimensions. The dimensions of each WRA of the DMTS shall be as specified in EI-680.

3.3.15 Color. The finish on all equipment shall be in accordance with MIL-E-5400. The finish used for numbers and letters shall be a black nonreflecting enamel in accordance with MIL-E-5558. The units shall be painted lusterless Gray (No. 36231) in accordance with FED-STD-595.

3.3.16 Mounting. All equipment shall be hard mounted.

3.3.17 Generated acoustic noise. The sound power level shall not exceed the values specified in Table 1 of MIL-STD-740 for type 3 equipment airborne, grade B. The sound pressure spectrum level for any pure tone within any octave band shall not exceed a level which is 10 dB less than the specified level for the octave band.

3.3.18 Human performance/human engineering. Human performance/human engineering design criteria for the DMTS shall be in accordance with MIL-STD-1472. MIL-H-46855 shall be used as a guide.

3.3.19 Control panels. The DMTC and DMTU operator panels shall meet the requirements specified herein, unless specifically approved by the procuring activity (see 6.2.2). Edge lighting shall not be required for these panels. The configuration, panel layout, switch lighting, switch legends and switch types for these panels shall be reviewed and approved by the procuring activity prior to fabrication of the first preproduction model.

3.3.19.1 DMTC operator panel indicator and switch legends. Indicator and switch legends shall be white translucent on black opaque background. Legends shall be visible without back-lighting. Legends shall appear colored in accordance with the following requirements only when back-lighting is energized.

3.3.19.1.1 DMTC operator panel indicator and switch legend lighting. Lighting for operator panel indicators and switches shall adhere to the following color requirements:

a. Switches - Switch legends shall illuminate green when function is available for selection. Switch legends shall illuminate amber when function is active. Switch legends shall be off if function is not available.

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b. Advisory or status indicators - Advisory or status indicator legends shall illuminate amber when conditions exist. Indicator legend back-lighting shall be off when condition does not exist.

c. DMTS OVER TEMP and AYK-14 OVER TEMP Indicators - These warning indicators legends shall be illuminated red when an overtemperature condition exists. Indicator back-lighting shall be off when an overtemperature condition does not exist.

The chromaticity coordinates shall be as specified in MIL-C-25050 for Type I aviation colors.

3.3.19.1.2 DMTC lamp intensity control. Intensity controls shall be provided to regulate the voltage applied to the lamps. Under no circumstances shall the voltage applied to any lamp be greater than its rated voltage. A separate intensity control is required for projection readout (PRO) type indicators.

3.3.19.2 DMTS switch indicator access. All switch and indicator housings shall be positively secured. The switch/indicator light capsules or projection type readouts containing the lamps and lens assemblies shall be removable from, and installable into, their respective housing from the front of the unit without the use of any tools.

3.3.20 Safety. Personal safety requirements shall conform to MIL-STD-454, Req. 1.

3.4 Performance. The Digital Magnetic Tape Set (DMTS) shall consist of a Digital Magnetic Tape Controller (DMTC), and one to four Digital Magnetic Tape Units (DMTU). The DMTS shall interface with various avionics subsystems via the following interfaces:

a. One CP-901 (ANEW) channel (MIL-STD-1397)

b. Two PROTEUS Digital Channels (PDC) (A30-15590)

c. One AYK-14 Data Multiplex Data Bus (DMDB) (MIL-STD-1553)

The subsystem interfaces shall be in accordance with AD-1404. The DMTS block diagram presented in Figure 8 illustrates one of several possible equipment configurations. (See Tables III and IV.) The DMTC shall contain two independent control sections which are controlled from a single operator's panel (OP).

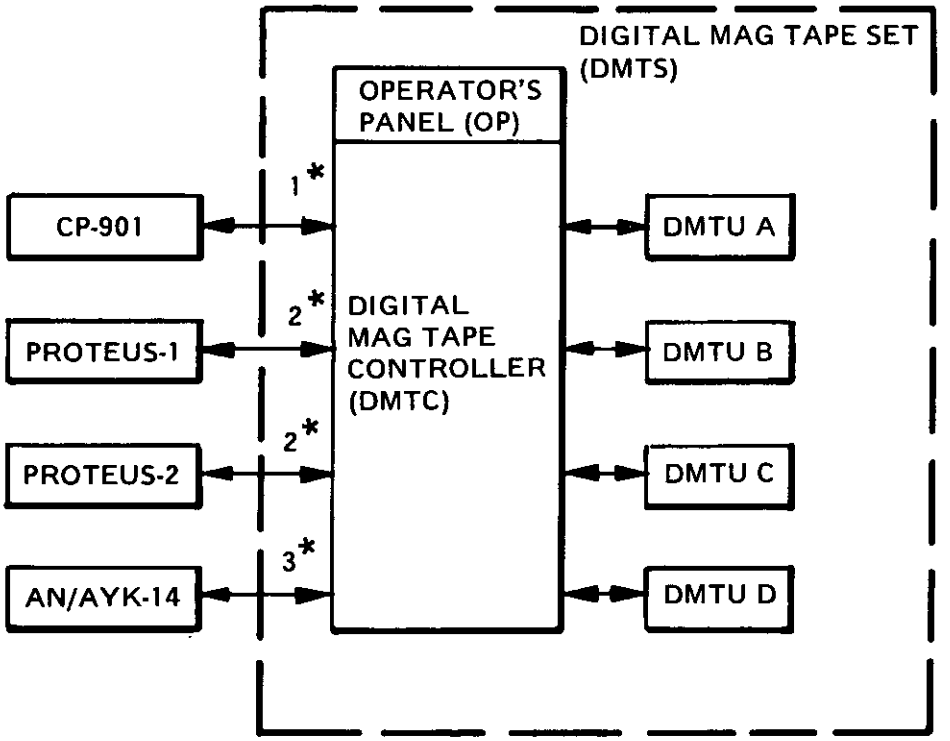
3.4.1 Operation. Each control section of the DMTC shall operate in one of three modes:

a. Initialization

b. Off Line

c. On Line

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*NOTE:

1. ANEW INTERFACE (MIL-STD-1397)
2. PDC (A30-15590)
3. DUAL MULTIPLEX DATA BUS (DMDB) (MIL-STD-1553)

FIGURE 8. Digital magnetic tape set (DMTS)

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TABLE III. DMTC interfaces

	Delivered	With 1/ Expansion	Total (Max)
ANEW Channel	1	2/	2/
PROTEUS Digital Channel (PDC)	2	2/	2/
Dual Multiplex Data Bus (DMDB)	1	2/	2/
DMTC/Operator Panel	2	2	2
DMTU/DMTC	4	4	4

NOTES: 1/ Unit Wiring and power supply capacity shall be provided for maximum configuration. DMTU's shall have separate power supplies.

2/ See Table IV for maximum mixes.

TABLE IV. Digital channel interface mixes

	ANEW	PDC	DMDB
Maximum Requirement for Three I/F Types	1 1	2 1	1 2
Maximum Requirement for Two I/F Types	1 1 0 0	3 0 3 1	0 2 1 2
Maximum Requirement for One I/F Type	1 0 0	0 4 0	0 0 3
Maximum Requirement for Standard I/F Types	1	2	1

I/F = Interface

3.4.1.1 Initialization. Each controller shall have an initialization switch on the OP which is utilized in Off Line Mode to Master Clear each control section prior to use. It is also used to initiate a BIT loop test on the selected I/O and DMTU configuration of each control section. If a fault is detected, amplifying messages will be displayed on the operator's alphanumeric readout (See Figure 9).

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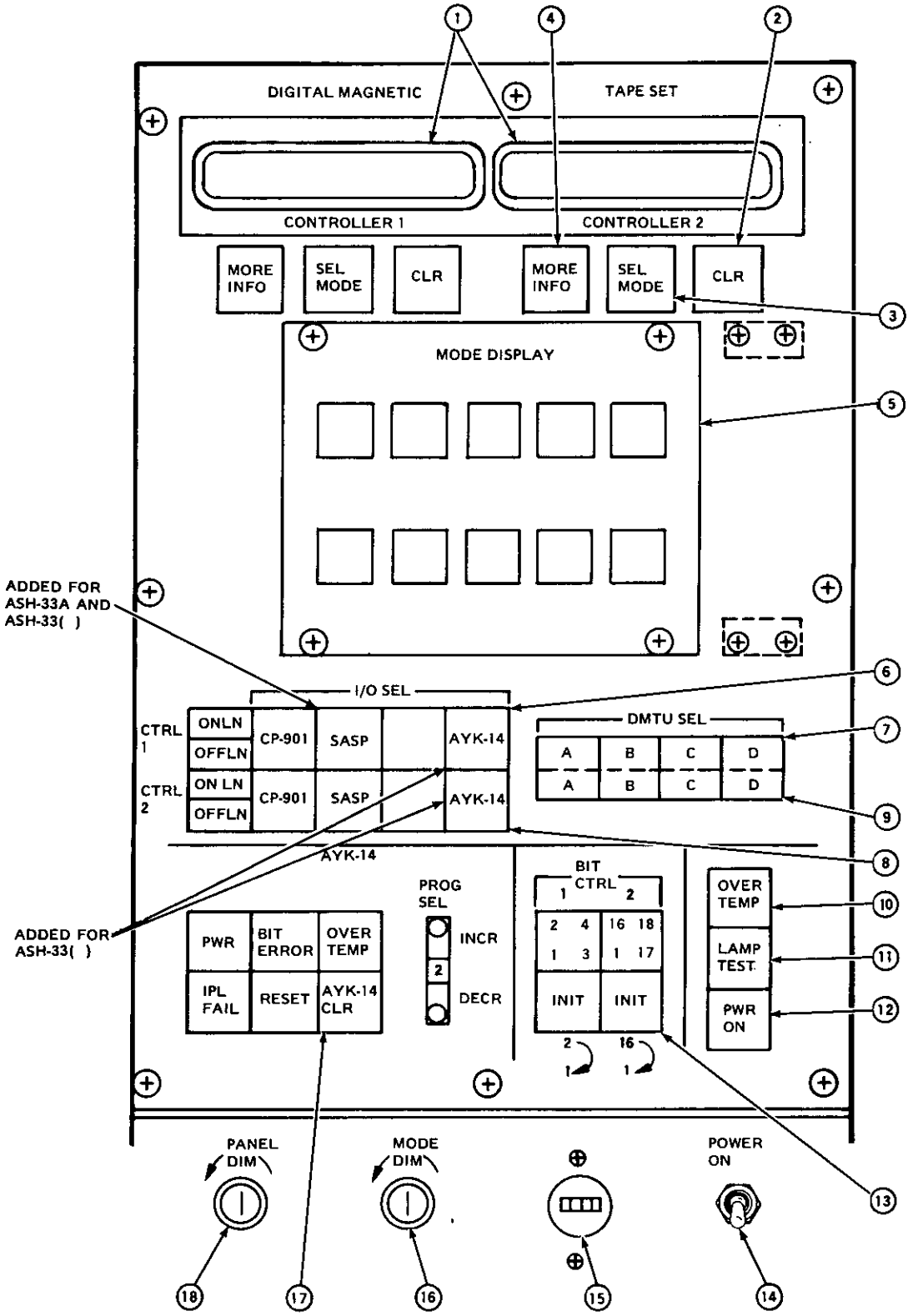


FIGURE 9. DMTC controls and indicators

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3.4.1.2 Off Line Mode. Either control section of the DMTC shall enter the Off Line Mode at any time upon the appropriate switch depressions by the operator at the OP or upon application of primary power if the particular control section mode select switch is set for Off Line Mode. When in the Off Line Mode the particular control section shall accept commands only from the operator at the OP, perform the necessary error checking to determine command validity, execute the command action and report progress and completion status as required to the operator via the OP. The only modes of operation which can be performed while Off Line are Initialization, Off Line BIT Tests, and DMTU Tests.

3.4.1.3 On Line Mode. Either control section of the DMTC shall enter the On Line Mode at any time upon the appropriate switch depressions by the operator at the OP or upon application of primary power if the particular On Line/Off Line select switch for that control section is set for On Line Mode. When in the On Line Mode, the control section shall accept commands on the assigned digital interface channels, perform the necessary error checking to determine command validity, execute the commanded action and report progress and completion status as required on the particular interface. The following modes of operation can be achieved only while in the On Line Mode:

3.4.1.3.1 CP-901 Mode. Either control section of the DMTC shall enter the CP-901 Mode upon appropriate switch (CP-901) selections by the operator at the OP. When in the CP-901 Mode, the particular control section shall accept and execute commands and provide responses as specified in 3.5.2.2.2.

3.4.1.3.2 PDC Mode. Either control section section of the DMTC shall enter a PDC Mode upon the appropriate switch (SASP) depressions by the operator at the OP. When in a PDC Mode, the particular control section shall accept commands from the operator via the OP which will initiate predefined Single Acoustic Signal Processor (SASP) System load sequences over the PDC assigned to the control section. The control section shall transmit and receive load and status information to/from the SASP System. Intermediate and completion status indications shall be displayed to the operator via the OP as specified in 3.5.1.3.2.1.

3.4.1.3.3 AYK-14 Mode. Either control section of the DMTC shall enter the AYK-14 Mode upon appropriate switch (AYK-14) depressions by the operator at the OP. When in the AYK-14 Mode the particular control section shall accept commands from the AYK-14 which will initiate predefined AYK-14 load sequences over the multiplex data bus designated for AYK-14 load. The control section shall transmit and receive load and status information to/from the AYK-14. Intermediate and completion status indications shall be displayed to the operator via the OP as specified in 3.5.1.3.2.2.

3.4.1.3.4 On Line built-in test Mode. Either control section of the DMTC shall enter the Built-In Test (BIT) Mode upon receipt of an Initiate BIT command from a computer when OP is On Line. The tests shall include at a minimum, but are not necessarily limited to:

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a. Internal end-around wraps on computer interfaces as configured from OP.

b. Complete internal DMTS checks for the selected data path.

c. Program and storage memory tests for the selected data path OP panel functions.

The tests shall be designed such that completion of testing of the DMTS from the OP shall be accomplished with a minimum of operator intervention. If feasible and cost effective, a completely automatic BIT is highly desirable. Intermediate and completion status results shall be provided by the DMTC to the device originating BIT. Completion status shall include a GO/NO GO status and the return of amplified status indications to the device originating BIT. The module(s) at fault, if the test was NO GO, shall be identified in the status results word. BIT shall be designed to accommodate digital interface interchangeability as specified in 3.4.2.1, and shall meet the requirements specified in 3.3.7.4.2.

3.4.2 Capability. The DMTC shall provide the following interfaces:

a. DMTC/CP-901 (ANEW)

b. DMTC/SASP (PDC)

c. DMTC/AYK-14 (DMDB)

d. DMTC/Operator Panel

e. DMTC/DMTU's

The DMTC/computer interfaces shall use independent circuitry on dedicated and independent circuit boards. Any DMTC/computer channel interface board(s) shall be removable without affecting DMTS operation for other channels.

3.4.2.1 Digital interface interchangeability. Digital interface interchangeability shall be provided as specified in Tables III and IV.

3.4.3 Programmability. A programmed implementation of the following DMTC features is required:

a. Digital Interface Control

b. Computer Command Interpretation

c. Tape Positioning Control

d. PDC Load Mode Control

e. AYK-14 Load Mode Control

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f. PDC Diagnostics Control

g. Off Line Mode Control

h. BIT Control

i. OP PRO Level and Alphanumeric Display Control

j. Discrete switches and indicators excluding DMTU oriented, Power On/Off, controller clear, overtemperature and lamp test.

3.4.3.1 Self-check requirements. ROM's utilized in conjunction with a microprocessor and a microcontroller shall have an architecture such that the contents of the ROM's can be checked for ROM failure by a checksum method employing a program or subprogram executed in a microprocessor and microcontroller.

3.4.3.2 DMTS program load. If a programmed controller is provided, under no circumstances shall the DMTS require a "program load" prior to normal DMTS operation. All programmed functions shall be in the nature of "firmware" in nonvolatile devices such as PLA or ROM.

3.4.4 Software. DMTS software shall be prepared in accordance with SECNAVINST 3560.1 and MIL-STD-1679 (see 6.2.2).

3.5 Detail requirements. The DMTS (Digital Magnetic Tape Set) shall consist of the following functional elements:

	Paragraph
a. DMTC Operator's Panel	3.5.1
b. DMTC (Digital Magnetic Tape Controller)	3.5.2
c. DMTU (Digital Magnetic Tape Unit)	3.5.3

3.5.1 Operators panel (OP). The OP shall contain the controls, switches, displays and indicators available to an operator. The arrangement of controls, switches, displays and indicators is shown in Figure 9 and the functions are specified in Table V. (On the OP and in Figure 9, the identifications Controller 1, CTRL 1, Controller 2, and CTRL 2, refer to control section 1 and control section 2, respectively.) The following general requirements shall also be satisfied.

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TABLE V. DMTC controls and indicators

Fig 9 Item No.	Control or Indicator	Description
1	CONTROLLER 1/ CONTROLLER 2 (display)	Alphanumeric display.
2	CLR (momentary switch)	Normally green; always available; clears alphanumeric display (A/N) when pressed.
3	SEL MODE (momentary switch/indicator)	Alternate action between CTRL 1 and CTRL 2, both can be green but only one can be amber (active); available only for PROTEUS (SASP) operations On Line or for any Off Line Mode.
4	MORE INFO (momentary switch/indicator)	Normally off; when green, indicates more information is available for display; data is displayed when pressed.
5	MODE DISPLAY (projection readout switches)	Will be off for any control section selected ON LN with CP-901 or AYK-14; otherwise displays are as shown in Figures 10a, 10b and 11.
6	CTRL 1 (Controller 1)	Controls and indicators for Controller 1.
	ON LN/OFF LN (alternate action switch/indicator)	OFF LN lights green; when pressed OFF LN goes out and ON LN lights amber indicating that Controller 1 is On Line.
	I/O SEL	Input/Output selections for Controller 1.
	CP-901 (alternate action switch/ indicator)	Normally amber; goes off when SASP or AYK-14 is pressed.
	SASP, AYK-14 (alternate action switch/indicators)	Lights green when active; only one may be active at a time.

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TABLE V. DMTC controls and indicators (continued)

Fig 9 Item No.	Control or Indicator	Description
7	DMTU SEL	Selection of DMTU.
	A, B, C, D (alternate action switch/ indicator)	Selects DMTU; if any DMTU is disconnected, its associated indicator will be off; up to four DMTU's can be selected for a control section but cannot have same logical address; selected DMTU lights amber for CTRL 1 and green for CTRL 2 (alternate action precludes a DMTU from being used by both control sections).
8 & 9	CTRL 2 (Controller 2)	Controls and indicators for Controller 2 (Descriptions the same as for CTRL 1).
10	OVER TEMP (indicator)	Lights red for overtemperature condition of 5 volt power supply or loss of cooling air flow.
11	LAMP TEST (momentary switch/indicator)	Normally green; when pressed, causes all other panel indicator lamps to light (except PWR ON). Also causes question mark (?) to be displayed on A/N display and causes 12 MODE option groups to be displayed in sequence.
12	PWR ON (indicator)	Lights amber when POWER switch is set to ON and all power supplies are normal.
13	BIT (Built-In-Test)	
	CTRL 1 (Controller 1)	
	INIT (momentary switch/indicator)	Normally lights green; otherwise, a GO indication is displayed on alphanumeric display. Initiates built-in-test when pressed; changes to amber when active. A detected failure is indicated by four-segment light or by a readout on alphanumeric display.

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TABLE V. DMTC controls and indicators (continued)

Fig 9 Item No.	Control or Indicator	Description
13 (Cont'd)	2/4/3/1 (indicator)	Normally off; during BIT, an illuminated number means that board number has failed.
	CTRL 2	(Description same as CTRL 1 except BIT sequence is 16/18/17/1.)
14	POWER ON/OFF (switch/circuit breaker)	Applies power to DMTC; provides automatic power cutoff in case of excessive current demand.
15	Elapsed time indicator	Activated when POWER switch is set to ON; indicates total accumulated DMTC power-on time.
16	MODE DIM (control)	Adjusts brightness level at MODE DISPLAY projection readouts.
17	AYK-14	Controls and indicators for AYK-14 computer.
	PWR ON (indicator)	Lights amber if AN-AYK-14 discrete lines are connected to DMTC.
	BIT ERROR (indicator)	Normally off; lights amber if an AYK-14 bit error is detected.
	OVER TEMP (indicator)	Normally off; lights red if an AYK-14 overtemperature condition occurs.
	IPL/FAIL (momentary switch/indicator)	IPL portion is normally green; initiates program load when pressed. FAIL portion is normally off; lights amber if IPL fails.
	RESET (momentary switch/indicator)	Normally off; lights green if an AK-14 reset is needed.
	AYK-14 CLR (momentary switch/indicator)	Normally off; lights green if AN/AYK-14 is to be cleared.

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TABLE V. DMTC controls and indicators (continued)

Fig 9 Item No.	Control or Indicator	Description
17 (Cont'd)	PROG SEL (digital switch)	Used to select program to be loaded. Program number is incremented one step with each depression of upper pushbutton. Program is decremented similarly with lower pushbutton.
18	PANEL DIM (control)	Adjusts brightness level of DMTC panel indicators (except MODE DISPLAY, see item 11).

3.5.1.1 General functions. The OP shall provide the basic man-machine interface for the DMTS. The major functions provided shall be:

a. DMTC power ON/OFF control and a power ON indication.

b. On Line/Off Line operator control and associated indicators.

c. Assignment of DMTU's to specific computer interfaces.

d. Initialization of assigned DMTU's and interface logic.

e. Off Line DMTS self test.

f. Operator control for loading selectable software into either the SASP or AYK-14 system.

g. Display for the operator of Alerts, Cues, Function Identifiers, Status and Fault Isolation reports.

3.5.1.2 Operator panel (OP) operations.

3.5.1.2.1 DMTU oriented functions. The OP shall provide for the selection and test of any of four DMTU's selected for either control section.

3.5.1.2.2 DMTS/DMTC oriented functions. The OP shall provide discrete pushbutton/indications, indicators, switches, and other devices to satisfy the following requirements.

3.5.1.2.2.1 POWER ON switch operation. The OP shall have a POWER ON switch and an indicator that illuminates amber when power is applied to the DMTC.

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3.5.1.2.2.2 Initiate BIT/master clear (CLR) pushbuttons. Depression of either of the CLR pushbuttons when in the Off Line Mode shall cause the corresponding control section (1 or 2) to enter a predetermined initialized state. It shall be equivalent to the MASTER CLEAR function. The Init BIT/Master Clear function shall be characterized by:

a. Initialzation of the assigned computer interface logic.

b. Initialization of the assigned DMTU(s) to a stopped, ready condition.

c. Termination of any pending operations.

d. The clearing of the OP display.

e. BIT of the selected configurations.

The Initiate BIT/Master Clear pushbuttons shall be available when Off Line Mode is selected.

3.5.1.2.2.3 Overtemperature Indicator operation. An Overtemperature (OVER TEMP) indicator shall be provided to warn the operator that the DMTC temperature is 85°C or higher or a fan has failed. The indicator shall illuminate red to indicate that the temperature of the DMTC has reached the warning level. In addition to the indicator, there shall be a relay closure signal pair available at a connector for external overtemperature warning applications. The relay closure signal pair and relay shall be capable of transferring 500 milliamps current, either AC or DC, through the closed (overtemperature activated) circuit. This relay closure is to be used in accordance with 3.3.13.1.

3.5.1.2.2.4 Time totalizing meter. A time totalizing meter of a type in accordance with 3.3.2.4 shall be provided on the DMTC to indicate accumulated on-time for the DMTC.

3.5.1.2.2.5 LAMP TEST. Depression of a LAMP TEST switch shall cause all indicator lamps to illuminate and shall cause the projection readout (PRO) switches to cycle through all legends (levels). Each PRO level shall be illuminated for a half-second before changing to the next level. The PRO levels shall cycle redundantly as long as the pushbutton is depressed. Refer to Figures 10, 11 and 12 for the various PRO levels.

3.5.1.2.2.6 Operator displays and controls. The OP shall have the necessary display and control capability to allow an operator to fully implement all features of DMTS operating modes and as a minimum shall provide the following features.

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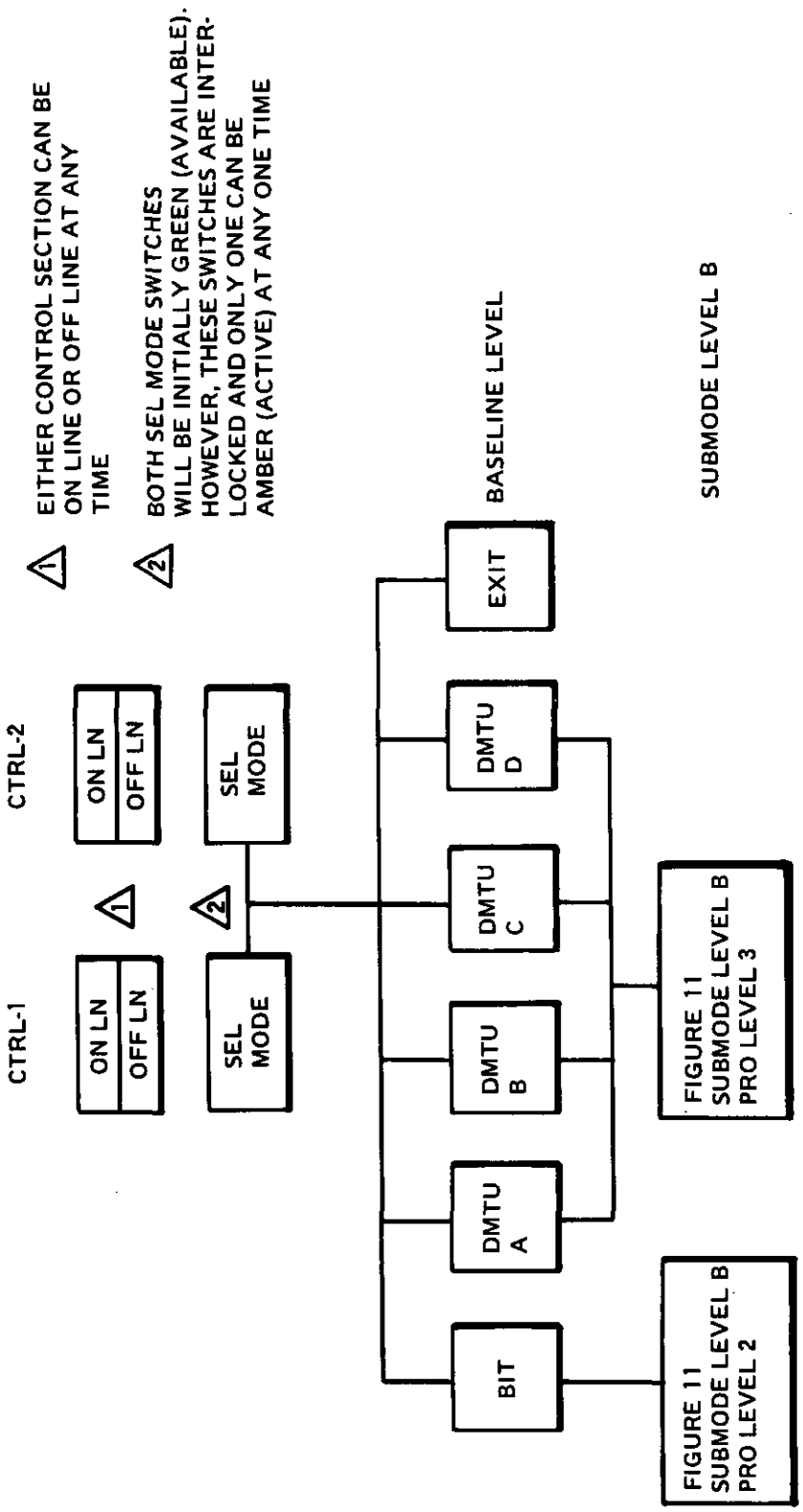


Figure 10. Off Line Mode PRO displays

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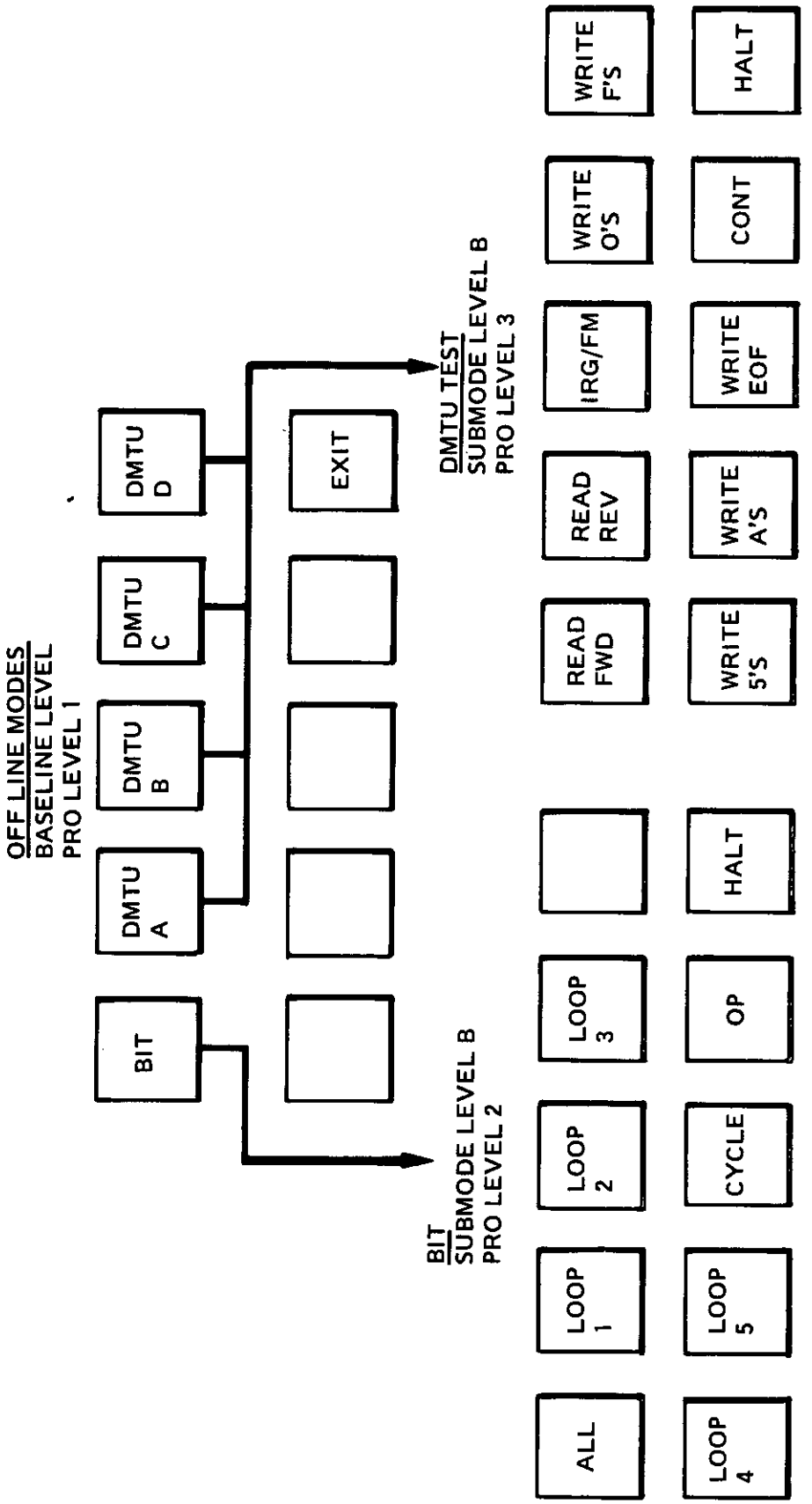


Figure 11. Off Line Mode PRO displays

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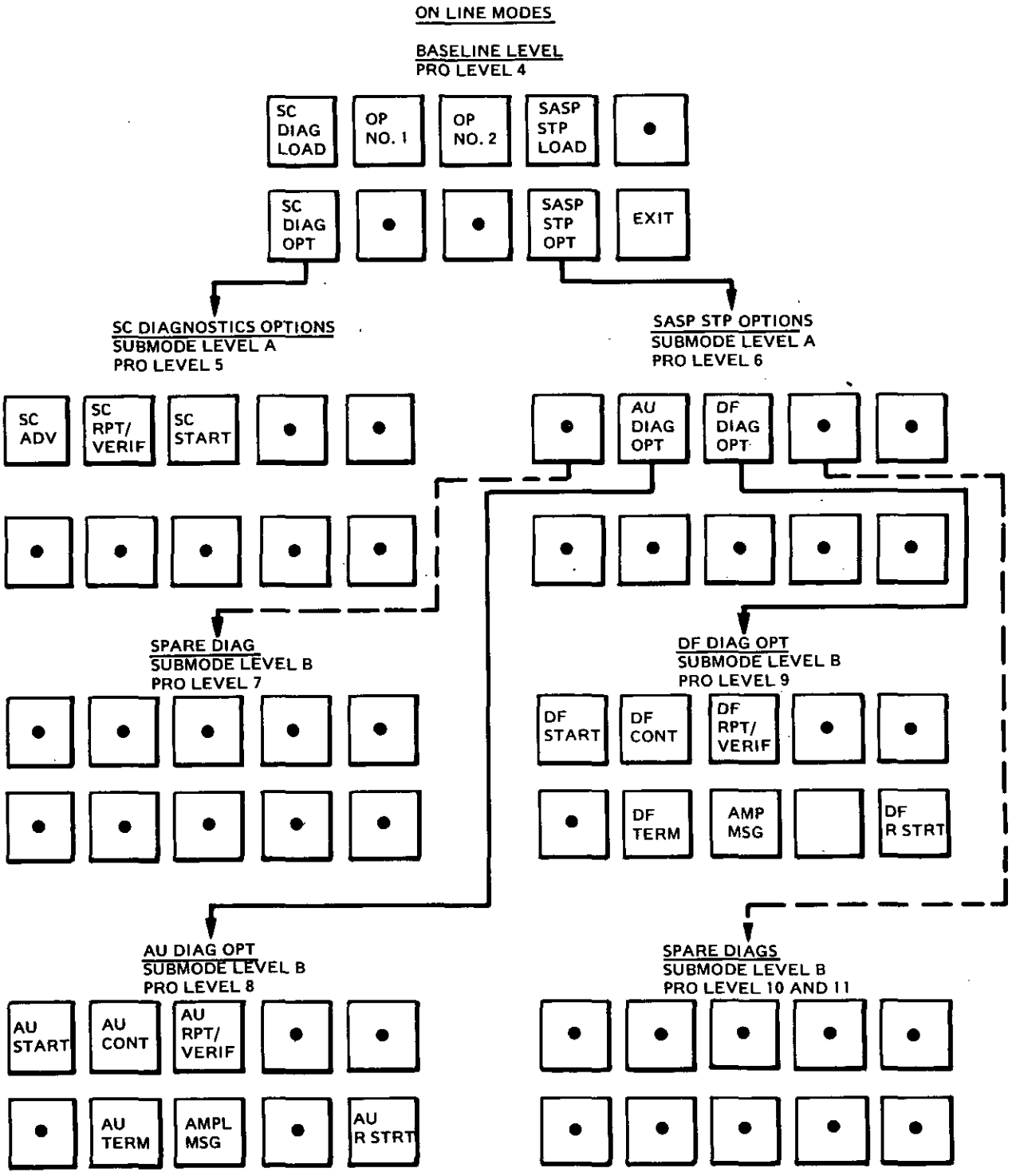


Figure 12. On Line Mode PRO displays

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3.5.1.2.2.6.1 Alphanumeric display. The OP shall provide independent but simultaneous display of alphanumeric (hexadecimal/octal) data associated with each control section. This display feature shall remain blank at all other times. Types of data to be displayed shall include status and data words and DMTS SRA fault isolation reports received from both control sections. Data displayed shall remain displayed until replaced by new control section-entered data or completely cancelled (blanked) by the CLR data entry function.

3.5.1.2.2.6.2 Alerts, cues, and function identifiers. The OP shall provide English language alerts, cues, function identifiers, and associated function initiation mechanisms to aid an operator in monitoring, executing functions, repeating or halting functions, and interpreting results of operations performed in all DMTS Modes. All alerts, cues, and function identifiers associated with subject modes shall be activated by either control section. The alerts, cues and function identifiers shall be implemented in a manner which simplifies the procedural sequencing of single and multi-step manual operations required to accomplish actions allowed by DMTS Modes. Alerts, cues, and function identifiers when in an active condition shall be highlighted to aid in visual recognition and differentiation from those which are not active. Switches associated with active function identifiers, when depressed, shall be recognized by the selected control section and shall initiate implementation of subject function. Depression of function identifier switches associated with functions which are not active shall not be recognized by the control section unless allowed as part of an operator option selection.

3.5.1.2.2.7 Configuration selection. The OP shall provide the switches and indicators whereby the operator can select and have displayed continuously the assignment of computer interface channels and DMTU's to control sections. Configuration selection switches shall be active in all DMTS Modes. Upon application of power, either by the deliberate power turn-on by operator or following power transients, the DMTS shall restore system to the previous DMTS configuration assignments, as specified in 3.3.12.1.5. See Table VI for illumination alternatives for the OP switch/indicators when initialized or reinitialized.

3.5.1.2.2.7.1 Control section/DMTU. The OP shall provide the switches and indicators whereby the operator can assign from one to four DMTU's to a particular control section and continuously monitor the selected configuration. However, the operator shall be prevented from assigning the same DMTU to both control sections at the same time.

3.5.1.2.2.7.2 Control section/computer interfaces. The OP shall provide the switches and indicators whereby the operator can assign any one of the four computer channels to a particular control section and continuously monitor the selected configuration. However, the operator shall be prevented from assigning the same computer channel to both control sections at the same time.

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TABLE VI. Indicator and switch functions

Function	Legend	Type Device	Indicator Color Meaning			
			Dark	Green	Amber	Red
AYK-14	PWR	Indicator	Off	---	On	---
	BIT ERROR	Indicator	OK	---	ERROR	---
	OVER TEMP	Indicator	OK	---	---	Overtemp
	IPL FAIL	Ind/Switch-Mon	---	Normal	---	---
		Indicator	OK	---	FAIL	---
	RESET	Ind/Switch-Mon	---	Normal	---	---
	AYK-14 CLR	Ind/Switch-Mon	---	Normal	---	---
	PROG SEL	Digital Switch	---	---	---	---
BIT CTRL 1	INIT	Ind/Switch-Mon	---	Avail	Active	---
	2/4/3/1	4-Segment Ind	Off	---	Fail	---
BIT CTRL 2	INIT	Ind/Switch-Mon	---	Avail	Active	---
	16/18/17/1	4-Segment Ind	Off	---	Fail	---
MISC.	OVERTEMP	Indicator	OK	---	---	Overtemp
	LAMP TEST	Ind/Switch-Mon	---	Avail	---	---
	PWR ON	Indicator	Off	---	On	---
CTRL 1	ON LN/ OFF LN	Ind/Switch-Alt	---	Avail	Active	---
CTRL 2	ON LN/ OFF LN	Ind/Switch-Alt	---	Avail	Active	---
I/O SEL - CTRL 1	CP901	Ind/Switch-Mon	Not connected	Avail	Active	---
	SASP	Ind/Switch-Mon	Not connected	Avail	Active	---
	AYK-14	Ind/Switch-Mon	Not connected	Avail	Active	---
I/O SEL CTRL 2	CP901	Ind/Switch-Mon	Not connected	Avail	Active	---
	SASP	Ind/Switch-Mon	Not connected	Avail	Active	---
	AYK-14	Ind/Switch-Mon	Not connected	Avail	Active	---

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TABLE VI. Indicator and switch functions (continued)

Function	Legend	Type Device	Indicator Color Meaning			
			Dark	Green	Amber	Red
DMTU SEL CTRL 1/2	A/A	Ind/Switch-Alt	Not connected	Avail	Active	---
	B/B	Ind/Switch-Alt	Not connected	Avail	Active	---
	C/C	Ind/Switch-Alt	Not connected	Avail	Active	---
	D/D	Ind/Switch-Alt	Not connected	Avail	Active	---
DISPLAY CONTROLS-CTRL 1	CLR	Ind/Switch-Mon	---	Avail	---	---
	SEL MODE	Ind/Switch-Mon	---	Avail	---	---
	MORE INFO	Ind/Switch-Mon	None available	Avail	---	---
DISPLAY CONTROLS-CTRL 2	CLR	Ind/Switch-Mon	---	Avail	---	---
	SEL MODE	Ind/Switch-Mon	---	Avail	---	---
	MORE INFO	Ind/Switch-Mon	None available	Avail	---	---

3.5.1.2.2.8 DMTS Mode selection. The OP shall provide the necessary switches and indicators to allow manual and independent selection of the following DMTS Modes for each control section at any time.

a. On Line (ON LN)
b. Off Line (OFF LN)
c. PDC Load (SASP)
d. AYK 14 Load
e. BIT

DMTS Mode selection shall be as specified in Figures 10, 11 and 12, and AD-1404.

3.5.1.3 DMTS operating Modes. The DMTS shall provide for two operating modes (Off Line and On Line Modes) and an Initialization Mode for each control section as described below and in AD-1404. Selection of control section modes shall be accomplished only through operator action at the OP. At DMTS POWER ON or return of normal DMTS operating power following a power interruption, each control section shall set its assigned computer interface lines to their idle status, reset and initialize the particular DMTU(s) interface lines, and be prepared to operate in accordance with the requirements of the assigned mode. A control section in any mode shall not interfere with the operation of the alternate control section, including use of the OP. A control section shall only be in one mode at a time as last selected/configured at the OP.

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3.5.1.3.1 Off Line Mode. The DMTS shall provide Off Line BIT and DMTU test options for each control section as specified in AD-1404 and as shown in Figures 10 and 11.

3.5.1.3.2 On Line Mode. The DMTS shall provide an On Line Mode for each control section. This mode shall allow the control section to be independently controlled by the computer command sets associated with the computer interfaces assigned to the control section. DMTU/control section status and data transfer shall be directed to/from the selected computer interface. The On Line Mode shall be characterized by allowing only the assigned computer to have complete control over its assigned control section and DMTU(s). Both control sections shall be allowed to operate simultaneously in the On Line Mode. One control section in the On Line Mode shall not prohibit the alternate control section from operating in a different mode.

3.5.1.3.2.1 CP-901 load Mode. CP-901 load Modes are initiated from a remote maintenance panel which is not a part of the DMTS. The CP-901 has no access to either the DMTC alphanumeric readout or mode display during On Line operations.

3.5.1.3.2.2 PDC load Mode. The DMTC shall provide the necessary functions to initiate and control loading of programs into the SASP system as well as initiating and controlling diagnostic programs within the SASP system, as shown in Figure 12. There are two major function groups associated with PDC Load Mode:

a. SASP Load

b. SASP Diagnostics

Prior to initiating any PDC Load, DMTU address "0" must be selected on the control panel of the active DMTU. If not selected, an "Address Error" message shall be displayed. Normal commands of the type Fxxxxxxxx (excluding INIT BIT CMD) shall be executed as specified in 3.5.2.2.3 during the PDC load Modes. Channel Reset (CRS) Control Frames shall be executed as specified in 3.5.2.2.3.3.1. These F type commands and CRS Control Frames may be dispersed throughout the load sequences and the DMTS shall be capable of executing these commands without disturbing the load sequence in progress. Exception Control Intercept Words (CIW's) are defined to be anything other than the expected CIW's and shall cause an error message. Data Words (DW's) following exception CIW's shall be accepted until a stop is received.

3.5.1.3.2.2.1 SASP Load Mode. The DMTS shall provide load selection, initiation and control which is selectable on the OP only when the control section is assigned to a PDC. The SASP Load Mode shall provide for English text readouts of load status during the loading of the selectable software packages into the selected SASP system in accordance with AD-1404 and Tables VII through X. Load operations once initiated at the OP shall be performed automatically without further operator participation. The DMTS

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TABLE VII. System controller (SC) diagnostic load sequence

Normal	Display	Exception	Display
Send SRL Control Frame Rewind Tape Wait 1 Second	SRL		
Send IMPL Command CIW = F0040000	IMPL	Start Time Out t = 10 Seconds (t Based on Time to Complete IMPL Process)	
		DMTC Times Out	IMPL TO
Receive Good IMPL Response CIW = E0000060	IMPL CMP	Receive Bad IMPL Response CIW = C00N0000 or C01N0000 N DW's $0 \leq N \leq 6$ *Press AMP SW . . . *Press AMP SW *Press AMP SW	IMPL ERR CIW C00N0000 or C01N0000 . . . DW WD N IMPL ERR
Receive Good IPL Response CIW = E0000020	IPL CMP	Receive Bad IPL Response CIW = C0020000 2 Data Words *Press AMP SW *Press AMP SW *Press AMP SW *Press AMP SW	IPL ERR CIW C0020000 DW 1 DW 2 IPL ERR
Send SC LD Command CIW = 01040001 DC = 00041101	SC LD		
Receive Good TC Response CIW = 01800000	SC LD	Receive Bad TC Response CIW = See Note Below *Press AMP SW *Press AMP SW	SC LD ERR CIW = See Note Below SC LD ERR

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TABLE VII. System controller (SC) diagnostic load sequence (Continued)

Normal	Display	Exception	Display
Receive Good PC Response CIW = 01C00000	SC LD CMP	Receive Bad PC Response CIW = 01C00003 3 DW's *Press AMP SW *Press AMP SW *Press AMP SW *Press AMP SW *Press Amp SW	SC LD ERR CIW 01C00003 DW WD 1 DW WD 2 DW WD 3 SC LD ERR

NOTES:

*Indicates operation action
AMP = "More Info"
TC = Transfer Complete (Handshake)
PC = Processing Complete (Operation Complete or Load Complete)
SRL = System Reset and Load
IMPL = Initiate Microprogram Load
CIW = Control Interrupt Word
DW = Data Word
IPL = Initiate Program Load
ERR = Error
CMP = Complete
N = Number of
SW = Switch
WD = Word
LD = Load

In the Exception column, a bad response is defined to be any CIW other than the good response listed in the Normal column or FXXXXXXX type commands or CRS Control Frames that may be dispersed throughout the load sequence. The IDS contains anticipated command sequences that are meshed throughout the load sequence. If a bad response is received as shown in the Exception column, the anticipated bad response CIW is shown. However, this does not preclude the possibility of other bad responses. In any case, the DMTS shall display the actual CIW received.

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TABLE VIII. ASP STP load sequence

Normal	Display	Exception	Display
Send SRL Control Frame Rewind Tape Wait 1 Second	SRL		
Send IMPL Command CIW = F0040000	IMPL	Start Time Out t = 10 Seconds (t Based on Time to Complete IMPL Process)	
		DMTC Times Out	IMPL TO
Receive Good IMPL Response CIW = E0000060	IMPL CMP	Receive Bad IMPL Response CIW = C00N0000 or C01N0000 N DW's $0 \leq N \leq 6$ *Press AMP SW . . . *Press AMP SW *Press AMP SW	IMPL ERR CIW C00N0000 or C01N0000 . . . DW WD N IMPL ERR
Receive Good IPL Response CIW = E0000020	IPL CMP	Receive Bad IPL Response CIW = C0020000 2 Data Words *Press AMP SW *Press AMP SW *Press AMP SW *Press AMP SW	IPL ERR CIW C0020000 DW 1 DW 2 IPL ERR
Send SC LD Command CIW = 01040001 DW = 00042101	SC LD		
Receive Good TC Response CIW = 01800000	SC LD	Receive Bad TC Response CIW - See Note Below *Press AMP SW *Press AMP SW	SC LD ERR CIW = See Note Below SC LD ERR

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TABLE VIII. ASP STP load sequence (Continued)

Normal	Display	Exception	Display
Receive Good PC Response CIW = 01C00000	SC LD CMP	Receive Bad PC Response CIW = 01C00003 3 DW's	SC LD ERR
		*Press AMP SW	CIW 01C00003
		*Press AMP SW	DW WD 1
		*Press AMP SW	DW WD 2
		*Press AMP SW	DW WD 3
		*Press AMP SW	SC LD ERR

NOTES:

*Indicates operation action
AMP = "More Info"
TC = Transfer Complete (Handshake)
PC = Processing Complete (Operation Complete or Load Complete)
SRL = System Reset and Load
IMPL = Initiate Microprogram Load
CIW = Control Interrupt Word
DW = Data Word
IPL = Initiate Program Load
ERR = Error
CMP = Complete
N = Number of
SW = Switch
WD = Word
LD = Load
SASP = Accoustic System Processor
STP = System Test Program

In the Exception column, a bad response is defined to be any CIW other than the good response listed in the Normal column or FXXXXXXX type commands or CRS Control Frames that may be dispersed throughout the load sequence. The IDS contains anticipated command sequences that are meshed throughout the load sequence. If a bad response is received as shown in the Exception column, the anticipated bad response CIW is shown. However, this does not preclude the possibility of other bad responses. In any case, the DMTS shall display the actual CIW received.

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TABLE IX. OP No. 1 load sequence

Normal	Display	Exception	Display
Send SRL Control Frame Rewind Tape Wait 1 Second Send IMPL Command CIW = F0040000	SRL IMPL	Start Time Out t = 10 Seconds (t Based on Time to Complete IMPL Process)	
		DMTC Times Out	IMPL TO
Receive Good IMPL Response CIW = E0000060	IMPL CMP	Receive Bad IMPL Response CIW = C00N0000 or C01N0000 N DW's $0 \leq N \leq 6$ *Press AMP SW . . . *Press AMP SW *Press AMP SW	IMPL ERR CIW C00N0000 or C01N0000 . . . DW WD N IMPL ERR
Receive Good IPL Response CIW = E0000020	IPL CMP	Receive Bad IPL Response CIW = C0020000 2 Data Words *Press AMP SW *Press AMP SW *Press AMP SW *Press AMP SW	IPL ERR CIW C0020000 DW 1 DW 2 IPL ERR
Send SC LD Command CIW = 01040001 DW = 00043101	SC LD		
Receive Good TC Response CIW = 01800000	SC LD	Receive Bad TC Response CIW = See Note Below *Press AMP SW *Press AMP SW	SC LD ERR CIW = See Note Below SC LD ERR

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TABLE IX. OP No. 1 load sequence (Continued)

Normal	Display	Exception	Display
Receive Good PC Response CIW = 01C00000	SC LD CMP	Receive Bad PC Response CIW = 01C00003 3 DW's *Press AMP SW *Press AMP SW *Press AMP SW *Press AMP SW *Press AMP SW	SC LD ERR CIW 01C00003 DW WD 1 DW WD 2 DW WD 3 SC LD ERR

NOTES:

*Indicates operation action
AMP = "More Info"
TC = Transfer Complete (Handshake)
PC = Processing Complete (Operation Complete or Load Complete)
SRL = System Reset and Load
IMPL = Initiate Microprogram Load
CIW = Control Interrupt Word
DW = Data Word
IPL = Initiate Program Load

ERR = Error
CMP = Complete
N = Number of
SW = Switch
WD = Word
LD = Load
TO = Time Out
OP NO. 1 = Operational Program No. 1

In the Exception column, a bad response is defined to be any CIW other than the good response listed in the Normal column or FXXXXXXX type commands or CRS Control Frames that may be dispersed throughout the load sequence. The IDS contains anticipated command sequences that are meshed throughout the load sequence. If a bad response is received as shown in the Exception column, the anticipated bad response CIW is shown. However, this does not preclude the possibility of other bad responses. In any case, the DMTS shall display the actual CIW received.

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TABLE X. OP No. 2 load sequence

Normal	Display	Exception	Display
Send SRL Control Frame Rewind Tape Wait 1 Second Send IMPL Command CIW = F0040000	SRL IMPL	Start Time Out t = 10 Seconds (t Based on Time to Complete IMPL Process)	
		DMTC Times Out	IMPL TO
Receive Good IMPL Response CIW - E0000060	IMPL CMP	Receive Bad IMPL Response CIW = C00N0000 or C01N0000 N DW's $0 \leq N \leq 6$ *Press AMP SW . . . *Press AMP SW *Press AMP SW	IMPL ERR CIW C00N0000 or C01N0000 . . . DW WD N IMPL ERR
Receive Good IPL Response CIW = E0000020	IPL CMP	Receive Bad IPL Response CIW = C0020000 2 Data Words *Press AMP SW *Press AMP SW *Press AMP SW *Press AMP SW	IPL ERR CIW C0020000 DW 1 DW 2 IPL ERR
Send SC LD Command CIW = 01040001 DW = 00044101	SC LD		
Receive Good TC Response CIW = 01800000	SC LD	Receive Bad TC Response CIW = See Note Below *Press AMP SW *Press AMP SW	SC LC ERR CIW = See Note Below SC LD ERR

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TABLE X. OP No. 2 load sequence (Continued)

Normal	Display	Exception	Display
Receive Good PC Response CIW = 01C00000	SC LD CMP	Receive Bad PC Response CIW = 01C00003 3 DW's *Press AMP SW *Press AMP SW *Press AMP SW *Press AMP SW *Press AMP SW	SC LD ERR CIW 01C00003 DW WD 1 DW WD 2 DW WD 3 SC LD ERR

NOTES:

*Indicates operation action
AMP = "More Info"
TC = Transfer Complete (Handshake)
PC = Processing Complete (Operation Complete or Load Complete)
SRL = System Reset and Load
IMPL = Initiate Microprogram Load
CIW = Control Interrupt Word
DW = Data Word
IPL = Initiate Program Load

ERR = Error
CMP = Complete
N = Number of
SW = Switch
WD = Word
LD = Load
OP No. 2 = Operational Program No. 2
TO = Time Out

In the Exception column, a bad response is defined to be any CIW other than the good response listed in the Normal column or FXXXXXXX type commands or CRS Control Frames that may be dispersed throughout the load sequence. The IDS contains anticipated command sequences that are meshed throughout the load sequence. If a bad response is received as shown in the Exception column, the anticipated bad response CIW is shown. However, this does not preclude the possibility of other bad responses. In any case, the DMTS shall display the actual CIW received.

shall provide for simultaneous loading of any available software packages into the SASP system with the stipulation that each control section be assigned a separate PDC and a separate DMTU. Load status during simultaneous load operations shall be displayed simultaneously. The OP shall provide the capability to select from four mutually exclusive SASP load options. A function shall be provided to initiate the selected load option. Once initiated, the command sequences which are transferred between the control section and the SASP shall be as specified in AD-1404 and Tables VII through X. A function to terminate a previously initiated load operation that is in progress shall be provided. The terminate function shall cause a System Reset and Load (SRL) control frame to be sent across the selected PDC interface. The SASP Load Options shall include (Figure 12):

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a. System Controller (SC) Diagnostics Load - Table VII

b. SASP System Test Program (STP) Load - Table VIII

c. Operational Program No. 1 (OP No. 1) - Table IX

d. Operational Program No. 2 (OP No. 2) - Table X

The SASP Load Options, the load initiation and load termination functions shall all be available simultaneously if the SASP STP Load Mode is selected. Selecting any of the SASP Load options shall cause the corresponding functions to be set active. Selecting an alternate SASP Load option shall render other load option functions inactive and activate those functions associated with the selected load option. The Initiate function once activated shall remain active until normal completion of the load function or deactivated by the Terminate function. The words identified in Tables VII, VIII, IX, and X are defined in AD-1404.

3.5.1.3.2.2.1.1 MORE INFO switch. The MORE INFO (More Information) switch shall be used to support BIT tests, load sequences and diagnostic interpretation results. The MORE INFO switch shall cause sequential data words to be displayed on the alphanumeric readouts. The DMTS shall be capable of inputting data words from SASP Display Control Set (DCS) via Data Word (DW) sequences in support of the load modes and the execution of SC diagnostics for displaying amplifying error information on the OP. All amplifying data words shall be outputted from the SASP via a single buffer transfer, e.g., until the SASP issues a STOP control frame. The amplifying data words shall be stored in the DMTS allowing for a cyclic redisplay as requested by the operator by depressing the MORE INFO switch. The MORE INFO switch shall be illuminated green whenever new amplifying data is available. If MORE INFO is lit as a result of a BIT test, selecting On Line Mode shall extinguish it and return control section to On Line Mode.

3.5.1.3.2.2.1.2 SC diagnostics load (SC DIAG LOAD) option. Initiating the SC DIAG LOAD Option shall initiate the command sequence specified in Table VII. Receipt of the CIW sequences and DW sequences specified in the table shall cause the corresponding messages to be displayed to the operator.

3.5.1.3.2.2.1.3 SASP system test program (STP) load option. Initiating the SASP STP Load Option shall initiate the command sequence specified in Table VIII. Receipt of the CIW sequences and DW sequences specified in the table shall cause the corresponding messages to be displayed to the operator.

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3.5.1.3.2.2.1.4 Operational Program No. 1 (OP No. 1) load option. Initiating the Operation Program No. 1 Load Option shall initiate the command sequence specified in Table IX. Receipt of the CIW sequences and DW sequences specified in the table shall cause the corresponding messages to be displayed to the operator.

3.5.1.3.2.2.1.5 Operational Program No. 2 (OP No. 2) load option. Initiating the Operation Program No. 2 Load Option shall initiate the command sequence specified in Table X. Receipt of the CIW sequences and DW sequences specified in the table shall cause the corresponding messages to be displayed to the operator.

3.5.1.3.2.2.2 SASP diagnostics modes. The DMTS shall provide SASP diagnostics test selection, initiation and control which is selectable at the OP only when the control section is assigned to a POC channel. The SASP Diagnostics Control functions shall be organized into the following function groups (Figure 11):

a. System Controller (SC) Diagnostics (DIAG) Option (OPT)

b. SASP STP Option

(1) Display Formatter (DF) Diagnostics Option

(2) Analyzer Unit (AU) Diagnostics Option

(3) Spare Diagnostics Options

Each of these function groups shall consist of a set of functions, defined in the following subparagraphs. Selection of either the DF DIAG OPT, or AU DIAG OPT control functions shall cause processing as specified in AD-1404. Selection of the SC DIAG control function shall not cause a function code to be transmitted to the computer. The spare diagnostic selections are not defined in this specification.

3.5.1.3.2.2.2.1 SC DIAG OPT. The operator shall be prohibited from initiating SC DIAG OPT functions unless the SC DIAG Load Option (see 3.5.1.3.2.2.1.2) has previously been completed. Selection of the SC DIAG OPT function shall make the following control functions available to the operator (Figure 12):

a. SC ADVANCE (ADV)

b. SC REPEAT/VERIFY (RPT/VERIF)

c. SC START

Selection of any of these functions shall not cause a CIW function code to be transmitted to the computer. The DMTC shall assume complete control of the active/inactive status indicators for these functions. The DMTC shall also interpret and execute the control functions when selected.

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3.5.1.3.2.2.2.1.1 SC DIAG tests. SC Diagnostics tests shall be all those tests included in the SC Diagnostics Program, and shall be commanded to operate in the mode where test execution stops upon detection of each error. The DMTC shall be capable of accepting all messages sent by the SC Diagnostics Program while it is executing. Also, the SC Diagnostics Program may issue a series of Read and Skip commands for the purpose of loading a program overlay. The SC Diagnostics Program will provide test results using CIW and DW sequences. The DMTC shall provide the translation to English readable text as required. See AD-1404 for the specific firmware requirements imposed upon the DMTS.

3.5.1.3.2.2.2.1.2 SC Advance. The SC advance (SC ADV) function shall not be available unless tests have been started previously and have stopped. After receipt of a Processing Complete CIW, selection of the SC ADV function shall cause the function to be set to the active state and the SC Diagnostics Advance command to be transmitted to the computer. (See AD-1404.) Test results shall be returned to the DMTC after the tests and shall be displayed to the operator. The function shall be set inactive upon receipt of the tests results. If the SC ADV function is selected while the tests are already active, the request shall be ignored.

3.5.1.3.2.2.2.1.3 SC Repeat/Verify. The SC Repeat/Verify (RPT/VERIF) function shall not be available unless tests have been started previously and have stopped. After receipt of a Processing Complete CIW, selection of the SC RPT/VERIF function shall cause the function to be set to the active state and the SC Diagnostic RPT/VERIF command to be transmitted to the computer. (See AD-1404.) Test results shall be returned to the DMTC after the tests and shall be displayed to the operator. The function shall be set inactive upon receipt of the test results. If the SC RPT/VERIF function is selected while tests are already active, the request shall be ignored.

3.5.1.3.2.2.2.1.4 SC START. If the SC START function is selected while no tests are active, it shall be set to the active state and the SC Diagnostics START command to initiate all tests shall be transmitted to the computer. (See AD-1404.) Test results shall be returned to the DMTC after the tests and shall be displayed to the operator. The function shall be set inactive upon receipt of the test results. If the SC START function is selected while tests are already active, the request shall be ignored. Each subsequent selection of the SC START function shall cause SC DIAG to begin with the first test in its test sequence, and to run as many tests as possible until either the test completes successfully or stops on an error. The DMTC shall reset any error flags each time the SC START function is initiated.

3.5.1.3.2.2.2.2 Display Formatter (DF) Diagnostics Option. The selection of the DF DIAG OPT function shall make the following functions available to the operator (Figure 12):

a. DF Start

b. DF Continue (CONT)

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c. DF Repeat/Verify (RPT/VERIF)

d. DF Terminate (TERM)

e. Amplify Message (AMP MSG)

f. DF Restart (R STRT)

Selection of any of these functions shall cause a CIW function code to be transmitted to the computer via a switched data PDC command as defined in AD-1404. In addition, the computer may also send command words to the DMTC to set one of more of those functions in an active state, or to set previously active functions in an inactive state. The computer shall assume complete control of the active/inactive status of the functions. The computer shall also interpret and execute the functions when selected. When the computer sends a command word to the DMTC indicating the status of a function, the function status shall be visibly obvious to the operator on the OP. During the execution of the DF DIAG OPT functions, messages will be displayed to the operator, using the PRINT command defined in AD-1404.

3.5.1.3.2.2.2.3 Analyzer Unit (AU) Diagnostics Option. The selection of the AU DIAG OPT function shall make the following functions available to the operator (Figure 12):

a. AU Start

b. AU Continue (CONT)

c. AU Repeat/Verify (RPT/VERIF)

d. AU Terminate (TERM)

e. Amplify Message (AMP MSG)

f. AU Restart (R STRT)

Selection of any of these functions shall cause a CIW function code to be transmitted to the computer via a switched data PDC command as specified in 3.5.2.2.3.2.7. In addition, the computer may also send command words to the DMTC to set one or more of those functions in an active state, or to set previously active functions in an inactive state. The computer shall assume complete control of the active/inactive status of the functions. The computer shall also interpret and execute the functions when selected. When the computer sends a command word to the DMTC indicating the status of a function, the function status shall be visibly obvious to the operator on the OP. During the execution of the AU DIAG OPT functions, messages will be displayed to the operator, using the PRINT command defined in AD-1404.

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3.5.1.3.2.3 AYK-14 load Mode. The DMTS shall provide an AYK-14 Load Mode for each control section which is selectable from the OP only when the control section is assigned to the AYK-14. The AYK-14 Load Mode shall provide for control and status readouts during loading of the AYK-14 operational program. The loading of programs and execution of programs, as applicable, once initiated at the OP shall be performed automatically with current status information pertaining to the progress of the operation being displayed continuously. Implementation of the AYK-14 Load Mode shall be that required to minimize operator intervention to a level where manuals, publications and other supporting documentation are not required. The control section shall automatically execute the required functions, interpret AYK-14 responses and guide an operator through the steps requiring manual intervention at the highest level possible. The operator shall also be provided with appropriate status/progress reports at the OP. The OP shall provide the support for:

a. AYK-14 discretes (switches and indicators)

b. AYK-14 bootstrap load (See AD-1404.)

3.5.1.3.2.3.1 AYK-14 discrete support

3.5.1.3.2.3.1.1 PWR normal indicator. An indicator shall be provided on the DMTS OP that shall light amber when the power normal discrete from the AYK-14 is in the active state.

3.5.1.3.2.3.1.2 BITE ERROR indicator. An indicator shall be provided on the DMTS OP that shall light amber whenever the BIT ERROR discrete from the AYK-14 is in the active state.

3.5.1.3.2.3.1.3 IPL FAIL indicator. An indicator shall be provided on the DMTS OP panel. The indicator shall light amber whenever the IPL FAIL discrete is in the active (logical 1) state.

3.5.1.3.2.3.1.4 OVER TEMP indicator. An indicator shall be provided on the DMTS OP panel that shall illuminate red whenever the overtemperature discrete from the AYK-14 is in the active state.

3.5.1.3.2.3.1.5 Remote terminal (RT) Master Clear discrete. Whenever the RT Master Clear discrete from the AYK-14 is in the active (logical 1) state and Controller 2 (Control Section 2) is assigned to the DMDB, the DMTS shall perform a Master Clear sequence on Controller 2 and the DMDB interface as defined in 3.5.1.2.2.2 subparagraphs a through d only. (Applicable for Controller 2 only when DMDB selected for Controller 2.)

3.5.1.3.2.3.1.6 RESET switch. A momentary active switch shall be provided on the OP to activate the AYK-14 system RESET discrete. Activation of the switch to its momentary position (ON) shall cause the system RESET discrete to make a transition from the normally inactive (logical 0 output) to the active state (logical 1 output) for a period of time between 1 and 2 microseconds. Debouncing of the switch closure is required.

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3.5.1.3.2.3.1.7 AYK-14 CLR switch. A momentary action switch shall be provided on the OP to activate the AYK-14 Master Clear discrete. Activation of the switch to its momentary position (ON) shall cause the Master Clear discrete to make a transition from the normally inactive (logic 0 output) to the active state (logical 1 output) for a period of time between 1 and 2 microseconds. Debouncing of the switch closure is required.

3.5.1.3.2.3.1.8 IPL FAIL switch. A momentary action switch shall be provided on the OP to activate the IPL FAIL discrete. Activation of the switch to its momentary position (ON) shall cause the IPL FAIL discrete to make a transition from the normally inactive (logical 0 output) to the active state (logical 1 output) for a period of time between 1 and 2 microseconds. Debouncing of the switch closure is required.

3.5.1.3.2.3.1.9 Program Select (PROG SEL) switch. A four-position PROG SEL switch shall be provided on the OP. The switch positions shall be labeled zero through three. The position of the switch shall control the state of the two Program Select discretes interfaced to the AYK-14 specified in 3.5.2.3.5. The state of the two discretes shall be a binary encoding of the labeled switch position with Program Select bit 0 as the least significant bit.

3.5.1.3.2.4 AYK-14 Load support. The DMTS shall provide an AYK-14 load support for each control section which is selectable from the OP only when the control section is assigned to the DMDB. The AYK-14 Load shall provide for control and status readouts during loading of the AYK-14 programs. The loading of program and execution of programs, as applicable, once initiated at the OP shall be performed automatically with current status information pertaining to the progress of the operation being displayed continuously. Implementation of the AYK-14 Load Mode shall be that required to minimize operator intervention to a level where manuals, publications and other supporting documentation is not required. The control section shall automatically execute the required functions, interpret AYK-14 responses and guide an operator through the steps requiring manual intervention at the highest level possible. The operator shall also be provided with appropriate status/progress reports at the OP.

3.5.1.3.3 Built-in-test Modes. The DMTS shall provide BIT Modes which are available from the OP in the Off Line Mode. Operation of these BIT Modes shall be in accordance with 3.4.1.1, 3.4.1.2, and 3.4.1.3.4.

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3.5.2 Digital Magnetic Tape Controller. Each control section of the DMTC shall provide for the control of 1 to 4 DMTU's. The DMTC shall:

a. accept and execute commands received on the digital interface channel(s) when in the On Line Mode,

b. accept and execute operator entered commands received via the OP when in the Off Line Mode,

c. control the loading of specific system diagnostic and operational software programs while providing the display of intermediate and completion status of the load processes to the operator when in a computer program load mode,

d. and perform self diagnostic fault detection and isolation when in the BIT Mode.

3.5.2.1 Function description. The DMTC shall contain any combination of the following digital channel types within the maximum constraints specified in Table IV, Digital Channel Interface Mixes:

a. ANEW (MIL-STD-1397)

b. PROTEUS Digital Channel (PDC) (A30-15590)

c. Dual Multiplex Data Bus (DMDB) (MIL-STD-1553)

The DMTC shall contain two independent but identical control sections. Each control section shall have access to any of the four identical DMTU's. DMTU address selection shall be in accordance with Figure 13. Not more than one DMTU with the same address code can be selected for a particular control section at any one time. An "address error" message shall be displayed if more than one DMTU has the same address for any one control section. If address error is subsequently corrected by the operator, the "address error" message shall be automatically cleared. Each control section shall have access to either the parallel ANEW Channel, the serial PDC, or the serial DMDB, but only one at any one time. Each control section shall interpret the various commands received on each of the possible channel types and initiate the commanded function. DMTC design shall not require a program load to make it operational. A functional block diagram of the DMTC is presented in Figure 14.

3.5.2.1.1 DMTC power supply. The DMTC shall have self contained power supply capability which shall have the capacity to provide all DMTC DC power needs.

3.5.2.2 Digital Interfaces. The description of the fields, words and messages transferred across the ANEW, PDC, and DMDB interfaces shall be as described in AD-1404. A basic DMTS command set which is common to the three interfaces is described in the following paragraphs.

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ANEW Command Word Bits

		19	18	17	16
DMTU	S	B	B	B	B
A	0	X	0	0	X
B	2	X	0	1	X
C	4	X	1	0	X
D	6	X	1	1	X

A. ANEW DMTU Identifier Codes

PDC Command Word Bits

		21	20	19	18
DMTU	S	B	B	B	B
A	0	X	X	0	0
B	1	X	X	0	1
C	2	X	X	1	0
D	3	X	X	1	1
1	8	1	0	0	0

B. PDC DMTU Identifier Codes

DMDB Command Word Bits

		3	2	1	0
DMTU	S	B	B	B	B
A	0	X	0	0	X
B	2	X	0	1	X
C	4	X	1	0	X
D	6	X	1	1	X
1	8	1	0	0	0

C. DMDB DMTU Identifier Codes

S = HEX CODE

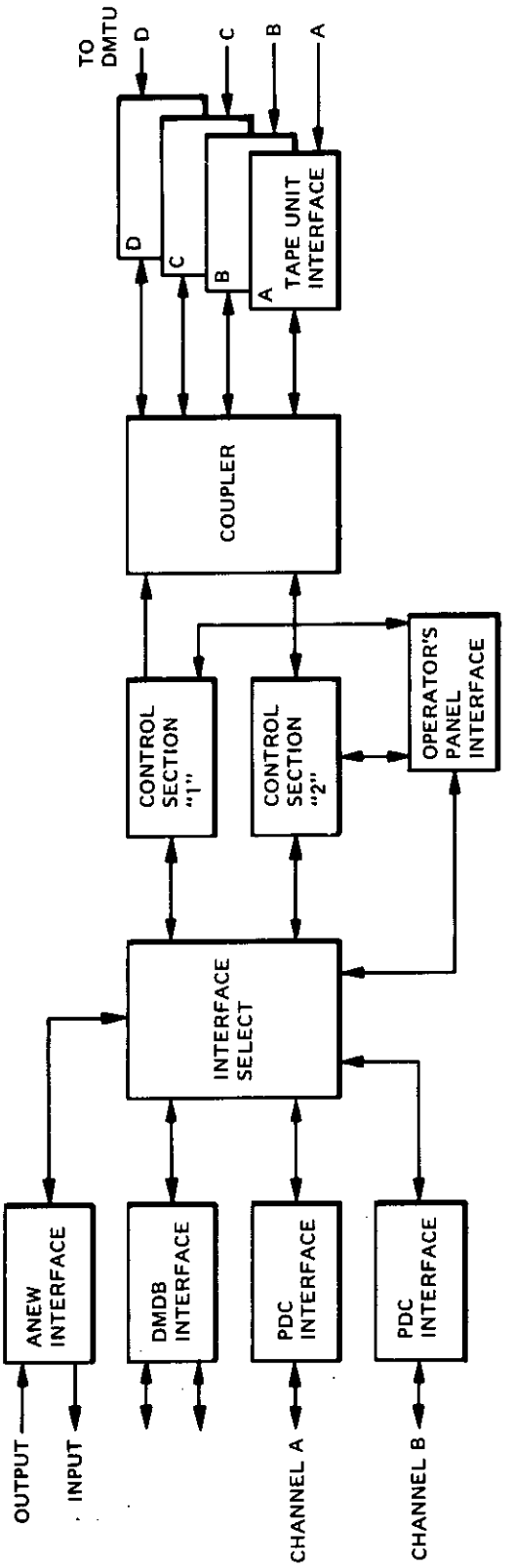
B = BINARY CODE BITS

X = DON'T CARE

1 = RESERVED FOR PRINT COMMANDS

FIGURE 13. DMTU identifier code

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NOTE:
NO ONE DMTU CAN BE ASSIGNED TO MORE THAN ONE CONTROL SECTION AT ANY ONE TIME. FURTHERMORE, NO MORE THAN ONE COMPUTER CHANNEL CAN BE ASSIGNED TO A PARTICULAR DMTU AT ANY ONE TIME.

FIGURE 14. DMTC functional block diagram

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3.5.2.2.1 Basic DMTS command set. The DMTS shall accept and execute the basic command set:

a. Write Record

b. Write Record-Extended IRG

c. Write File Mark

d. Write File Mark-Extended IRG

e. Read Record

f. Read Record Reverse

g. Read File

h. Search File/Read Record

i. Search File Reverse/Read Record

j. Rewind

k. Skip Forward Record

l. Skip Forward File

m. Skip Reverse Record

n. Skip Reverse File

o. Master Clear

p. Halt

q. Status Request

r. Loop Test

s. Initiate BIT

The DMTS shall maintain status checking circuitry. The status shall be reported as specified in each particular interface section of AD-1404.

3.5.2.2.1.1 Write Record. Receipt of a Write Record command shall cause the receiving control section to initiate the writing of data on the tape of the commanded DMTU. First a normal Inter-Record Gap (IRG) shall be written. Then data received via data word transfer sequences shall be disassembled and parity generated. The data and its associated parity bits shall be recorded on tape. When the recorded data passes over the read head, parity shall be checked. The DMTS shall permit the writing of records

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of any length. For the ANEW and the PDC interfaces, the record length will be an integer multiple of 32 bit words. For the DMDB interface the record length will be an integer multiple of 16 bit words.

3.5.2.2.1.2 Write Record - Extended IRG. Receipt of a Write Record-Extended IRG command shall cause the receiving control section to initiate the writing of data on the tape of the commanded DMTU. First an extended IRG shall be written, then the data shall be written as specified in 3.5.2.2.1.1.

3.5.2.2.1.3 Write File Mark. Receipt of a Write File Mark command shall cause the receiving control section to initiate the writing of a file mark on the tape of the commanded DMTU. First a normal IRG shall be written. Then the file mark shall be written.

3.5.2.2.1.4 Write File Mark - Extended IRG. Receipt of a Write File Mark-Extended IRG command shall cause the receiving control section to initiate the writing of a file mark on the tape of the commanded DMTU. First an extended IRG shall be written. Then the file mark shall be written as specified in 3.5.2.2.1.3.

3.5.2.2.1.5 Read Record. Receipt of a Read Record command shall cause the receiving control section to initiate the reading of data from the tape of the commanded DMTU in the forward direction. The data read from the tape shall be assembled into data words and transferred via data word sequences to the computer. During the Read operation, parity shall be checked.

3.5.2.2.1.6 Read Record Reverse. Receipt of a Read Record Reverse command shall cause the receiving control section to initiate the reading of data from the tape of the commanded DMTU in the reverse direction. The data read from tape shall be assembled such that each data word is the same as when written. Data words assembled from the record shall be transmitted to the computer via data word sequences in the reverse order as received when written. During the Read operation, parity shall be checked. The tape shall be stopped in the IRG preceding the record read.

3.5.2.2.1.7 Read File. Receipt of a Read File command shall cause the receiving control section to initiate successive Read Record forward operations from the tape of the commanded DMTU. The Read Record forward operation shall be as specified in 3.5.2.2.1.5. The tape shall be stopped in the IRG following the file mark.

3.5.2.2.1.8 Search File/Read Record. Receipt of a Search File/Read Record command shall cause the receiving control section to initiate the reading, in the forward direction, of the first word in each record within the particular file on the tape of the commanded DMTU. The command word will be followed by an Identifier word. The first word of each record shall be compared with the Identifier word. Upon a successful comparison all data in the record, including the Identifier word, shall be read and transferred to the computer as specified in 3.5.2.2.1.5. If no compare is found within the file, then tape shall be positioned at the next file mark. If no file mark is found, then tape shall be positioned at the End of Tape (EOT).

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3.5.2.2.1.9 Search File Reverse/Read Record. Receipt of a Search File Reverse/Read Record command shall cause the receiving control section to initiate the reading of each word in each record in a specific file in the reverse direction. The command word will be followed by an Identifier word. The first word of each record (the word nearest the Beginning of Tape (BOT) shall be compared with the identifier word. Upon a successful comparison, all data in the record, including the identifier word, shall be read in the forward direction and transferred to the computer as specified in 3.5.2.2.1.5. If no compare is found within the file, then tape shall be positioned at the next file mark. If no file mark is found, then tape shall be positioned at BOT.

3.5.2.2.1.10 Rewind. Receipt of a Rewind command shall cause the receiving control section to initiate high speed rewind. Upon detection of the Beginning of Tape (BOT) mark, tape shall stop and the BOT mark shall be positioned over the BOT mark sensor.

3.5.2.2.1.11 Skip Forward Record. Receipt of a Skip Forward Record command shall cause the receiving control section to position the tape on the commanded DMTU at the next IRG in the forward direction. The tape shall be properly positioned for the initiation of a read or write operation. The time to execute a Skip Forward Record command shall not exceed 200 milliseconds for the skipping over of a 1024 word (32 bits each) record (assuming DMTU at rest in a normal IRG).

3.5.2.2.1.12 Skip Forward File. Receipt of a Skip Forward File command shall cause the receiving control section to move the tape forward and position the tape on the commanded DMTU at the IRG following the next File Mark on the forward direction. The tape shall be properly positioned for the initiation of a read or write operation.

3.5.2.2.1.13 Skip Reverse Record. Receipt of a Skip Reverse Record command shall cause the receiving control section to position the tape on the commanded DMTU at the next IRG, in the reverse direction. The tape shall be properly positioned for the initiation of a read or write operation. The time to execute a Skip Reverse Record command shall not exceed 200 milliseconds for the skipping over of a 1024 word (32 bits each) record (assuming DMTU at rest in a normal IRG).

3.5.2.2.1.14 Skip Reverse File. Receipt of a Skip Reverse File command shall cause the receiving control section to position the tape on the commanded DMTU at the IRG on the BOT side preceding File Mark. The tape shall be properly positioned for the initiation of a read reverse or write operation.

3.5.2.2.1.15 Master Clear. Receipt of a Master Clear command shall immediately cause the receiving control section to terminate all current operation, clear status, set all computer interface lines to their idle states, and initialize the control section and all assigned DMTU's. If tape is in motion at the time of receipt of a Master Clear command the tape shall be stopped immediately, unless the DMTU is in a REWIND Mode.

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3.5.2.2.1.16 Halt. Receipt of a Halt command during twenty-five inches per second (ips) tape movement operation shall cause the receiving control section to stop tape motion on the commanded DMTU at the next IRG. Data transfers associated with the DMTU shall be terminated immediately upon receipt of the Halt command. If there is no "next IRG" then tape shall run to EOT or BOT depending upon direction of tape motion when Halt command is received. Halt command is to be ignored during a rewind operation.

3.5.2.2.1.17 Status Request. Receipt of a Status Request command shall cause the receiving control section to return the requested status to the computer. Execution of a Status Request command shall not be permitted during any associated control section operations currently in progress except Rewind. The detail requirements of status conditions to be reported and the method by which they should be reported are specified in each particular interface section of AD-1404.

3.5.2.2.1.18 Loop test. For detailed requirements refer to each particular interface section of AD-1404.

3.5.2.2.1.19 Initiate BIT. Receipt of an Initiate BIT command shall cause the receiving control section to initiate a self test on itself and its associated digital interface and DMTU. The receiving control section shall report a GO/NO GO condition. The faulty module(s), (if a NO GO condition), shall be reported in the status data word(s) upon the completion of the BIT.

3.5.2.3 Electrical interfaces

3.5.2.3.1 DMTC/DMTU Interface. The DMTC/DMTU Interface shall be designed to support execution of all the DMTS operations specified in 3.5 and its associated subparagraphs. The DMTC/DMTU Interface shall also provide the necessary lines to support all the OP DMTU control and status indicators specified in 3.5.1 and its associated subparagraphs. The DMTC/DMTU Interface shall operate properly with the units separated by a cable of any length up to a maximum length of one hundred feet.

3.5.2.3.2 ANEW interface. The ANEW interface shall be a thirty-two-bit parallel, MIL-STD-1397, TYPE C (ANEW positive) channel operating in the Category I mode, with the DMTS performing a peripheral function except as modified herein.

3.5.2.3.2.1 Driver circuits. Whenever power to a control line driver circuit is removed, the driver shall present not less than 100,000 ohms impedance to the line and its return, with the restriction that the line voltage (due to the input circuit on the other end of the line) is within the range of plus three to plus seven volts DC and the return voltage (due to the input circuit on the other end of the return) is within the range of plus two to minus five volts DC.

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3.5.2.3.2.2 Channel connectors. The channel connectors shall conform to MIL-C-38999, Series I (100 percent scoop-proof). There shall be two receptacle type connectors with pin contacts for each ANEW channel. The output subchannel (i.e., the subchannel with drivers at the data line interface) connectors shall be keyed differently from the input subchannel (i.e., the subchannel with receivers at the data line interface) connectors.

3.5.2.3.3 PDC interface. All requirements for the PDC shall be in accordance with A30-15590. The time-out window of 1.6 us, referenced in this document, has been changed to 3.2 us for sink responses to source transmissions.

3.5.2.3.3.1 NADC DCS interface descriptions. NACD DCS Documents LAB/007, 008, 013, and 015 contain interface descriptions for the PDC sink and source logic devices. The intent of adding these descriptions is not to impose the NADC design on the contractor, but rather to amplify/clarify the functional implementation of these interfaces. For example, the sink and source status words (ES0-ES3) meanings are amplified in these documents.

3.5.2.3.4 Dual multiplex data bus. The DMTS shall interface and operate as a remote terminal on two serial data buses. Each serial data bus shall be in accordance with MIL-STD-1553 as amplified herein. The remote terminal shall be implemented as decimal address thirty-one. The DMTS shall monitor, receive and transmit control commands, data and status over either data bus (bus A or bus B). Monitoring shall be defined as the ability to receive a command sync on either bus and respond on that bus. The DMTS shall continuously monitor both buses and respond on the bus upon which it detects a legal sync. Only one bus will be utilized for a particular transfer at any given time. That is, active transfers will not occur simultaneously on both buses. Both buses shall operate under control of a bus controller (external to the DMTS) in a command/response mode. No transfer of data shall take place unless commanded by the bus controller. Each bus shall contain disable circuitry such that the bus can be set to a disabled state by receipt of a bus controller command. A new bus controller command shall not be issued until 10 usec minimum after the previous command sequence has been completed.

3.5.2.3.4.1 *I/O requirements*

3.5.2.3.4.1.1 Fault isolation resistors. The isolation resistors (R_1, Figure 15) shall be 56 ohms, plus or minus ten percent.

3.5.2.3.4.1.2 RT output levels. An RT peak signal output voltage, as measured on the bus side of the isolation resistors, of between 3.0 and 4.0 volts line-to-line shall be considered acceptable.

3.5.2.3.4.1.3 RF output waveform. Peak amplitude variations within a message shall be less than plus or minus fifteen percent of the average peak amplitude. This requirement shall apply to all pulse top distortion such as overshoot, undershoot, droop and ripple. The output waveform shall be essentially a straight line in the region of zero voltage crossing. The

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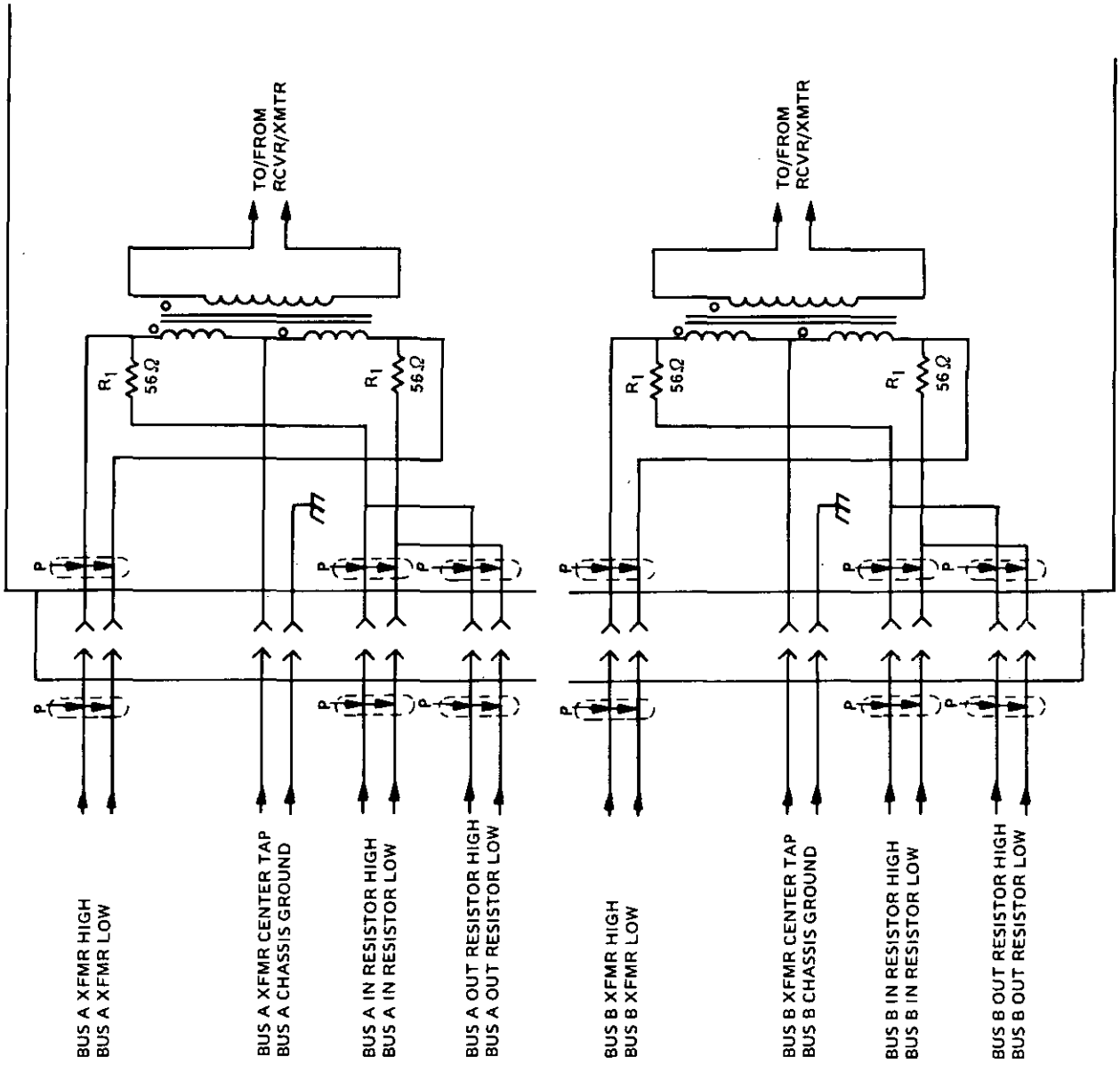


FIGURE 15. Dual multiplex data bus cable assembly diagram

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output of valid Manchester code observed at any point along a properly terminated cable up to 300 feet in length shall have a zero crossing that deviates from an ideal waveform by no more than plus or minus 15 nanoseconds. The waveform shall be symmetrical such that an energy buildup does not occur on the transmission line. The bus shall be returned to a voltage magnitude of less than 200 millivolt within one microsecond after transmission of the last bit of the last word sent to the controller. Harmonics above 1.0 MHz shall essentially be removed from the waveform prior to insertion onto the data bus. Harmonic content of the waveform shall be limited such that frequency components from 0.01 to 1.0 MHz do not deviate more than 0.6 dB referenced to average peak value. Output signal components at 1.5 MHz shall be attenuated at least 3 dB referenced to average peak value. Similarly, the component at 2.5 MHz shall be attenuated at least 13.5 dB. Attenuation of spectral components above 4.0 MHz shall be 25.5 dB or greater. Filtering shall be implemented so as to minimize phase distortion of the output waveform. Harmonic content and phase distortion requirements shall include the effects of the output driver and shall be met with its output loaded with a purely resistive 35 ohm load.

3.5.2.3.4.1.4 RT input waveform. The RT shall receive and operate properly with an input waveform varying from a square wave to a sine wave. The RT shall respond to an input signal whose positive or negative peak amplitude, line-to-line, is within the range of 10.0 to 0.5 volts measured at point "C" in Figure 7 of MIL-STD-1553. For testing purposes, the sync waveform shall be assumed to have the same rise time characteristic as a worst case valid data bit (ten percent to ninety percent of 0.5 volts in 250 nanoseconds).

3.5.2.3.4.1.5 Receiver common mode rejection. RT receiver design (and component selection) shall preserve common mode rejection beyond the differential response frequency limit of the data detector. Balanced line-to-ground impedance to the data bus conductor pair shall be maintained. Filtering shall be employed at the receiver input to decrease noise susceptibility in the presence of externally caused imbalances in line-to-ground impedances.

3.5.2.3.4.2 Coupling and termination. All connections to the data bus shall be wired as specified in Figure 15. The wiring from the RT receptacle to its associated DC isolation/coupling transformer shall not exceed 25 centimeters in length. The wiring shall be shielded twisted pair to the greatest extent possible with an overall characteristic impedance of 70 ohms line-to-line (±10 percent).

3.5.2.3.4.2.1 Connectors. The connector utilized to provide connection to the data bus shall be in accordance with Figure 15.

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3.5.2.3.5 AYK-14 special discrete signals. The DMTS shall interface with the following discrete input signals to the AYK-14 digital computer:

a. IPL (Initial Program Load)

b. System Reset

c. Master Clear (Central Processing Unit)

d. Program Select Bit 0

e. Program Select Bit 1

The DMTC shall utilize an Advanced Micro Devices AMD26LS31, or equal, differential line driver for the interfacing of these input signals. The DMTC shall interface with the following discrete output signals from the AYK-14 digital computer:

a. IPL Fail

b. BIT (Built-in-Test) Error (CPU)

c. AYK-14 Over Temperature

d. Power Normal

e. RT Master Clear

The DMTC shall utilize an Advanced Micro Devices AMD26LS33, or equal, differential line receiver for the interfacing of these output signals. The line driver and line receiver circuits shall be configured as depicted in Figure 16. The special discrete signal interface shall be as specified in Figure 17. Twisted pair wiring shall be used to the maximum extent possible. At the interface the logic signals shall be defined as follows:

a. Logic 0 (inactive level). Signal line (Signal +) at low voltage level, complement line (Signal -) at high voltage level.

b. Logic 1 (active level). Signal line (Signal +) at high voltage level, complement line (Signal -) at low voltage level.

3.5.2.3.6 DMTC/Operators Panel Interface. The DMTC/OP interface shall be designed to support all the OP control and display functions specified in 3.5.1 and its associated subparagraphs. The DMTC/OP interface shall operate properly with the units separated by a cable of any length up to a maximum of 100 feet, unless the OP is included as an integral part of DMTC.

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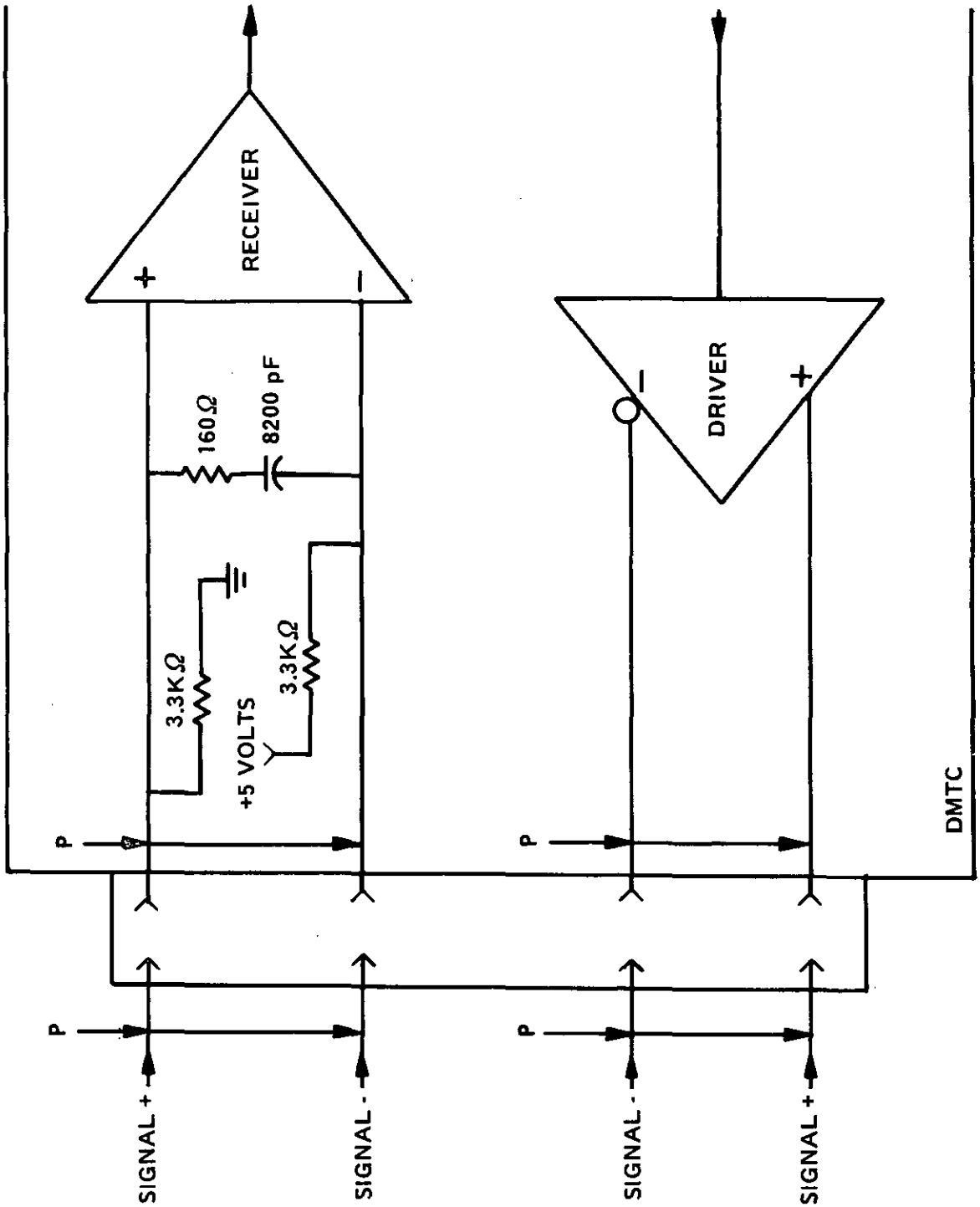


FIGURE 16. AYK-14 special discrete signals line driver and line receiver circuits

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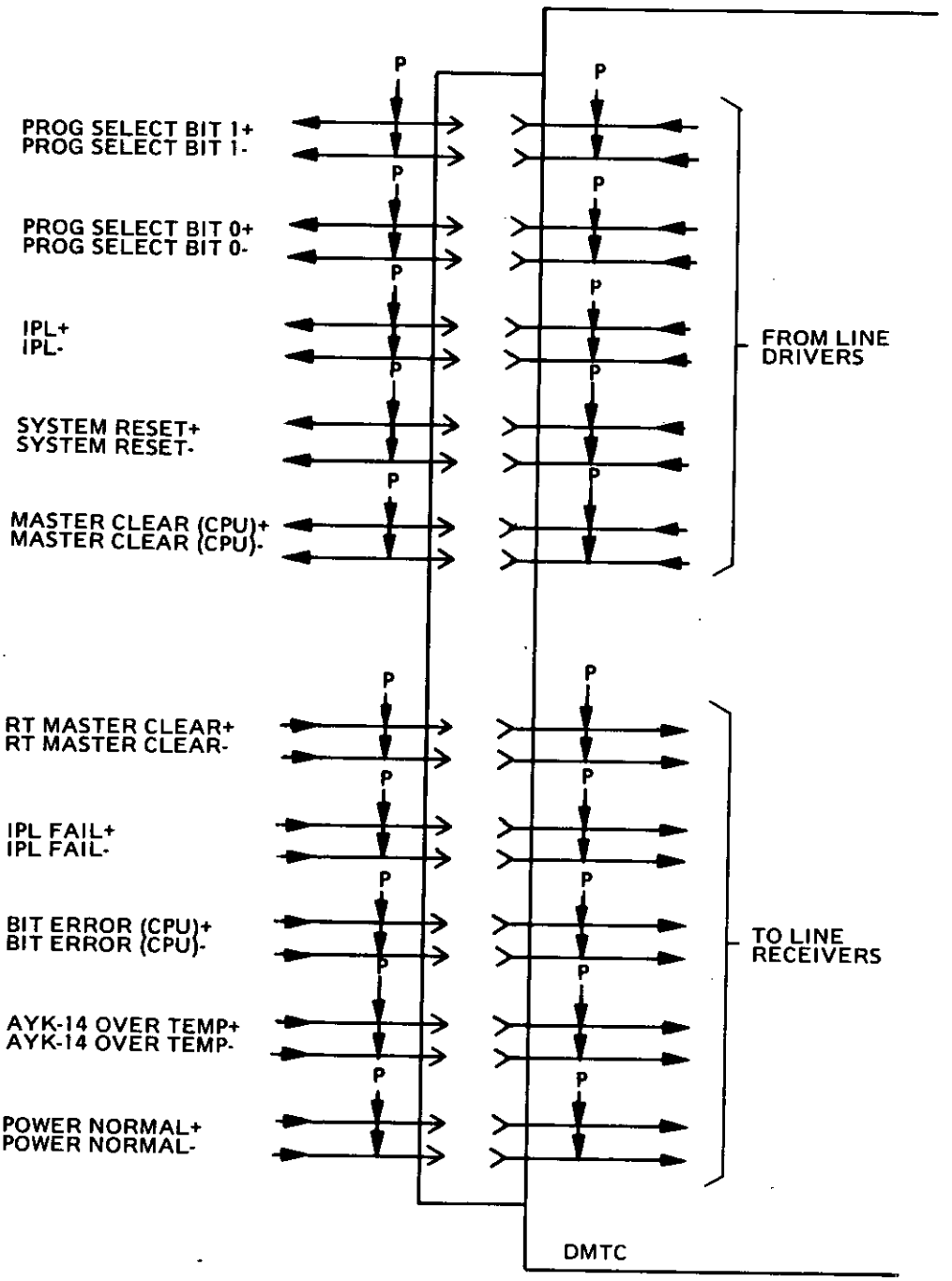


FIGURE 17. Special discrete signals cable assembly diagram

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3.5.3 igital magnetic tape unit (DMTU) description

3.5.3.1 General functional description. The DMTU shall be capable of receiving, storing and transmitting digital data with control implemented by the DMTC. The DMTU shall be designed with the necessary logic and electronic control circuitry to meet all requirements specified herein.

3.5.3.1.1 Tape Format. The DMTU specified herein shall meet all the requirements of ANSI X3.39-1973 as modified herein.

3.5.3.1.1.1 EOT/BOT markers. The tape cartridge shall contain metalized markers for BOT and EOT. The markers shall be 3.3 inches minimum in length and their width shall not exceed the magnetic tape width. Location of the markers shall be as specified in Figure 18.

3.5.3.1.1.2 Density identification area. The DMTS will not require or generate a density identification burst of recording immediately after BOT and before the first data block. The identification area shall be DC erased.

3.5.3.1.2 Performance. Unless otherwise specified herein, values set forth to establish the requirements of satisfactory performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions shall be specified.

3.5.3.1.2.1 Operation. The DMTU shall transport magnetic tape (contained within a cartridge) past a nine track read/write and erase head under control of the DMTC and perform the following functions:

a. Receive information from the DMTC and record it on magnetic tape

b. Read data from the magnetic tape and transmit it to the DMTC.

c. Receive and store instruction signals from the DMTC and modify its own operation or change its condition in accordance with the signals received.

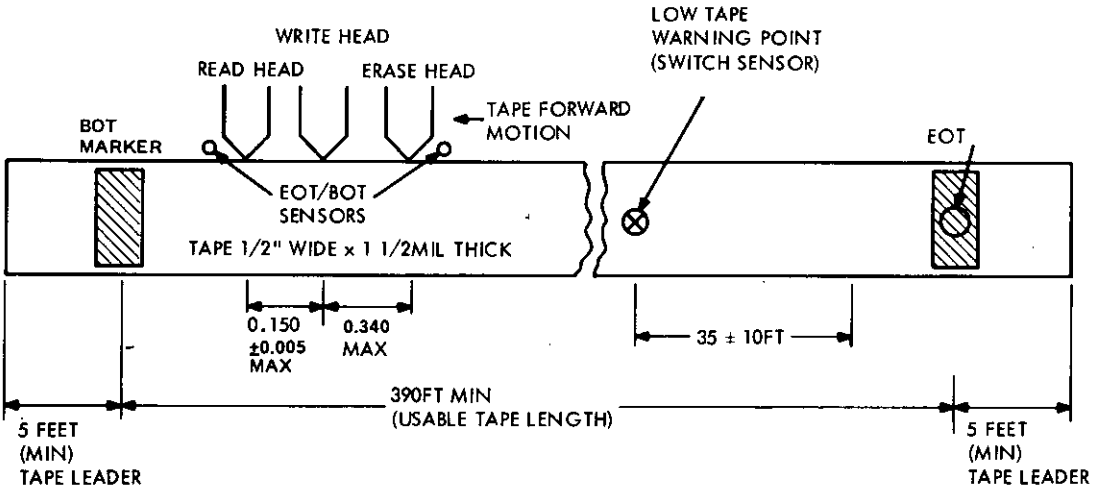
d. Transmit signals to the DMTC reflecting the DMTU's operating mode or condition.

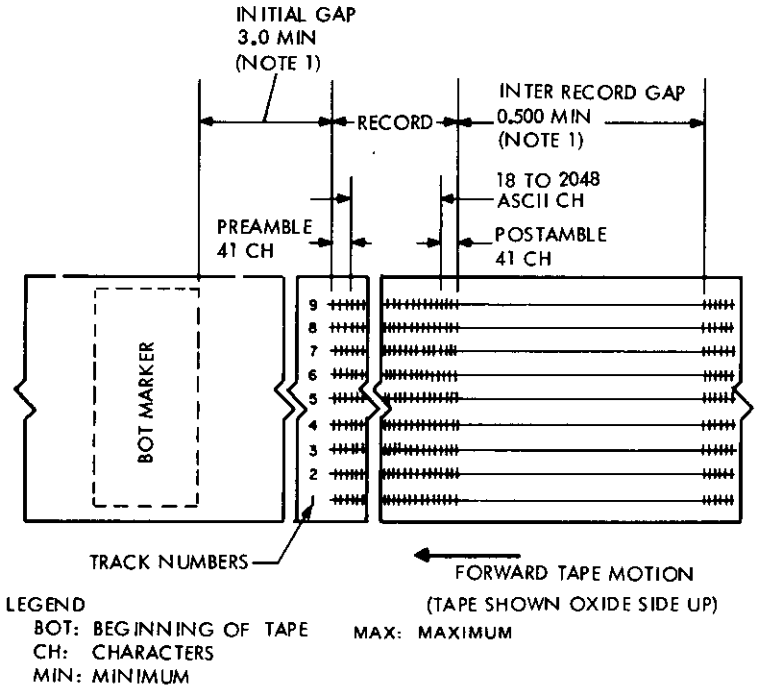
3.5.3.1.3 Detail requirements

3.5.3.1.3.1 Digital magnetic tape unit (DMTU) RF-450/ASH-33

3.5.3.1.3.1.1 General functional description. The DMTU shall receive, store, and transmit digital data with control implemented by the DMTC. The DMTU shall meet the recorded tape interchange requirements of ANSI X3.39 - 1973, except as modified herein.

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NOTES:

(1) TAPE TO BE FULLY SATURATED IN THE ERASED DIRECTION IN THE INTERBLOCK GAP AND THE INITIAL GAP.

(2) ALL DIMENSIONS ARE GIVEN IN INCHES.

(3) TAPE MARKER STRIPS ARE 3.3 INCHES LONG (MIN)

RECORDING FORMAT

FIGURE 18. DMTU Tape marker location and format

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3.5.3.1.3.1.2 Contents. The DMTU shall contain the following functional systems:

3.5.3.1.3.1.2.1 Power supply. The DMTU power supply shall consist of a DC to DC converter, over and under voltage sensors, and overload sensors. The DC to DC converter shall generate the DC voltages for DMTU operation. The overvoltage and undervoltage sensors shall provide protection for the DMTU by monitoring the DC voltages and shutting down the DC to DC converter if the sensed limits are exceeded.

3.5.3.1.3.1.2.2 Servo system. The DMTU servo system shall initiate and regulate the movement of the magnetic tape across the magnetic head in the mode commanded by the control system.

3.5.3.1.3.1.2.2.1 Normal speed. Normal tape speed shall be twenty-five inches per second.

3.5.3.1.3.1.2.2.2 Rewind speed. The tape rewind speed shall be fifty inches per second.

3.5.3.1.3.1.2.2.3 Speed accuracy. The maximum long-term speed variation shall be less than ±3 percent while the maximum peak-to-peak jitter shall be less than 10 percent. Jitter shall be any short-term speed fluctuations with respect to tape motion.

3.5.3.1.3.1.2.2.4 Start/stop time. Twenty-one milliseconds maximum at twenty-five inches per second.

3.5.3.1.3.1.2.2.5 Start/stop distance. 0.3 inches maximum at twenty-five inches per second.

3.5.3.1.3.1.2.3 Control system. The control system shall consist of the DMTU front panel, internal control switches, and the control circuit card assembly. The control shall monitor the status of the DMTU and generate logic signals to control all modes of tape motion. The DMTU shall be controlled remotely from the DMTC through the interface system. When the DMTU is set for manual mode of operation, control shall be transferred to the DMTU front panel controls.

3.5.3.1.3.1.2.3.1 DMTU front panel. The DMTU front panel, Figure 19, shall contain the control switches and indicators available to an operator. The switches and indicators are described in Table XI.

3.5.3.1.3.1.2.3.2 DMTU I/O interface. The DMTU I/O interface consists of differential line drivers (9614) and line receivers (9615) for input and output buffering (see Figure 20). These drivers and receivers shall provide interface capability with up to 100 feet of twisted pair shielded cable. The DMTU interface input and output logic definition shall be as shown in EI-680. The command inputs and status outputs are defined as follows:

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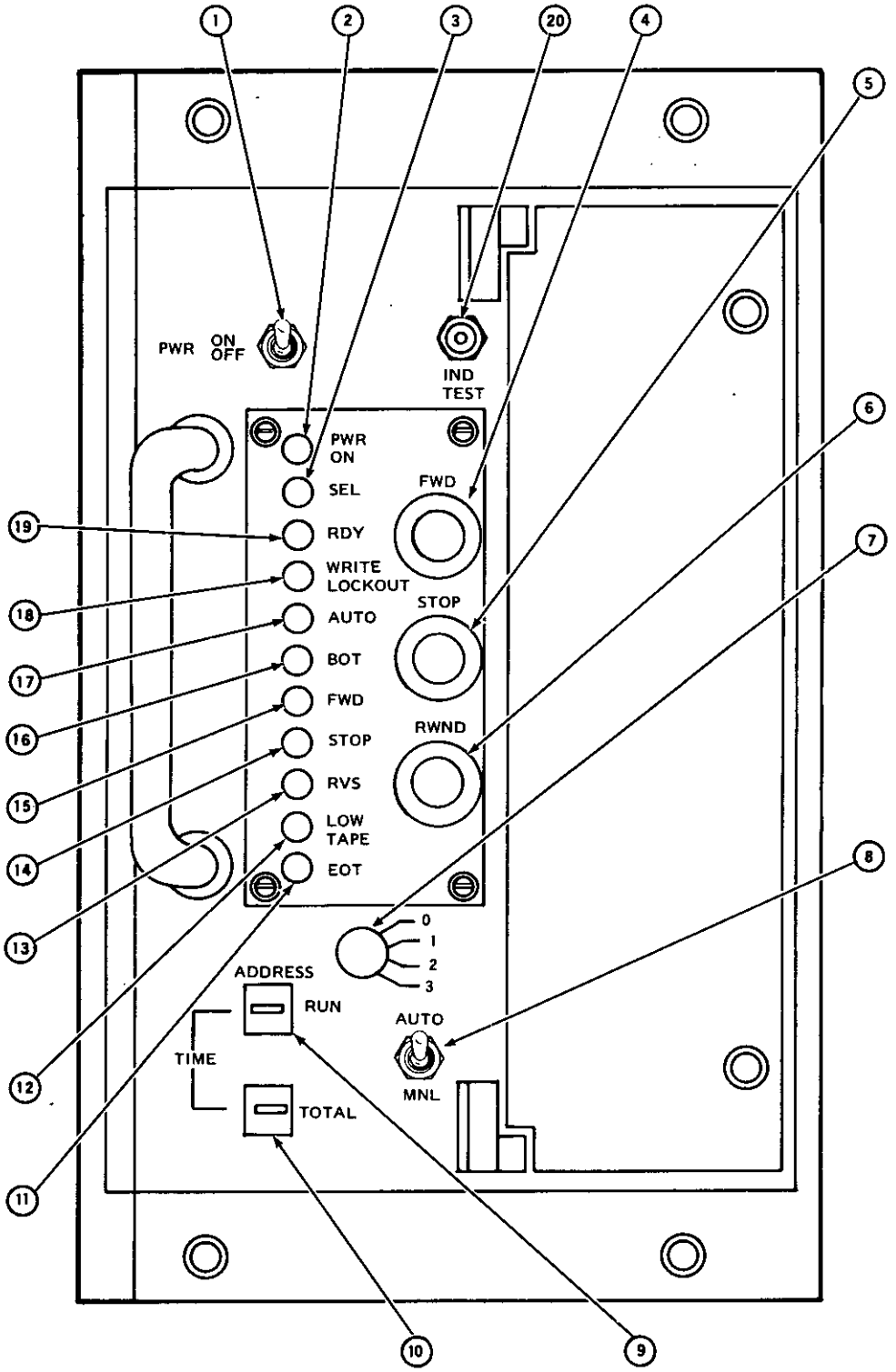


FIGURE 19. DMTU controls and indicators

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TABLE XI. DMTU controls and indicators

Figure 19 Item No.	Control or Indicator	Description
1	POWER ON/OFF (switch/ circuit breaker)	Controls primary power to DMTU.
2	PWR ON (indicator)	Indicates, when lighted, that power is on and power supply voltages are within limits.
3	SEL (indicator)	Indicates, when lighted, that DMTU is selected for On Line operation.
4	FWD (pushbutton switch)	Actuates DMTU forward tape drive when AUTO/MNL switch (8) is set to MNL.
5	STOP (pushbutton switch)	Initiates DMTU stop when AUTO/MNL switch (8) is set to MNL.
6	RWND (pushbutton switch)	Actuates DMTU high speed reverse tape drive when AUTO/MNL switch (8) is set to MNL.
7	ADDRESS (four position rotary switch)	Assigns one of four software identifier codes to DMTU. Code is 0 through 3.
8	AUTO/MNL (two position switch)	Controls DMTU On Line (AUTO)/Off Line (MNL) modes. When in AUTO position, DMTU is controlled by DMTC. When in MNL position, panel switches control tape motion.
9	RUN TIME (indicator)	Activated when tape drive is running. Indicates total tape accumulated motion time (0-9999 hours).
10	TOTAL TIME (indicator)	Activated when power is applied to DMTU. Indicates total accumulated DMTU power ON time (0-9999 hours).
11	EOT (indicator)	Lights when tape is at or beyond end-of-tape strip.
12	LOW TAPE (indicator)	Lights when tape is positioned between low tape warning point and end-of-tape strip.

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TABLE XI. DMTU controls and indicators (Continued)

Figure 19 Item No.	Control or Indicator	Description
13	RVS (indicator)	Lights when DMTU is moving tape in the reverse drive mode.
14	STOP (indicator)	Lights when tape is in stopped (no motion) condition.
15	FWD (indicator)	Lights when DMTU is moving tape in forward direction.
16	BOT (indicator)	Lights when tape is at beginning-of-tape strip (load point).
17	AUTO (indicator)	Lights when the DMTU is in AUTO mode.
18	WRITE LOCKOUT (indicator)	Lights when tape cartridge installed and tape cannot be written on.
19	RDY (indicator)	Indicates, when lighted, that DMTU power is within limits, cartridge is inserted, and access door is closed.
20	IND TEST (pushbutton switch)	Causes all panel indicator lamps to illuminate when depressed.

a. Select Command (SEL A/SEL B). Enables DMTU to receive and execute commands from the DMTC. The two select command lines are binary coded and contain the DMTU logical address. Select is acquired when the address on these lines matches the address preset by the DMTU address switch.

b. Set Write Command (SWC). This command (pulse) of twenty-four microseconds minimum width, shall enable the write electronics and cause write current to flow in the write and erase heads. This condition shall begin and continue unless or until WRITE is disabled by any of the following conditions:

1. Reset Write Command
2. Reverse Command
3. Rewind Command
4. Ready Condition is Lost

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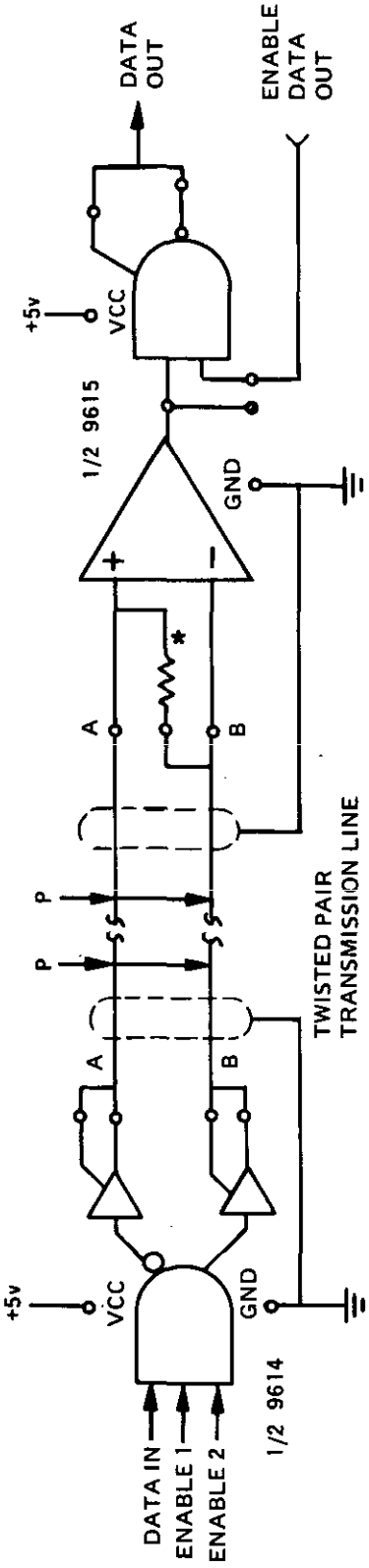


FIGURE 20. (M38510/10403)/(M38510/10404) - Fairchild 9614/9615 line driver/receiver circuit implementation simplex, balanced differential mode

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5. DMTU Manual Mode is Selected

6. Write Lockout is Set on Tape Cartridge

c. Reset Write Command (RWC). This command (pulse) of twenty-four microseconds minimum width, shall disable the write electronics and cause write current in the write and erase heads to stop.

d. Forward Command (FWC). This command (level) shall cause the DMTU to transport tape in the forward direction at 25 ips ±3 percent. Removal of this command shall cause the DMTU to execute a stop sequence.

e. Reverse Command (RVC). This command (level) shall cause the DMTU to transport tape in the reverse direction at 25 ips ±3 percent. Removal of this command shall cause the DMTU to execute a stop sequence.

f. Reverse Command (REW). This command (pulse) of twenty-four microseconds minimum width shall cause the DMTU to transport tape at 50 ips ±3 percent in the reverse direction until BOT is reached and a stop sequence is initiated.

g. Loop Test. This command (level) places the DMTU in the Loop Test Mode to check the operation of the Read and Write electronics. Tape motion is inhibited during this operation.

h. Select Acknowledge (SAK) Status. This status is sent in acknowledgment of a Select command, which matches the DMTU address switch code setting. All commands, except Select, are illegal until SAK is transmitted. SAK enables all commands, except select, and all status outputs.

i. Ready Status. Indicates cartridge inserted and locked in place, door is closed, power supply voltages are within tolerances and DMTU is ready for operation.

j. Beginning of Tape (BOT) Status. Indicates BOT marker sensed by unit and tape is positioned at load point. Any reverse/rewind command at BOT is illegal and shall not be executed. Once the BOT marker has passed the sensor in the forward direction, BOT status shall be reset.

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k. End of Tape (EOT) Status. Indicates EOT marker sensed by unit and tape is positioned at EOT. Any forward command at EOT is illegal and shall not be executed. Once the EOT marker has passed the EOT sensor in the reverse direction, EOT status shall be reset.

l. Low Tape (LT) Status. Indicates low tape has been sensed by unit and that 35 ±10 feet of tape remains before EOT is reached in the forward direction. Once the LT point has been passed in the reverse direction, LT status shall be reset.

m. Write Lockout (WLO) Status. Indicates that the cartridge is not write enabled. A set Write command in the presence of WLO status is illegal and shall not be executed.

n. Rewind (RWD) Status. Indicates the DMTU is engaged in a rewind operation.

o. Auto Status. Indicates that the DMTU is in the Auto mode and has been selected remotely. This condition is required for the DMTU to execute any input command from the DMTC.

3.5.3.1.3.1.2.3.3 DMTU data interface. The DMTU shall have erase-before-write and read-after-write capabilities. Write data shall be transmitted to the DMTU in phase encoded form of eight data bits plus parity bit. A 41-byte preamble shall precede each record to be written and a 41-byte postamble shall be sent at the conclusion of the record. Each record shall be separated by an Inter Record Gap (IRG) of 0.75 inches maximum. The maximum IRG is designed around the worst case start/stop times of the DMTU and is of sufficient distance to ensure that the DMTU will stop tape in the middle of the IRG when 25 ips ± 3 percent bidirectional read is initiated by the DMTC and the DMTC issues a Stop command at the conclusion of reading a record. Head spacing time out shall be provided by the controller. The following paragraphs describe the write and read data interface:

a. Write Data Strobe (WDS). This strobe shall be a pulse with a frequency of twice the data transfer rate. This strobe shall not be required for proper system operation and is to be used as a system test aid.

b. Write Data (nine lines). Write data in phase encoded form shall be transmitted by the DMTC after the following conditions are satisfied:

(1) Ready

(2) Select Acknowledge Status

(3) Forward Command

(4) Set Write Command

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(5) Not Write Lockout Status

(6) Not EOT Status

c. Read Data (nine lines). Data on tape shall be available to be read by the remote controller any time the DMTU is selected in Auto mode and in the Forward or Reverse Drive mode. Read shall be inhibited in the Rewind mode.

3.5.3.1.3.1.2.3.4 Data performance

3.5.3.1.3.1.2.3.4.1 Recording density. The DMTU shall record data at a density of 1600 bits per inch, assuming a record tape speed of 25 ips.

3.5.3.1.3.1.2.3.4.2 Data transfer rate. The data transfer rate between the DMTU and the DMTC during normal (25 ips) speed Read and Write operations shall be 40,000 bytes (eight bits per byte) per second ±6 percent.

3.5.3.1.3.1.2.3.4.3 Error rate. An error shall be defined as a single or multiple bit error detected by the DMTC during any Read or Read after Write operation which results in a data error at the DMTC computer output. The computer shall perform a bit-by-bit comparison of data input to data output. A transient error is defined as an error detected in which proper data can be read by repeating the Read operation a maximum of three times. Any error that causes improper data to be read on four consecutive Reads shall be considered a permanent error. As an absolute maximum, the transient (recoverable) error rate shall be one error in 10^7 bits. The maximum permanent (hard or nonrecoverable) error rate shall be one error in 10^8 bits. These error rate requirements shall apply over the useful tape life, over the operating temperature range of 0°C to 50°C. The specified error rate shall apply to any pattern of data recorded on the tape, whether it be a random, pseudorandom, or highly repetitive.

3.5.3.1.3.1.2.4 Magnetic heads. The DMTU shall contain a dual gap read and write head with an attached full width DC erase head. The read/write head shall be compatible with ANSI X3.39 - 1973. The head assembly with mounting plate attached shall be replaceable as an assembly without the need for mechanical or electrical adjustment.

3.5.3.1.3.1.2.4.1 Read/Write tracks. Nine tracks for parallel format.

3.5.3.1.3.1.2.4.2 Write gap to Read gap spacing. 0.150 ±.005 inch.

3.5.3.1.3.1.2.4.3 Total skew. The maximum skew of the combined read and write heads shall be ±25 microseconds maximum at twenty-five inches per second.

3.5.3.1.3.1.2.4.4 Erase gap to Write gap spacing. 0.340 inches maximum.

3.5.3.1.3.1.2.4.5 Erasure. The erase head shall provide DC erasure of previously recorded data to a -40 dB level minimum.

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3.5.3.1.3.1.2.4.6 Head life. 3,000 hours minimum of tape motion time, regardless of whether the tape speed is twenty-five ips or fifty ips.

3.5.3.1.3.1.2.5 Connectors. Connections to external equipment(s) shall be provided as follows:

	Reference Designator	Type Receptacle	Function
a.	2 J1	MS27656T17F08P	Power Input
b.	2 J2	MS27656T23F35P	I/O Interface

Connectors 2 J1 and 2 J2 shall meet the requirements of MIL-C-38999, Series I.

3.5.3.1.3.2 Digital magnetic tape cartridge

3.5.3.1.3.2.1 General description. The Digital Magnetic Tape Cartridge (TC) shall be a ruggedized carrier of half inch magnetic recording tape. The tape shall be contained on two coplanar hubs each of which is driven directly by capstans within the DMTU. The cartridge shall provide dust contamination protection when removed from the DMTU. Other features shall include a write lockout mechanism and hub brakes. The write lockout mechanism shall be screwdriver set. The brakes shall be engaged only when the cartridge is removed from the DMTU to prevent tape loosening.

3.5.3.1.3.2.2 Tape width. 0.498 $\pm$.002 inches.

3.5.3.1.3.2.3 Tape length. The TC shall contain 390 feet of usable tape between BOT and EOT markers. (See Figure 18.)

3.5.3.1.3.2.4 Tape markers. The TC shall contain two metalized markers 3.3 inches minimum in length for BOT and EOT sensing. The width of the markers shall not exceed the tape width and shall be applied to the back side (nonoxide) of the tape. (See Figure 18.)

3.5.3.1.3.2.5 Magnetic tape. The magnetic tape shall be certified to have no more than one permanent error per 10^6 bits.

3.5.3.1.3.2.6 Write lockout device. The cartridge shall contain a mechanical device which when set to the "safe" position will inhibit writing on or erasing of the protected magnetic tape.

3.5.3.1.3.2.7 Tape life. The tape utilized shall have a useful tape life corresponding to 5,000 cycles from BOT to EOT to BOT over the read/write heads, while retaining the specified error rate.

3.5.3.1.3.2.8 Tape/cartridge compatibility. Interchanging cartridges between DMTU's shall not cause any degradation of performance or reliability as specified herein.

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3.5.3.1.3.2.9 Weight. The weight of the TC, including tape, shall be not greater than 2.0 pounds.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 First article inspection. Unless otherwise specified in the contract, preproduction units which include design approval test (DAT) and reliability development test (RDT) units, shall be subjected to the inspection and test requirements specified in Table XII.

4.4 Quality conformance inspection. Unless otherwise specified in the contract, production equipments including first article, initial production, and acceptance test units shall be subjected to the inspection and test requirements specified in Table XIII.

4.5 Test methods

4.5.1 Presubmission testing. Each preproduction and production equipment unit shall undergo the following presubmission tests (see 6.2.2). Each spare, i.e., WRA, SRA and sub-SRA or repair part, shall undergo the following presubmission tests in accordance with Appendix A, AD-1350.

a. Examination of product (see 4.5.1.1)

b. Random vibration (see 4.5.1.2)

c. Burn-in (see 4.5.1.3)

d. Operational test (see 4.5.1.4)

4.5.1.1 Examination of product. No part, module, unit, or complete equipment shall be submitted by the contractor for any of the tests specified herein until it has been previously inspected and tested by the contractor and found to comply with all applicable requirements specified herein.

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TABLE XII. First article inspections

Inspection	Requirement Paragraph	Test Methods Paragraph
a. Presubmission	3.3	4.5.1
(1) Examination of product	3.3	4.5.1.1
(2) Random vibration	3.3	4.5.1.2
(3) Burn-in	3.3	4.5.1.3
(4) Operational	3.3	4.5.1.4
b. Thermal imaging (first model only)	3.2.4.2 3.3.2.3.3 3.3.13.7	4.5.2
c. Step stress	3.3.10.2	4.5.3
d. Design approval	3.3	4.5.4
(1) Electrical bench	3.4/3.5	4.5.4.1
(2) Temperature/altitude	3.3.10.2/ 3.3.10.3	4.5.4.2
(3) Vibration	3.3.10.1	4.5.4.3
(4) Shock	3.3.10	4.5.4.4
(5) Crash Safety	3.3.10	4.5.4.5
(6) Humidity	3.3.10.2.1	4.5.4.6
(7) Salt Fog	3.3.10.6	4.5.4.7
(8) Explosion	3.3.10	4.5.4.8
(9) Sand and dust	3.3.10.5	4.5.4.9
(10) Fungus	3.3.10	4.5.4.10
(11) Interference control	3.3.6	4.5.4.11
(12) Temperature shock	3.3.10	4.5.4.12
(13) Bench handling	3.3	4.5.4.13
e. Reliability development test (RDT)	3.3.2	4.5.5
f. Maintainability demonstration	3.3.7	4.5.6
g. Software demonstration	3.4,3.5	4.5.7

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TABLE XIII. Quality conformance inspection

Inspection	Requirement Paragraph	Test Methods Paragraph
a. Presubmission	3.3	4.5.1
(1) Examination of product	3.3	4.5.1.1
(2) Random vibration	3.3	4.5.1.2
(3) Burn-in	3.3	4.5.1.3
(4) Operational	3.3	4.5.1.4
b. First article	3.3	4.6
(1) Electrical bench	3.4/3.5	4.5.4.1
(2) Temperature/altitude	3.3.10.2/ 3.3.10.3	4.5.4.2
(3) Vibration	3.3.10.1	4.5.4.3
(4) Shock	3.3.10	4.5.4.4
(5) Crash safety	3.3.10	4.5.4.5
(6) Humidity	3.3.10.2.1	4.5.4.6
(7) Interference control	3.3.6	4.5.4.11
(8) Temperature shock	3.3.10	4.5.4.12
(9) Bench handling	3.3	4.5.4.13
c. Initial production		4.7
d. Equipment acceptance		4.8
(1) Individual		4.8.1
(2) Special tests		4.8.2
(3) Reliability assurance test	3.3.2.3	4.8.3

4.5.1.2 Random vibration. Each equipment shall undergo random vibration in accordance with Method 514.1, Procedure II, Part 3, curve AF of MIL-STD-810, as modified herein and by Figures 21 and 22. Each equipment shall undergo 10 minutes of random vibration. The vibration shall be applied perpendicular to the plane of the printed circuit boards. If different planes exist, the vibration shall be applied to each for ten minutes. After completion of the random vibration, each equipment shall be examined carefully to determine that material and workmanship requirements have been met and that no material failures have occurred.

4.5.1.3 Burn-in. Each equipment shall undergo a minimum of 50 equipment "ON" hours with the last two equipment "ON" cycles failure free. The combined thermal and vibration environment profile is shown in Figures 23 and 24. Temperature cycling shall encompass the temperature range of from -54°C to +85°C as shown in the burn-in cycle time-temperature profile of Figures 23 and 24. The vibration shall be in accordance

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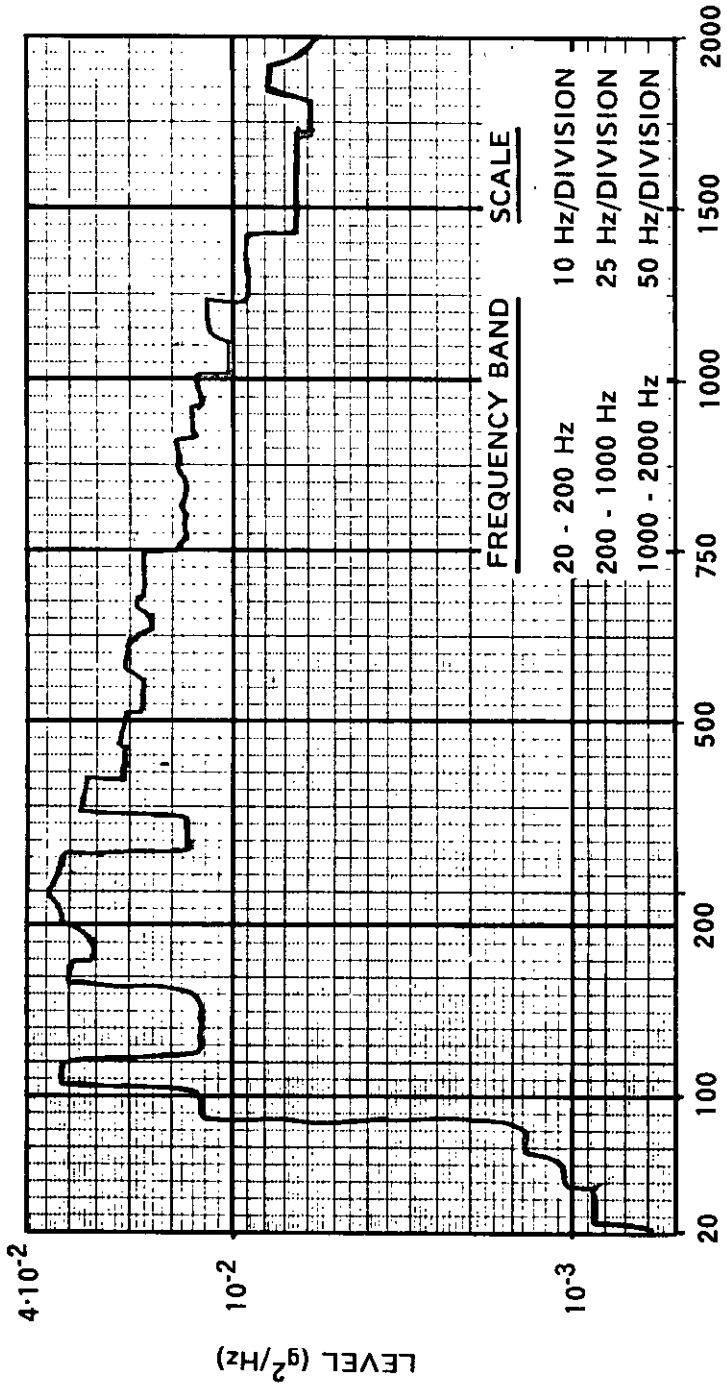


FIGURE 21. DMTU random vibration spectrum

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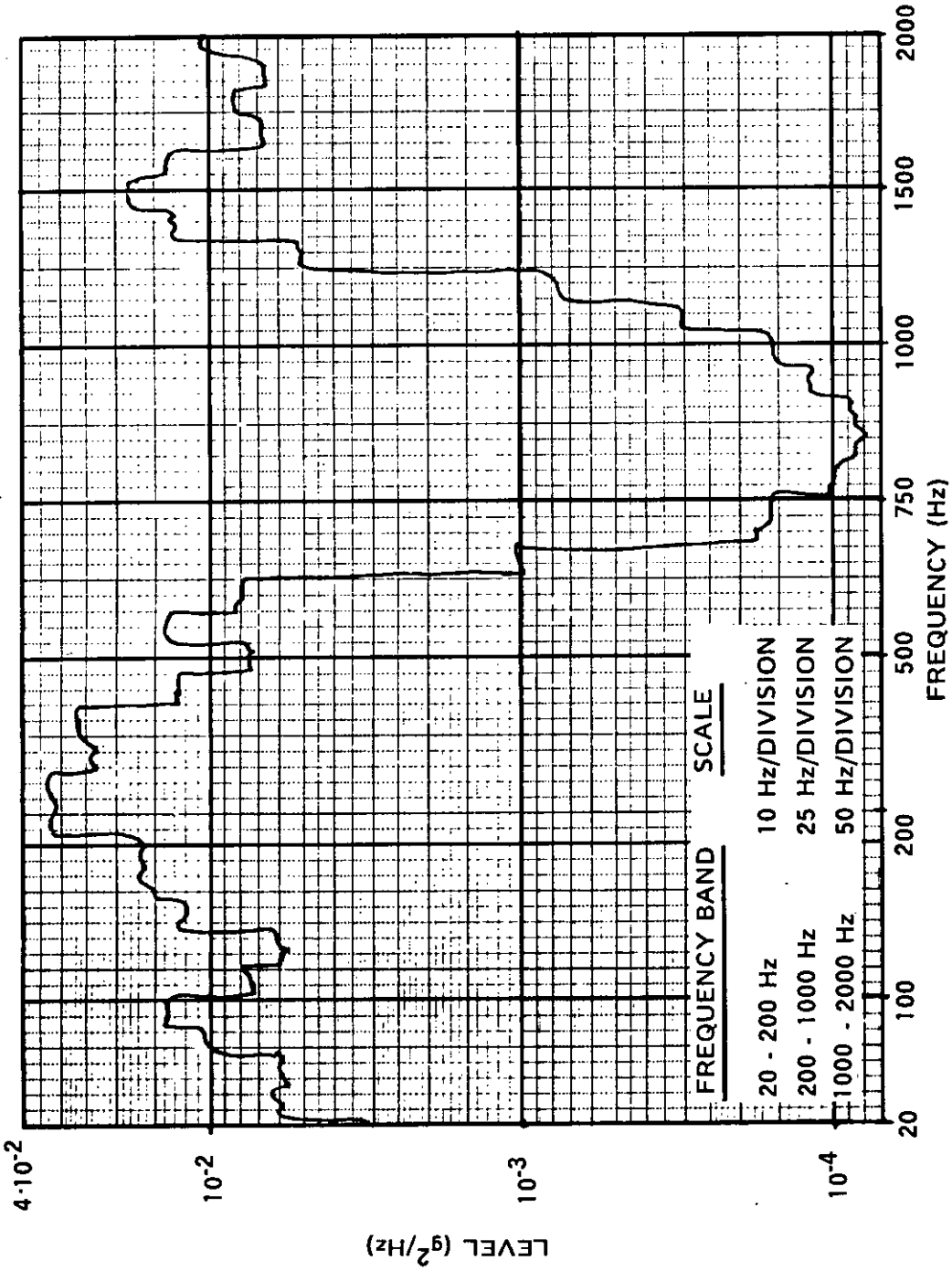


Figure 22. [MTC random vibration spectrum

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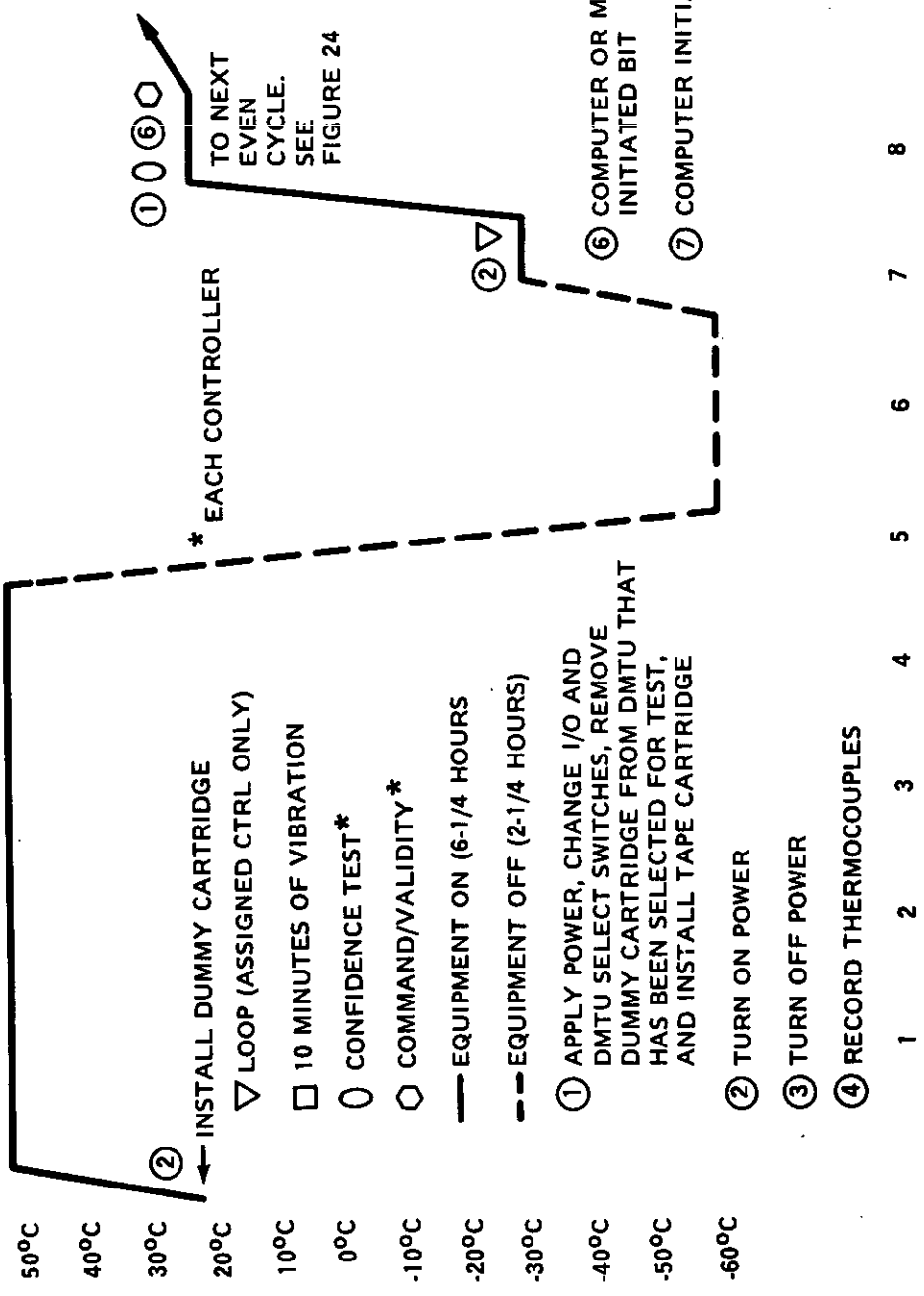


FIGURE 23. Burn-in cycle (odd cycles)

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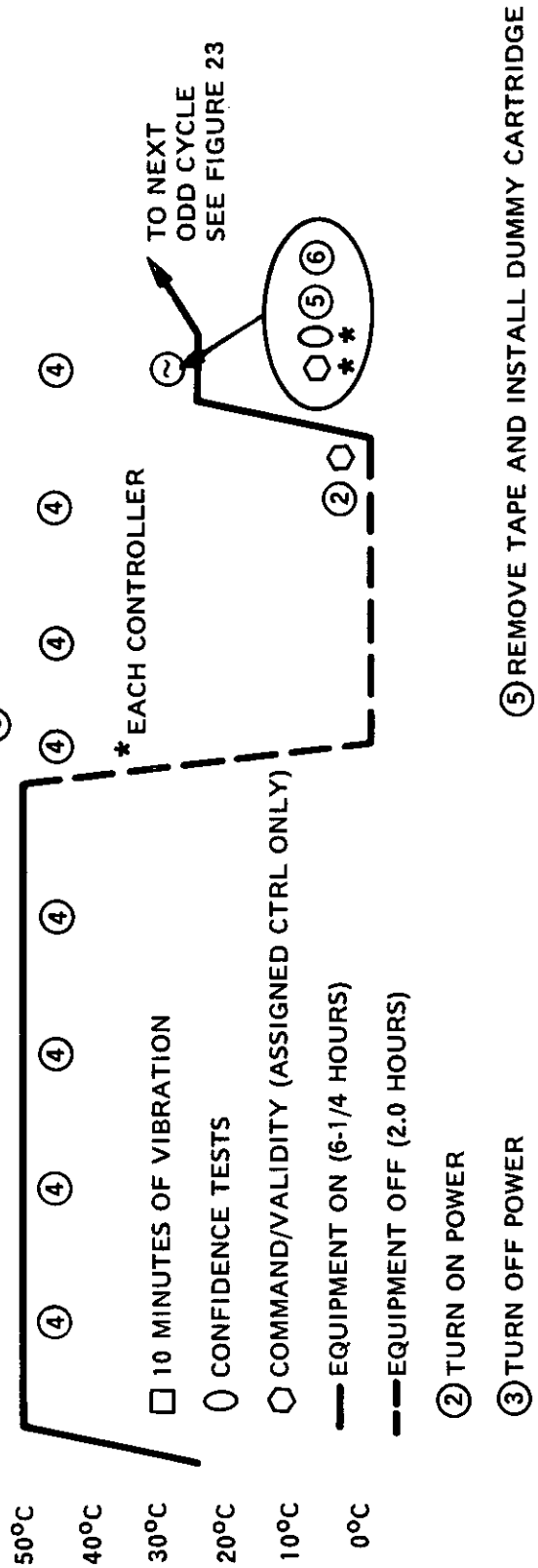


FIGURE 24. Burn-in cycle (even cycles)

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with 4.5.1.3.1. Vibration shall be applied for 10 minutes of each hour of equipment "ON" time. During each burn-in cycle, BIT shall be exercised. Also, the equipment shall be connected so that all interfaces are simulated such that all modes of operation and all inputs and outputs are exercised during each hour of equipment "ON" cycle.

4.5.1.3.1 Test cycle. Each test cycle shall be generally as defined in MIL-STD-1635 and shown in Figures 23 and 24. The following specific conditions apply.

a. Vibration: The equipment shall be vibrated for the periods shown in Figure 23 at 2 g ±10 percent peak-to-peak acceleration at 34 or 68 Hz.

b. High storage temperatures: +85°C (without tape)

c. High operating temperature: +55°C (with dummy cartridge)

d. High operating temperature: +50°C (with operational tape)

e. Low storage temperature: -54°C (without tape)

f. Low operating temperature: -25°C (with dummy cartridge)

g. Low operating temperature: 0°C (with operational tape)

h. Chamber temperature rate: 5°C per minute (minimum)

4.5.1.4 Operational test. Each equipment shall have power applied (equipment temperature stabilized) and sufficient characteristics shall be measured and recorded (including BIT) to insure satisfactory operation.

4.5.2 Thermal imaging. Thermal imaging of the first model shall be conducted prior to initiation of the first burn-in test. Each SRA shall undergo thermal imaging with an infrared thermographic imager to ensure that no major thermal problems exist in the equipment. Thermal imaging shall be used to verify the thermal analysis and to verify thermocouple locations for subsequent thermal survey and temperature measurements during environmental testing (see 6.2.2).

4.5.3 Stress test (ST). After instrumenting the components chosen during the thermal survey, the equipment shall be subjected to the following temperature tests. The tests shall be conducted with input power at the maximum values specified in 3.3.12 (see 6.2.2).

4.5.3.1 High ambient temperature. After reaching temperature stabilization at 55°C at maximum equivalent altitude, a complete functional test shall be run to verify that the equipment complies with the performance requirements of this specification. After stabilizing at 71°C at maximum equivalent altitude, verify that no component exceeds 85 percent (in °C) of its maximum rated temperature. This test may be performed simultaneously with the temperature/altitude tests of 4.5.4.2.

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4.5.3.2 Reduced air flow (forced air flow equipment only). With inlet air flow (at sea level) constricted by 80 percent, verify that no component case temperature increases by more than 5°C after stabilization.

4.5.3.3 No air flow (forced air flow equipment only). With the equipment operated for three hours with no air flow at an ambient of 27°C, verify that no component exceeds 85 percent (in °C) of its rated maximum temperature.

4.5.3.4 Corrective actions. If any temperatures reach or exceed 85 percent (in °C) of the component's maximum rated temperature in the tests of 4.5.3.1 and 4.5.3.3 or increases by more than 5°C in the test of 4.5.3.2, action shall be taken to reduce the thermal stress. Such action may consist of using a higher rated part, relocating the part, or reducing the part's ambient temperature. Corrective action must be taken for any hard failures during any of the above ST's.

4.5.4 Design approval tests (DAT's). The contractor shall conduct DAT's on development test models. Details of the test shall be defined in the test procedures (see 6.2.2) to be prepared for approval in accordance with the equipment specification and the contract data requirements. These tests shall consist of electromagnetic compatibility tests in accordance with MIL-STD-461 and MIL-STD-462 for Class 1A equipment as amended herein, a functional test including bench test in accordance with approved test procedures, and the following environmental tests in accordance with MIL-T-5422 for MIL-E-5400 Class 1A equipment, as amended herein, in the sequence presented below. (Only test details amplifying or modifying MIL-E-5422 procedures need be submitted.)

4.5.4.1 Electrical (bench) tests. These tests shall be conducted to demonstrate compliance with the specified performance parameters of 3.4 and 3.5 of this specification.

4.5.4.2 Temperature/altitude tests. These tests shall be conducted in accordance with MIL-T-5422. Temperature and altitude extremes to be used are specified in 3.3.10.2 and 3.3.10.3 herein.

4.5.4.3 Vibration tests. Tests shall be conducted in accordance with MIL-T-5422. In addition:

a. Vibration amplitude and frequency shall be in accordance with 3.3.10.1 herein.

b. Add between Steps 3 and 4 of MIL-T-5422, the equipment shall be vibrated along the axis chosen for Steps 1 and 2 with the frequency varying through the cycles of 15-19-15, then 60-76-60 and lastly, 120-152-120 Hz, at 0.036 inches double amplitude. One cycle (e.g., 15 to 19 and back to 15) shall be logarithmic with time per cycle. The equipment shall be exposed to each frequency band (15-19, 60-76, and 120-152 Hz) for 30 minutes (six cycles minimum).

c. Steps 4 and 5 shall include this new step.

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4.5.4.4 Shock test. Test shall be conducted in accordance with MIL-T-5422 (see 3.3.10).

4.5.4.5 Crash safety test. Test shall be conducted in accordance with MIL-T-5422 (see 3.3.10).

4.5.4.6 Humidity test. Test shall be conducted in accordance with MIL-T-5422 (see 3.3.10.2.1).

4.5.4.7 Salt fog test. Test shall be conducted in accordance with MIL-T-5422 (see 3.3.10.6).

4.5.4.8 Explosion test. An analysis in lieu of test is acceptable (see 3.3.10).

4.5.4.9 Sand and dust. Test shall be conducted in accordance with MIL-T-5422 (see 3.3.10.5).

4.5.4.10 Fungus test. Test shall be conducted in accordance with MIL-T-5422 (see 3.3.10).

4.5.4.11 Interference control. Equipments shall be tested to requirements of MIL-STD-461 as modified by 3.3.6 of this specification and according to the procedures specified in MIL-STD-462.

4.5.4.12 Temperature shock test. Test shall be conducted in accordance with MIL-T-5422 (see 3.3.10).

4.5.4.13 Bench handling test. Test shall be conducted in accordance with MIL-T-5422 (see 3.3).

4.5.5 Reliability development test (RDT). An RDT shall be conducted in accordance with MIL-STD-2068 for the tests and MIL-STD-1635 for plotting instantaneous data and to the requirements of this specification (see 6.2.2). Each equipment shall be subjected to the tests and inspection specified in 4.5.1 of this specification prior to RDT. RDT shall consist of the following:

a. The cumulative operating time shall be 3000 hours.

b. No sample unit shall be operated for less than 600 hours.

c. The unit under test shall be operating throughout the entire equipment ON time. Full operation shall be simulated and the equipment simulated to represent its function in the weapon system.

d. The equipment shall be judged to have satisfactorily completed the RDT when the total required operating time has been completed and upon the approval of the procuring activity.

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e. A minimum of three development equipments shall be tested.

f. Temperature cycling shall be as shown in Figures 23 and 24.

g. Vibration testing shall be in accordance with MIL-T-5422 except that the vibration levels of 4.5.1.3 shall apply. This vibration shall be applied for 10 minutes out of every hour of operating time.

4.5.5.1 Classification of failures. All problems and failures shall be classified as relevant unless determined otherwise by the procuring activity.

4.5.5.2 Problem and failure investigation. All problems, malfunctions, and failures shall be investigated to determine cause of each equipment or part problem or failure, including Government Furnished Property included in or as a part of the test item. The investigation shall consist of any applicable method, including laboratory analysis, necessary to determine the cause of failure.

4.5.5.3 Corrective action. Corrective action shall be developed and implemented for each problem or failure to prevent recurrence. Formal engineering change proposal procedures are not required, but the procuring activity will review the corrective actions at the scheduled status reviews prior to implementation. In all cases, the failure analysis and the resulting corrective actions shall be documented by the contractor. The effectiveness of the corrective action should be demonstrated by testing under the same controlled conditions that existed when the original failure occurred. Approval shall be obtained from the procuring activity when failures require corrective action.

4.5.6 Maintainability demonstration tests

4.5.6.1 Self-test acceptance tests. A self-test acceptance test (see 6.2.2) will be conducted in a laboratory environment on one or more sets specified by the contract prior to the maintainability demonstration. The maintainability requirements specified in 3.3.7.4.2 shall be demonstrated by verifying that the self-test procedures are adequate to detect and isolate faults as required. The primary purpose of the self-test acceptance test will be to demonstrate that the automatic fault detection and isolation capability intended for use at organizational levels are capable of performing as specified in 3.3.7.4.1.

4.5.6.1.1 Test conditions. Before beginning the test, all equipment shall be checked to verify proper operation. All equipment, tools, fault insertion devices, and spares shall be ready for the test.

4.5.6.1.2 Fault catalog. During the contractor's hardware and self-test design phases, a fault catalog shall be developed which reflects loss or degradation of function caused by or traceable to a specific piece part failure. Faults for the catalog may also be developed, in part, by shorting to ground or opening various connections made to nonrepairables, such as IC's and module connectors. The fault catalog shall consist of four hundred faults.

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4.5.6.1.3 Fault insertion. Sample size distribution and fault selection shall be in accordance with MIL-STD-471. Unless otherwise approved by the procuring activity, 200 faults shall be individually inserted followed by execution of the readiness test and self-test functions, execution of technician instructions (if any), and documentation of results (see 6.2.2). The contractor shall develop and utilize a special module(s), or fault insertion devices which permits faults to be inserted without damaging end item modules. The contractor may use these devices to aid in validation testing during the design phase. The device(s) shall be deliverable item(s).

4.5.6.1.4 BIT/BITE verification test. The BIT/BITE shall be judged to pass the acceptance test only if the detection and isolation requirements specified under 3.3.7.4.1 herein are satisfied. Failure to satisfy this requirement shall constitute grounds for rejection.

4.5.6.1.5 Rejection and retest. The BIT/BITE which has been rejected because of failure to meet any element of the acceptance criteria shall be corrected and resubmitted for acceptance test. Before submittal, full particulars concerning previous rejection, together with correction action implemented, shall be submitted to the contracting activity for approval. Any required retest shall begin with the random selection of faults or shall utilize the same criteria as applied to the initial test.

4.5.6.2 Maintainability demonstration. A maintainability demonstration test (see 6.2.2) shall be conducted in accordance with MIL-STD-471, Test Method 9 to prove all the requirements specified herein have been met. The maintainability demonstration shall not start until the BIT/BITE verification test has been passed.

4.5.6.2.1 Test conditions. The maintainability test shall be conducted in a laboratory environment. One or more equipments shall be used for this test. The equipment to be used shall have passed all applicable tests required for a typical production system. All dust covers shall be in place on the equipment prior to the initiation of this test. Before beginning the test, the equipment shall be checked to determine that it is operating normally. Test equipment, tools, spare parts and test programs shall be provided by the contractor. Documentation shall be included.

4.5.6.2.2 Repair actions. A repair action shall be as follows:

a. Recognition of a fault

b. Fault isolation to a module

c. Module replacement

d. Equipment checkout

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4.5.6.2.3 Simulated failures. Failures shall be randomly introduced, one at a time, throughout the equipment. The points of failure shall be approved by the procuring activity prior to initiation of test.

4.5.6.2.3.1 Inducing failures. Single failures shall be induced into the equipment by substituting modules that are known to be defective, or, where defective modules are not available, simulated failures may be induced by other methods when approved by the procuring activity. Any failure induced for demonstration purposes will consist of a single continuous malfunction and shall exclude those types of failures specified in 3.3.7.4.

4.5.6.2.4 Repair time measurements. The time required for the technician to perform each repair action, including the use of technical manuals and support equipment, shall be measured starting at the time the technician is made aware that a malfunction exists and ending at the time the equipment is restored to normal operation. Any delay, such as waiting for parts, obtaining test equipment, or administrative delays, shall be excluded.

4.5.6.2.5 Accept/reject criteria. The accept/reject criteria for the demonstration of corrective maintenance times shall be in accordance with MIL-STD-471, Test Method 9. An accept decision can only be made when an accept decision is made for all indices.

4.5.6.2.6 Technicians. The procuring activity shall provide and select the technicians to perform maintainability demonstration.

4.5.6.2.7 Technical documentation. Technical documentation, if required, shall be limited to the equipment technical manual typical of an organizational maintenance manual.

4.5.6.2.8 Acceptance. The maintainability demonstration shall demonstrate that the system has met the accept criteria, prior to contract delivery.

4.5.6.2.9 Rejection. If a reject decision is reached, the procuring activity concerned shall be immediately notified. The contractor shall, at no additional cost to the procuring activity:

a. Develop a plan for design or material changes adequate to correct all deficiencies, and

b. Upon approval of the plan, perform corrective action and repeat the demonstration until an accept decision is reached.

4.5.7 Software tests. Software tests shall be performed in accordance with the contract (see 6.2.2).

4.6 First article testing. The initial production units shall be used for the following tests (see 6.2.2):

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a.	Electrical bench	(see 4.5.4.1)
b.	Temperature/altitude	(see 4.5.4.2)
c.	Vibration	(see 4.5.4.3)
d.	Shock	(see 4.5.4.4)
e.	Crash safety	(see 4.5.4.5)
f.	Humidity	(see 4.5.4.6)
g.	Salt fog	(see 4.5.4.7)
h.	Explosion	(see 4.5.4.8)
i.	Interference control	(see 4.5.4.11)
j.	Temperature shock	(see 4.5.4.12)
k.	Bench handling	(see 4.5.4.13)

4.7 Initial production tests. Unless otherwise specified in the contract, one of the first ten production equipments shall be selected and sent at the contractor's expense to a designated government laboratory for test. This equipment shall be selected by procuring activity after having successfully passed the acceptance test in accordance with 4.9.5 of this specification.

4.7.1 Scope of tests. This equipment shall be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equal to the previously approved preproduction sample in design, construction, workmanship, performance, and quality, and that it meets all applicable requirements.

4.7.2 Accessory material. In addition to the complete equipment submitted for initial production tests, the contractor shall also submit such accessory material and include data necessary to test the equipment.

4.7.3 Initial production sample approval. Approval of the initial production equipments shall be by the procuring activity upon satisfactory completion of all tests. Any design, material, or performance defect made evident during the test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is requested by the procuring activity.

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4.8 Equipment acceptance tests. The contractor shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the representative(s) of the procuring activity. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall prepare, for each equipment test, reports showing qualitative results for all acceptance tests (see 6.2.2). Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

a. Individual test (see 4.8.1)

b. Special tests (see 4.8.2)

c. Reliability assurance (see 4.8.3)

4.8.1 Individual tests. Each equipment submitted for acceptance shall have passed, as a minimum, the presubmission tests of 4.5.1. These tests shall be adequate to determine compliance with the requirements of materials, workmanship, operational adequacy, and reliability.

4.8.2 Special tests. Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control.

4.8.2.1 Special test schedule. Selection of equipment for special tests shall be made as follows:

a. On an early equipment after an engineering or material change, and

b. Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

4.8.2.2 Scope of tests. Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure (see 6.2.2) for procuring activity approval prior to conducting the tests.

4.8.2.3 Equipment failure. Should a failure occur during either the individual or special tests, the following action shall be taken:

a. Determine the cause of failure and the equipment BIT performance in detecting and isolating the failure

b. Determine if the failure is an isolated case or design defect

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c. Prepare for procuring activity approval proposed corrective action intended to reduce the possibility of the same component or BIT failure occurring in future tests or documented justification where no corrective action has been proposed (see 6.2.2)

d. Where practical, include a step in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.8.3 Reliability assurance tests. Reliability assurance tests (see 6.2.2) shall be conducted in accordance with MIL-STD-781. Tests as required by both the qualification phase and the production reliability testing shall be conducted. Classification of failure shall be in accordance with MIL-STD-2074.

4.8.3.1 Qualification phase. Prior to the acceptance of equipment under the contract or order, a minimum of three equipments but no more than five shall be tested in accordance with MIL-STD-781, as modified herein. For the qualification phase, the environmental conditions of 4.5.1.3 shall be used along with the thermal cycle of Figures 23 and 24. Also, the vibration conditions specified in 4.5.1.3 shall be used. The accept/reject criteria shall be as specified in test plan IIIC of MIL-STD-781. No single equipment "ON" time shall be less than 60 percent of the average operating time of all the equipment under test.

4.8.3.2 Production reliability tests. The equipment throughout production shall be tested as outlined in MIL-STD-781 and as modified herein. Test Plan XVIII C of MIL-STD-781 shall be used, except that the requirements of 4.5.1.3 shall be used. The thermal cycle of Figures 23 and 24 shall apply. The classification of failures shall be in accordance with MIL-STD-2074.

4.9 General test requirements

4.9.1 Test details. The test details, such as the length of the test cycle, length of the heat portion of the cycle, performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures (see 6.2.2) to be approved by the procuring activity prior to the beginning of the qualification test phase of the reliability assurance tests.

4.9.2 Test plans and procedures. Test plans and procedures (see 6.2.2) shall be prepared by the contractor for procuring activity review and approval, in accordance with contract data requirements. MIL-T-18303 shall be used as a guide for preparation of test procedures. Approved test procedures from previous contracts may be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

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4.9.3 Presubmission testing. No item, part, or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply with all applicable requirements.

4.9.4 Time metering. WRA's time meters shall be energized by the contractor when the units receive their initial check. Time logs shall be maintained by the contractor to record the total operating time of components.

5. PACKAGING

5.1 General. All WRA's and parts of the DMTS shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract or purchase order, in accordance with MIL-E-17555, MIL-STD-794, Level A and MIL-P-116, Method 11A. In the event the equipment is not covered in MIL-E-17555, the method of presentation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended use. The DMTS is to be used on the P-3B and P-3C weapon systems.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this document.

b. The specific equipment configuration desired (see 1.1).

c. Levels of preservation-packaging and packing (see 5.1).

d. Reliability tracking, if required (see 3.3.2.3.2).

e. Software tests to be performed (see 4.5.7).

f. The material and data required for initial production tests. (see 4.7.2).

g. Modified test plans and procedures or additional tests, if required (see 4.9.2).

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6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contracts Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved CDRL incoporated in the contract. When provisions of DAR 7-104.9(n)(z) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data requirement title	Applicable DID No.	Options
3.2	Military Detail Specifications and Specification Sheets	DI-E-7029	-
	Specification Requirement Sheets	DI-E-1133	-
	Drawings, Engineering and Associated Lists	DI-E-7031	-
	Test Data for Nonstandard Parts	DI-E-7030	-
	Parts Control Program Plan (not required if MIL-STD-680 is contractually specified)	DI-E-7026	-
3.2.1	Requests, Nonstandard Parts, Materials and Non-repairable Parts Approval	DI-E-7028A	-
	Data, Microelectronic Devices and Assemblies, Approval Requests	UDI-E-21337	-
3.2.4.1	Microcircuit Material Item and Linkage Record	DI-E-2182	-
3.2.4.2	Reliability Stress Analysis Report	UDI-R-21423	-
3.2.5	Request for Deviation/Waiver	DI-E-3129	-
3.3.2	Reliability and Maintainability Program Report (Plan)	UDI-R-21131	-

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Paragraph No.	Data requirement title	Applicable DID No.	Options
3.3.2 (Cont'd)	Failure Mode, Effects, and Criticality Analysis Report	DI-R-7085	-
	Reliability Block Diagrams and Mathematical Models Report	DI-R-7094	-
3.3.2.2	Reliability and Maintainability Program Report (Plan)	UDI-R-21131	-
3.3.4	Rack or Console Mounted Control Drawing Requirements	UDI-E-21339	-
3.3.6.1	Electromagnetic Interference Control Plan	DI-R-7061	-
3.3.6.2.2	Electromagnetic Interference Control Plan	DI-R-7061	-
3.3.7	Maintainability Demonstration Plan	DI-R-2129	-
3.3.7.3	Request for Deviation/Waiver	DI-E-3129	-
3.3.7.3.1	Request for Deviation/Waiver	DI-E-3129	-
3.3.7.3.6	Requests, Nonstandard Parts, Materials, and Nonrepairable Parts Approval	DI-E-7028A	-
3.3.7.3.10	Maintenance Support Plan	DI-S-1813	-
3.3.7.4.2	System Component Maintenance Diagnostic Programs	DI-H-5073	-
	Report, Design Trade-off Studies	UDI-E-23978	-
3.3.10.1	Report, Vibration Survey	DI-R-7037	-
3.3.19	Rack or Console Mounted Control Drawing Requirements	UDI-E-21339	-

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Paragraph No.	Data requirement title	Applicable DID No.	Options
3.4.4	Interface Design Specification	DI-E-2135	-
	Software Development Plan	DI-A-2176	-
	System Operational Specification	UDI-E-21400	-
	System Operational Design Document	UDI-E-21401	-
	Data Base Design Document	DI-S-2140	-
	Program Performance Specification	DI-E-2136A	-
4.5.1	Test Procedures	UDI-T-21347	-
4.5.2	Test Procedures	UDI-T-21347	-
4.5.3	Test Procedures	UDI-T-21347	-
4.5.4	Test Procedures	UDI-T-21347	-
	Report, Design Approval Test	UDI-T-21350	-
4.5.5	Reliability Test Procedures	DI-R-7035	-
	Reliability Test and Demonstration Report	DI-R-7034	-
	System/Equipment Failures Analysis and Corrective Action Report	DI-R-5283	-
4.5.6.1	Test Procedures	UDI-T-21347	-
4.5.6.1.3	Maintainability Demonstration Report	DI-R-2130A	-
4.5.6.2	Maintainability Demonstration Report	DI-R-2130A	-
	Maintainability Demonstration Plan	DI-R-2129	-

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Paragraph No.	Data requirement title	Applicable DID No.	Options
4.5.7	Interface Design Specification	DI-E-2135	-
	Software Quality Assurance Plan	DI-R-2174	-
	System Operator's Manual	DI-M-2148	-
	Computer Program Test Specification	DI-E-2143	-
	Computer Program Test Procedures	DI-T-2144	-
	Computer Program Test Report	DI-T-2156	-
4.6	Test Procedures	UDI-T-21347	-
	Design Approval Test Report	UDI-T-21350	-
4.8	Acceptance Test Results	UDI-R-20338C	-
4.8.2.2	Test Procedures	UDI-T-21347	-
4.8.2.3(c)	System/Equipment Failure Analysis and Corrective Action Report	DI-R-5283	-
4.8.3	Reliability Test Procedures	DI-R-7035	-
	Reliability Test and Demonstration Report	DI-R-7034	-
4.9.1	Test Procedures	UDI-T-21347	-
4.9.2	Test Procedures	UDI-T-21347	-
	Software Quality Assurance Plan	DI-R-2174	-
	Maintainability Demonstration Plan	DI-R-2129	-

(Data item descriptions related to this specification, and indentified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractor in

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connection with specified acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the procuring activity.)

6.3 First article. When a first article inspection is required, the item will be tested and should be a preproduction unit(s). The first article should consist of five DMTS units plus two DMTU's. The contracting officer should include specific instructions in acquisition documents for examinations, test approval of the documents first article.

6.4 Supersession data. This specification includes the requirements of MIL-STD-461A(3), dated 1 May 1970, MIL-STD-462(2), dated 1 May 1970, MIL-STD-704A, dated 9 August 1966, MIL-STD-1553A, dated 30 April 1975, and MIL-E-5400R, dated 31 October 1975.

6.5 Performance objectives. Minimum size and weight, simplicity of operation, ease of maintenance, and improvement in the performance and reliability of the specified functions beyond the requirements of this specification are objectives which should be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval should be submitted to the procuring activity for consideration. Each request should be accompanied by complete supporting information.

6.6 Definitions

6.6.1 Useful life. The period of time from delivery of the specified item to the developing activity until its identity is destroyed by classifying it as salvage.

6.6.2 Operating life. The number of hours the specified item must operate satisfactorily in the required environment, maintenance being accomplished in accordance with applicable directives.

6.6.3 Duty cycle. The performance profile or time phase apportionment of modes of operation and functions to be performed by each equipment during the on-time portion of the environmental test cycle. It is intended that the duty cycle be representative of field operation.

6.6.4 Preproduction (equipment). An equipment suitable for complete evaluation of form, fit, and performance. It is in final form in all respects, employs standard parts (or nonstandard parts approved by the procuring activity).

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6.6.5 Nonrepairable subassemblies. As a general rule, nonrepairable subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high-voltage power supplies), the nonrepairable subassembly should evidence a Mean-Time-to-Failure greater than 50,000 hours.

6.7 Type designations. The parenthesis (), when used in the equipment type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon assignment of nomenclaure in accordance with 3.3.8. The complete type number will be used on nameplates, shipping records and instruction books, as applicable.

6.8 Associated equipment. The following are the equipments associated with the DMTS.

a. Avionics Unit Computer, CP-901/ASQ-114, MIL-C-81332

b. Computer Set, Standard Airborne, AN/AYK-14, AS-4197(AV)-1

c. Display and Control System, AN/USQ-78(V), MIL-A-85234

Preparing activity:
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